

Sri Lanka Institute of Information Technology



Network Design & Management

Lab Report 01

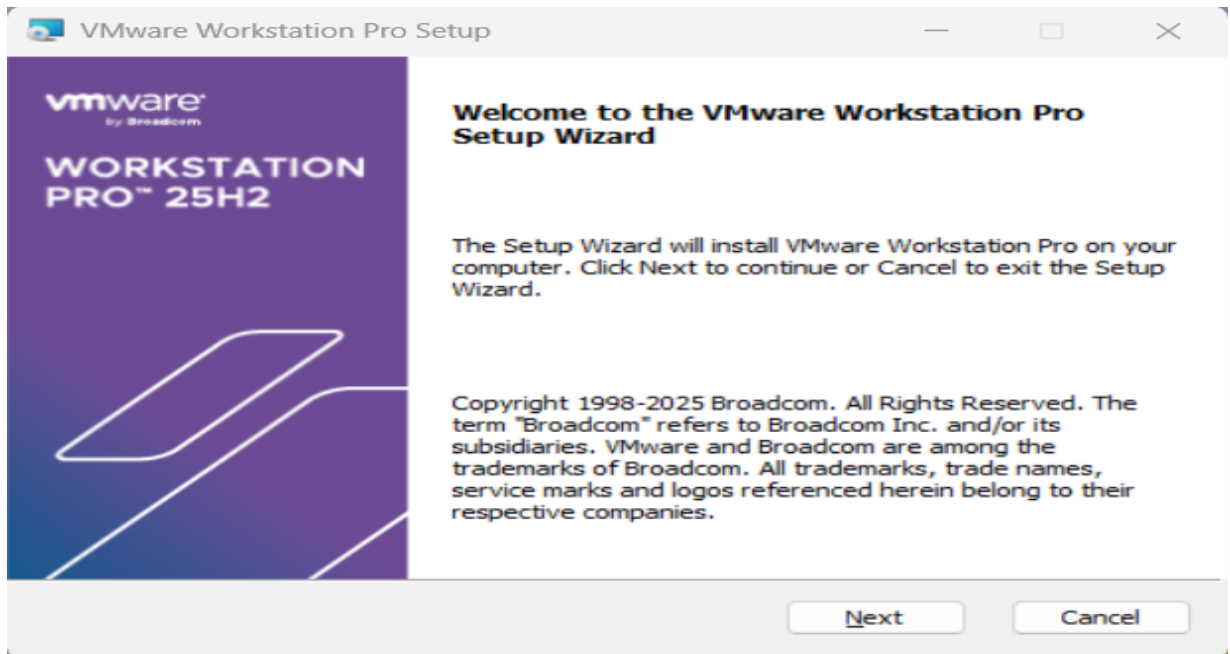
IT23624030 - W.M.R.S.K.Wijesinghe

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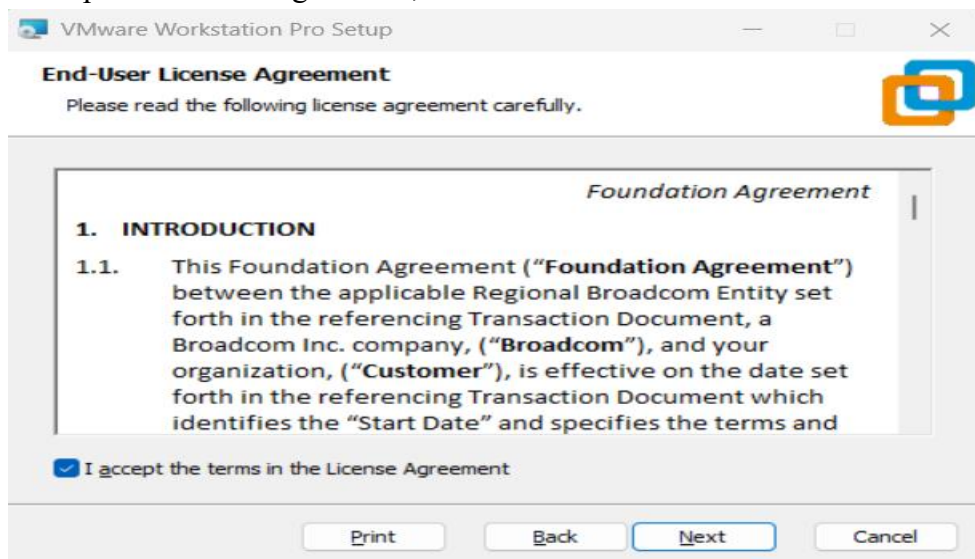
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Download and Install VMware

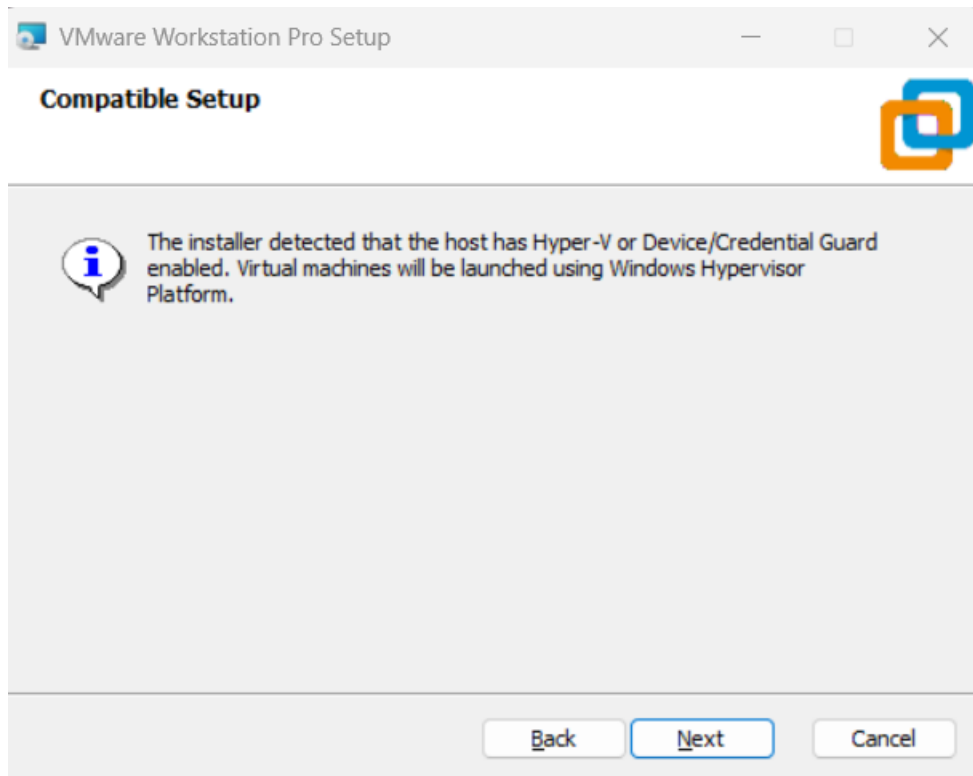
1. Download VMware
[VMware-Workstation-Full-25H2-2.exe](#)
2. Open the downloaded installer file.
3. Install the VMware in our device.
 - Click 'Next'.



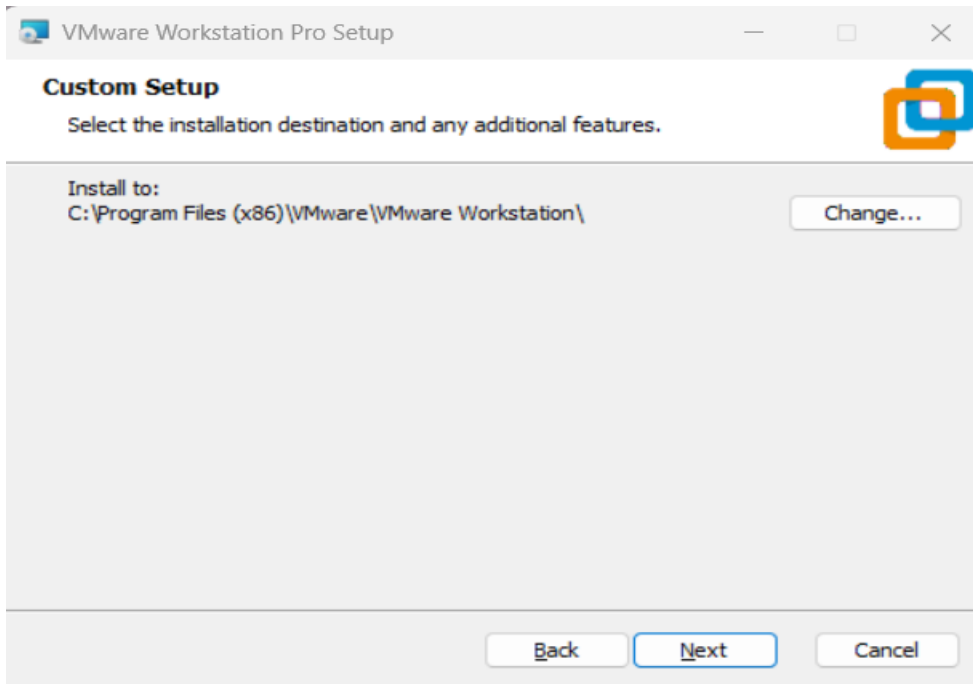
- Accept the License Agreement, then click 'Next'



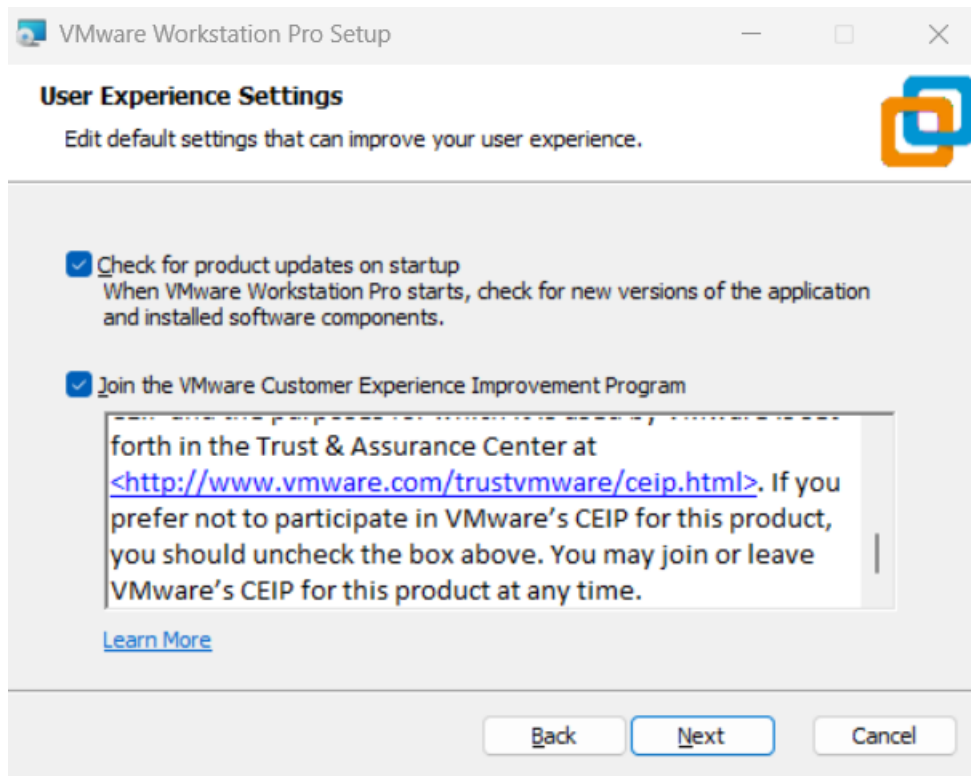
- Click 'Next'.



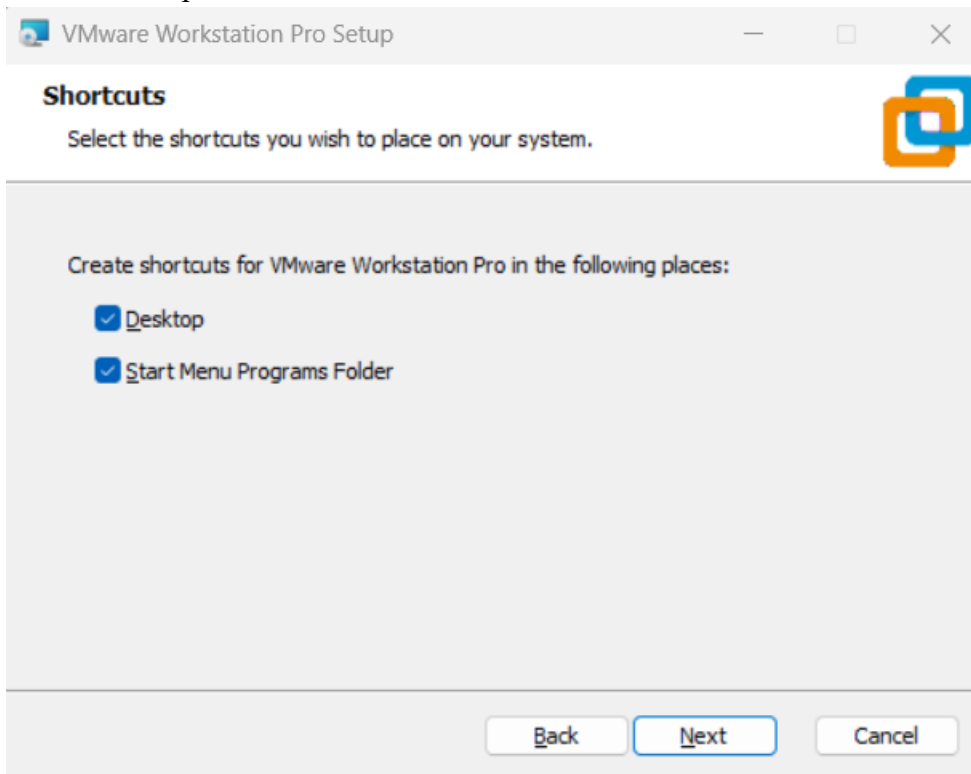
- Choose the installation location, then click 'Next'.



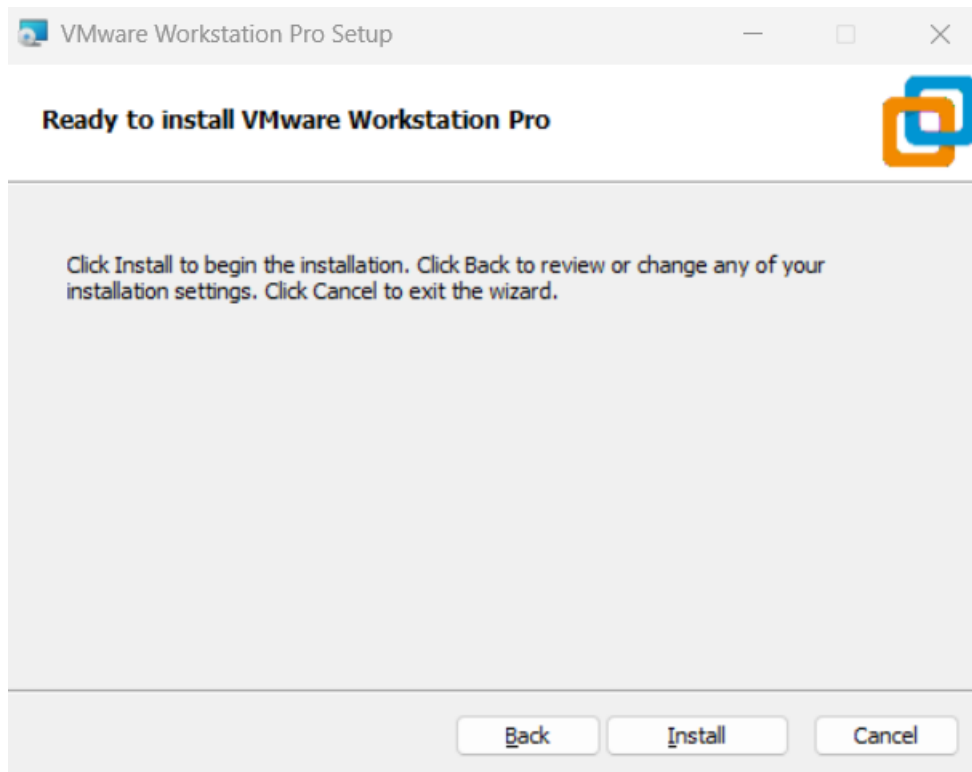
- Select two checkboxes, then click 'Next'.



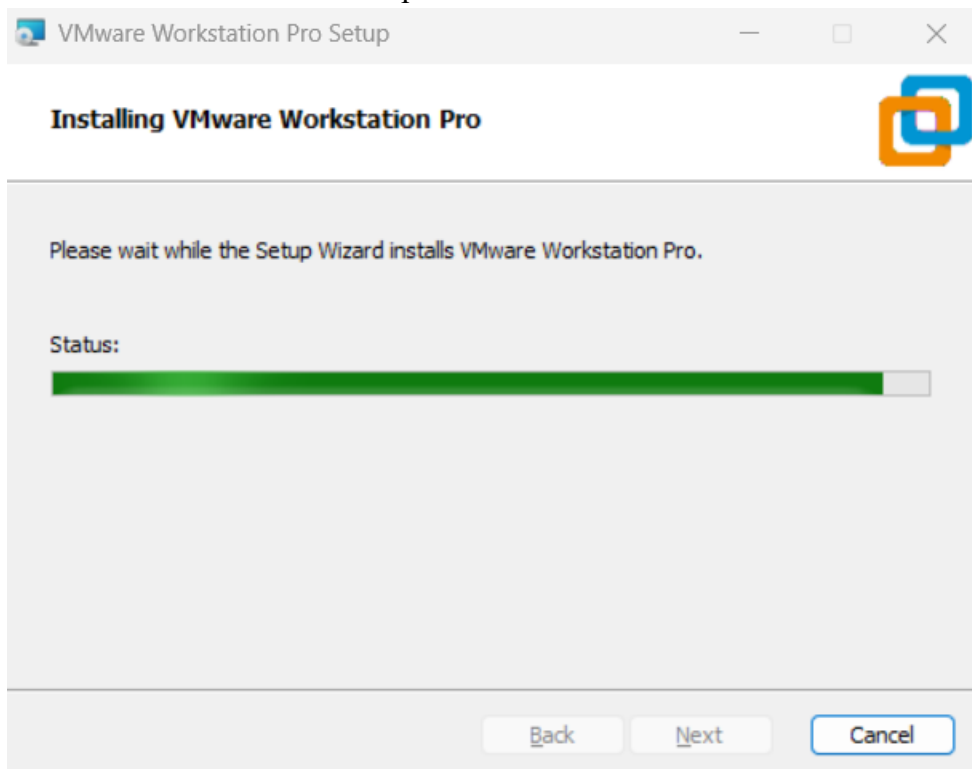
- Select both options, then click 'Next'.



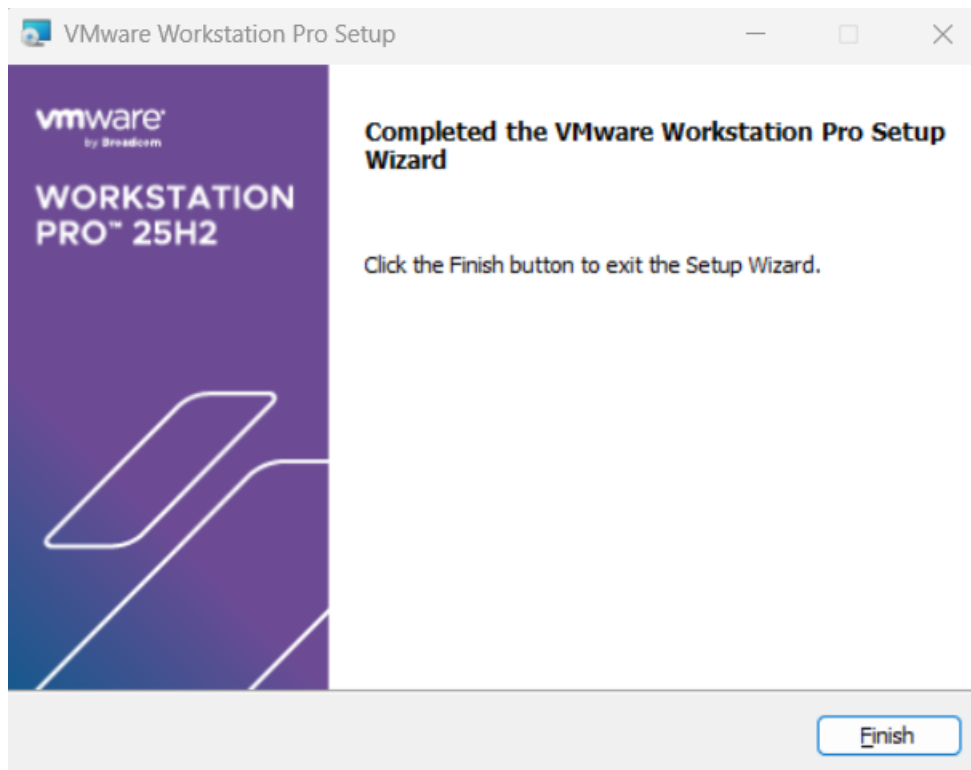
- Click 'Install'.



- Wait until the installation completes.

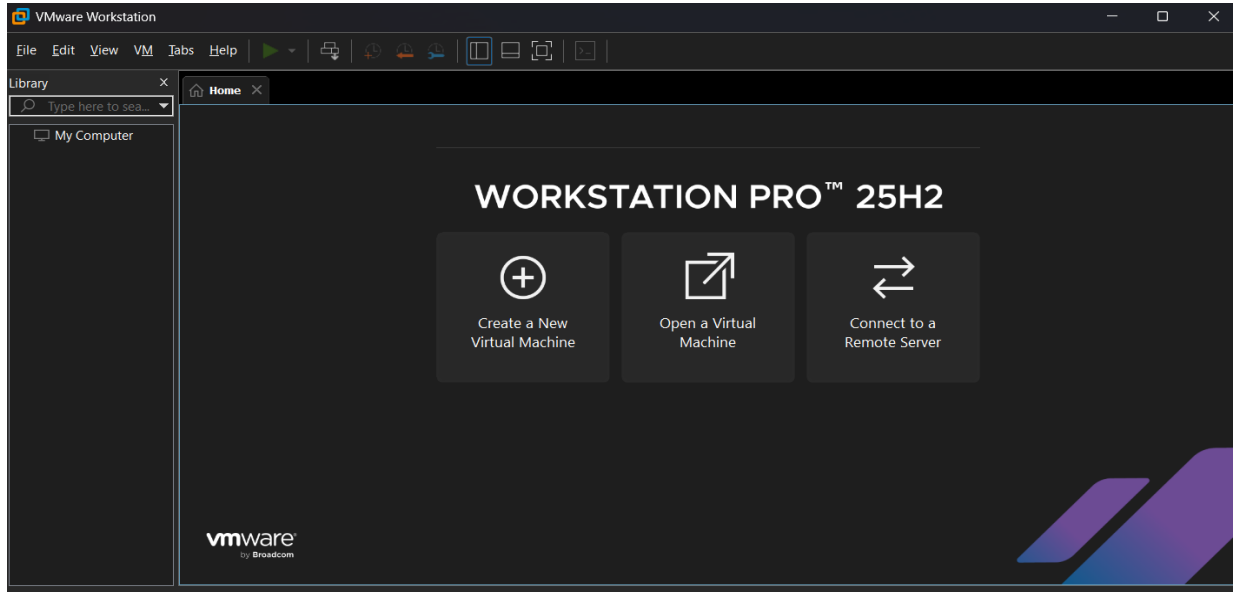


➤ Click 'Finish'.

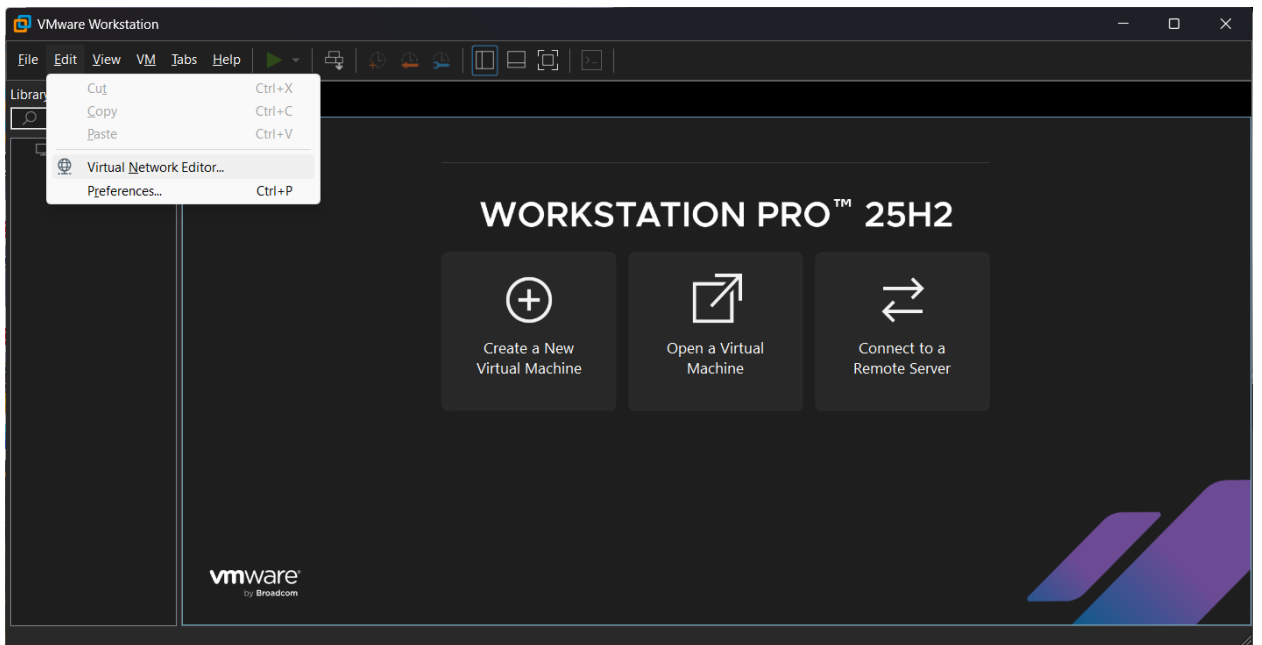


Creating the Server Specifications

1. Open VMware.



2. Go to 'Edit', then click on 'Virtual Network Editor'.



3. Open the Virtual Network Editor window, click 'Change Settings', and confirm by clicking 'Yes' if prompted by Windows.

The screenshot shows the 'Virtual Network Editor' window. At the top, there is a table with the following data:

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet1	Host-only	-	Connected	Enabled	192.168.88.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.126.0

Below the table are buttons: 'Add Network...', 'Remove Network', and 'Rename Network...'. The 'VMnet Information' section has three radio buttons: 'Bridged (connect VMs directly to the external network)', 'NAT (shared host's IP address with VMs)', and 'Host-only (connect VMs internally in a private network)'. The 'Host-only' option is selected. There are checkboxes for 'Connect a host virtual adapter to this network' (checked) and 'Use local DHCP service to distribute IP address to VMs' (checked). The 'Host virtual adapter name' is 'VMware Network Adapter VMnet1'. There are input fields for 'Subnet IP' (192.168.88.0) and 'Subnet mask' (255.255.255.0). At the bottom, there is a warning: 'Administrator privileges are required to modify the network configuration.' and a 'Change Settings' button. Other buttons at the bottom include 'Restore Defaults', 'Import...', 'Export...', 'OK', 'Cancel', 'Apply', and 'Help'.

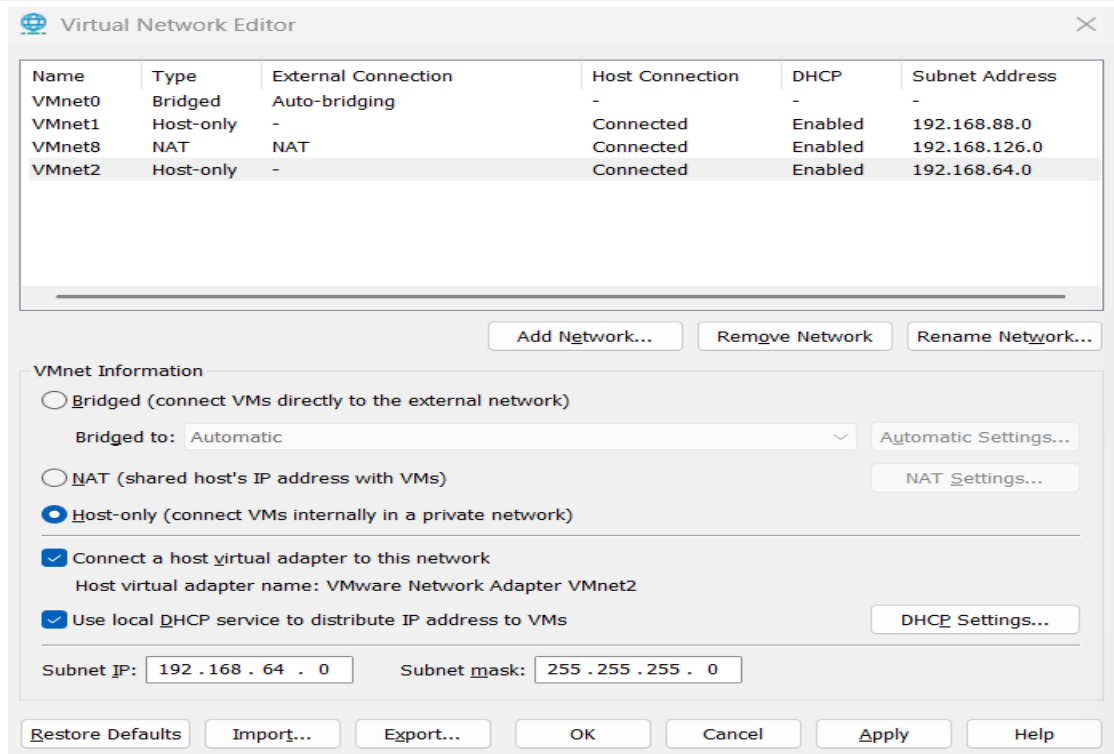
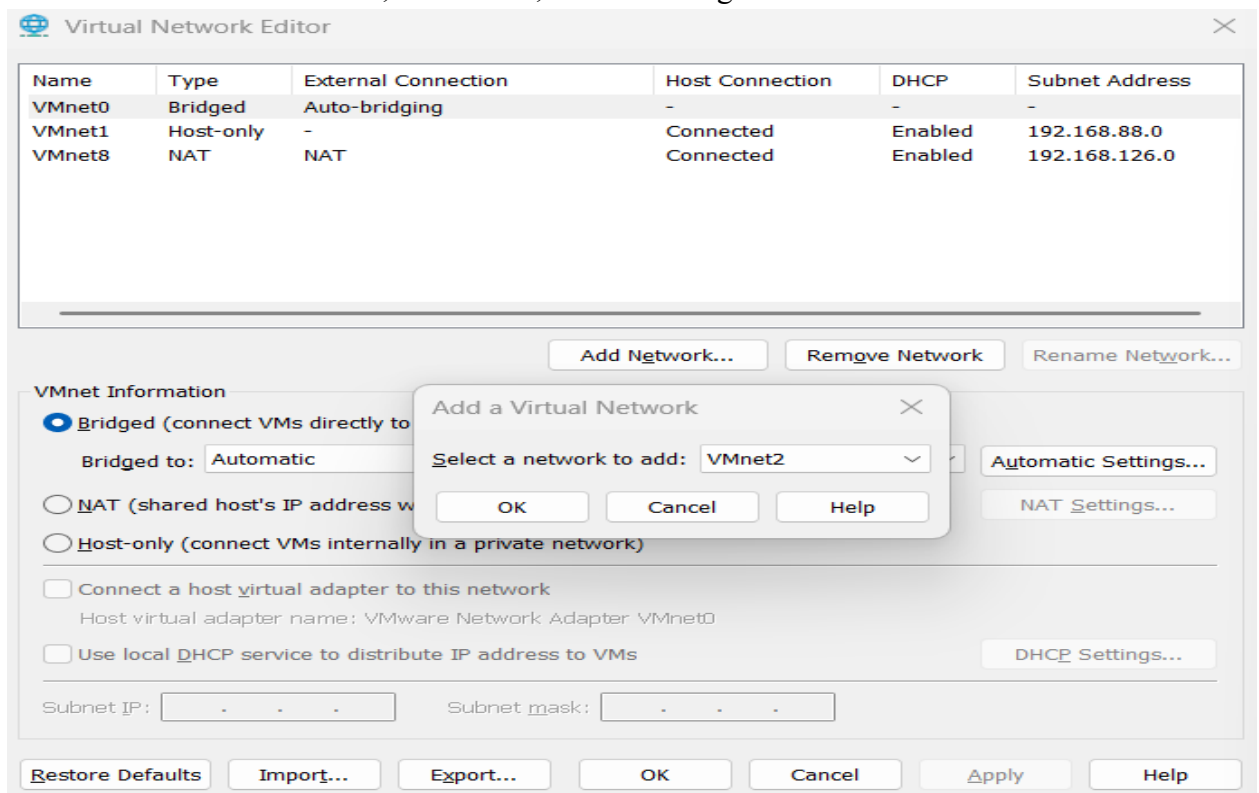
4. After giving permission, it opens like this. Then click 'Add Network'.

The screenshot shows the 'Virtual Network Editor' window after giving permission. The table at the top now includes VMnet0:

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.88.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.126.0

The 'VMnet Information' section now has the 'Bridged' radio button selected. The 'Bridged to' dropdown is set to 'Automatic'. The 'Connect a host virtual adapter to this network' checkbox is now unchecked, and the 'Host virtual adapter name' is 'VMware Network Adapter VMnet0'. The 'Use local DHCP service to distribute IP address to VMs' checkbox is also unchecked. The 'Subnet IP' and 'Subnet mask' fields are empty. The 'Change Settings' button is now disabled. The other buttons at the bottom remain the same.

5. Select network 'VMnet2', click 'OK', and then assign it.



6. Change the Subnet IP to 10.0.1.0 and the subnet mask to 255.255.255.0, then click 'Apply'.

The screenshot shows the 'Virtual Network Editor' window. At the top, there is a table listing four virtual networks: VMnet0 (Bridged), VMnet1 (Host-only), VMnet8 (NAT), and VMnet2 (Host-only). VMnet2 is highlighted. Below the table are buttons for 'Add Network...', 'Remove Network', and 'Rename Network...'. The 'VMnet Information' section shows three radio buttons: 'Bridged', 'NAT', and 'Host-only'. The 'Host-only' option is selected. Below these are checkboxes for 'Connect a host virtual adapter to this network' (checked) and 'Use local DHCP service to distribute IP address to VMs' (checked). The 'Subnet IP' field is set to '10 . 0 . 1 . 0' and the 'Subnet mask' field is set to '255 . 255 . 255 . 0'. At the bottom are buttons for 'Restore Defaults', 'Import...', 'Export...', 'OK', 'Cancel', 'Apply', and 'Help'.

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.88.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.126.0
VMnet2	Host-only	-	Connected	Enabled	10.0.1.0

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Automatic Settings...

☐ NAT (shared host's IP address with VMs) NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☒ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet2

☒ Use local DHCP service to distribute IP address to VMs DHCP Settings...

Subnet IP: 10 . 0 . 1 . 0 Subnet mask: 255 . 255 . 255 . 0

Restore Defaults Import... Export... OK Cancel Apply Help

7. After changing the subnet address, click 'OK'.

This screenshot is identical to the previous one, showing the 'Virtual Network Editor' window with VMnet2 selected. The 'Host-only' radio button is selected, and the 'Subnet IP' is '10 . 0 . 1 . 0'. In this version, the 'OK' button at the bottom is highlighted with a blue border, indicating it is the next step in the process.

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet0	Bridged	Auto-bridging	-	-	-
VMnet1	Host-only	-	Connected	Enabled	192.168.88.0
VMnet8	NAT	NAT	Connected	Enabled	192.168.126.0
VMnet2	Host-only	-	Connected	Enabled	10.0.1.0

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Automatic Settings...

☐ NAT (shared host's IP address with VMs) NAT Settings...

☒ Host-only (connect VMs internally in a private network)

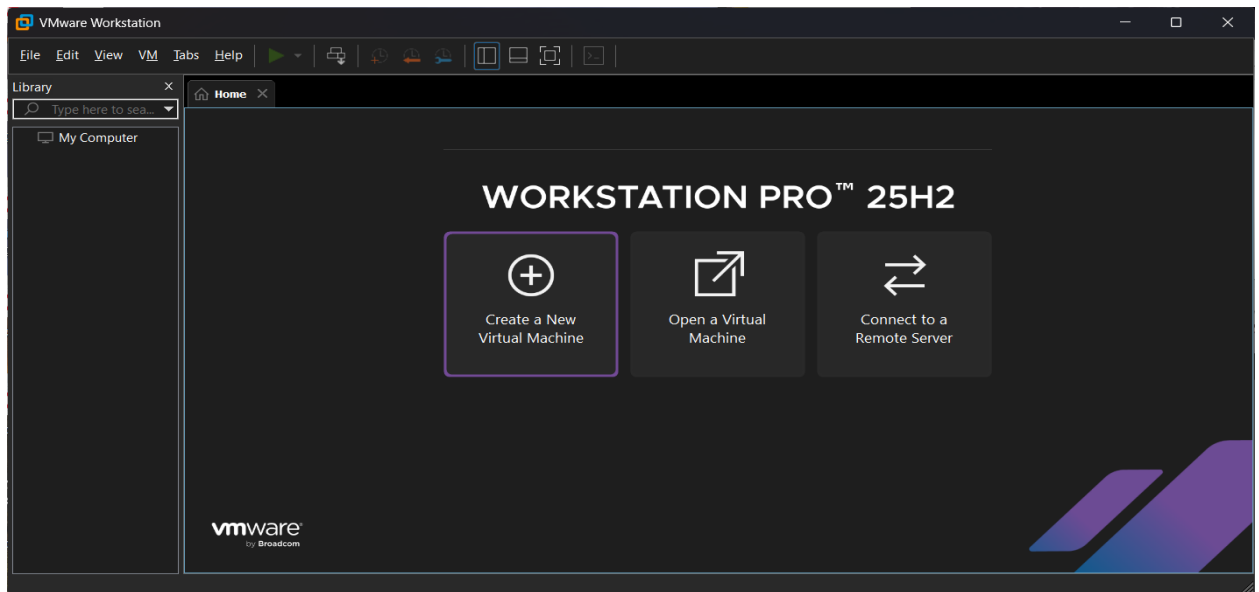
☒ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet2

☒ Use local DHCP service to distribute IP address to VMs DHCP Settings...

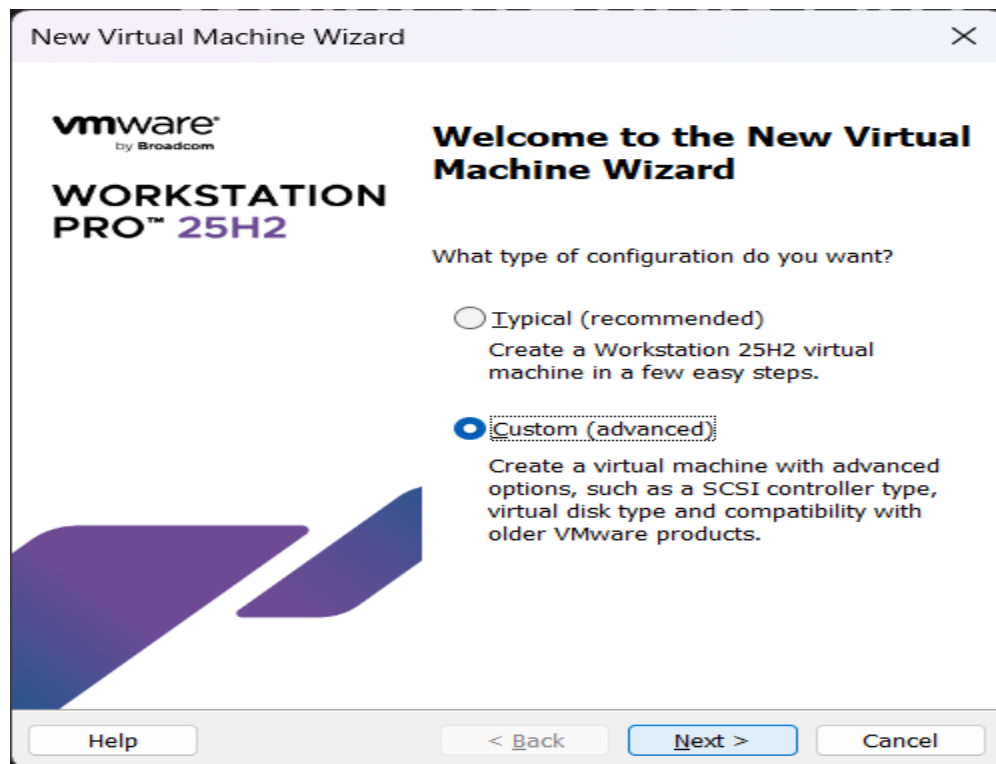
Subnet IP: 10 . 0 . 1 . 0 Subnet mask: 255 . 255 . 255 . 0

Restore Defaults Import... Export... OK Cancel Apply Help

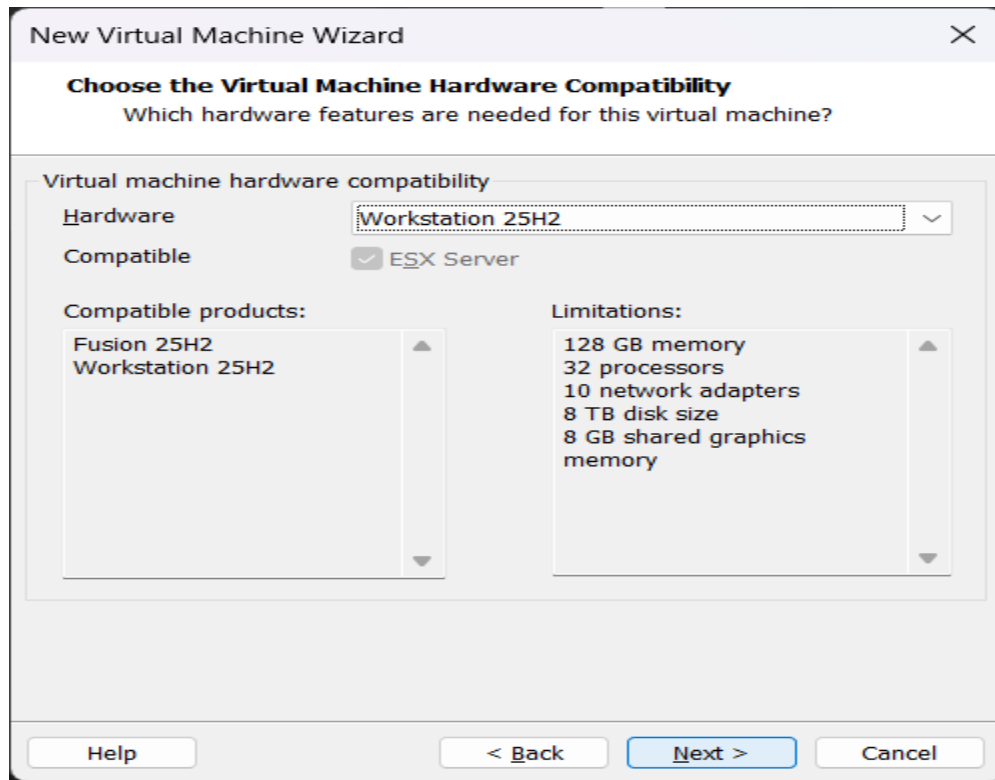
8. Create a new virtual machine.



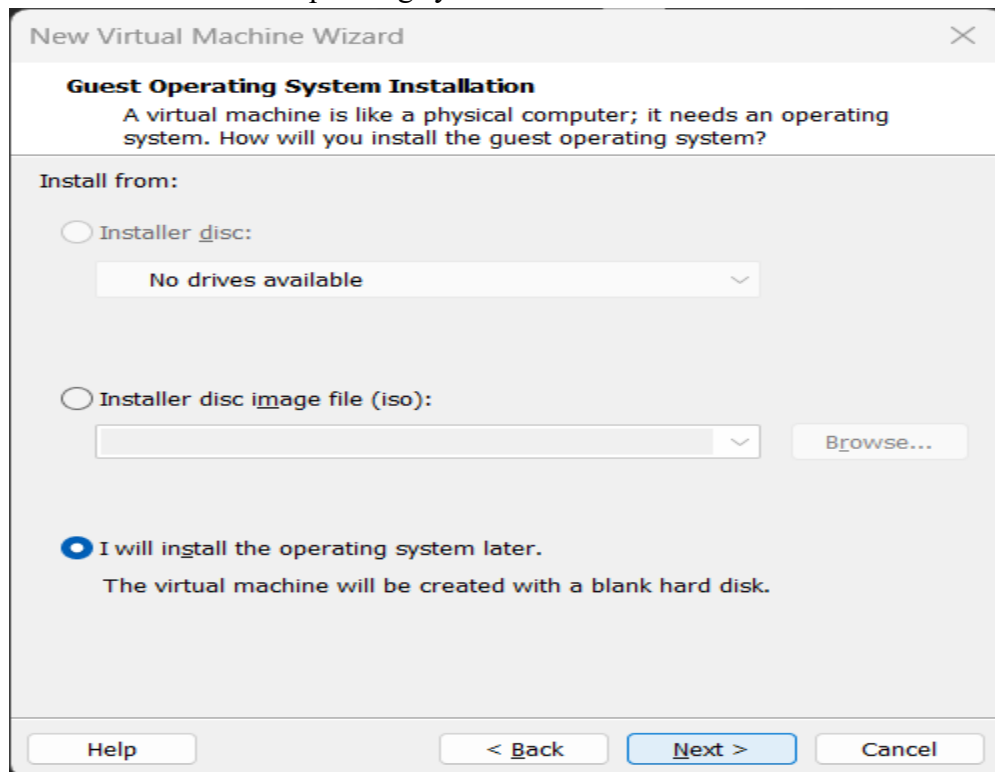
9. Choose 'Custom' and click 'Next'.



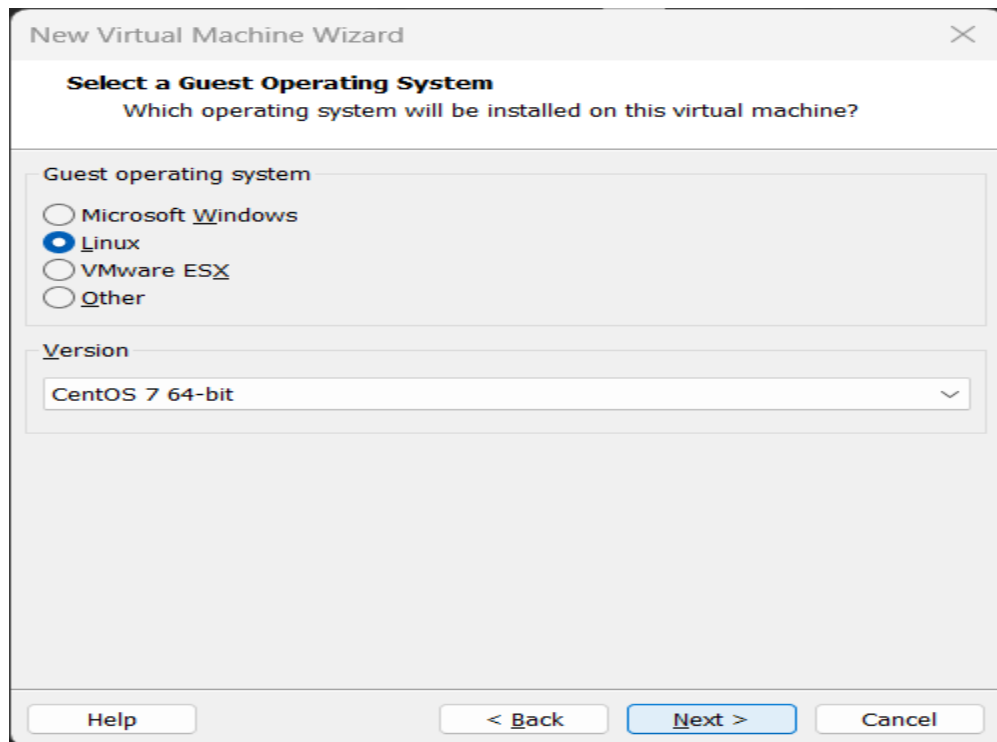
10. Click 'Next'.



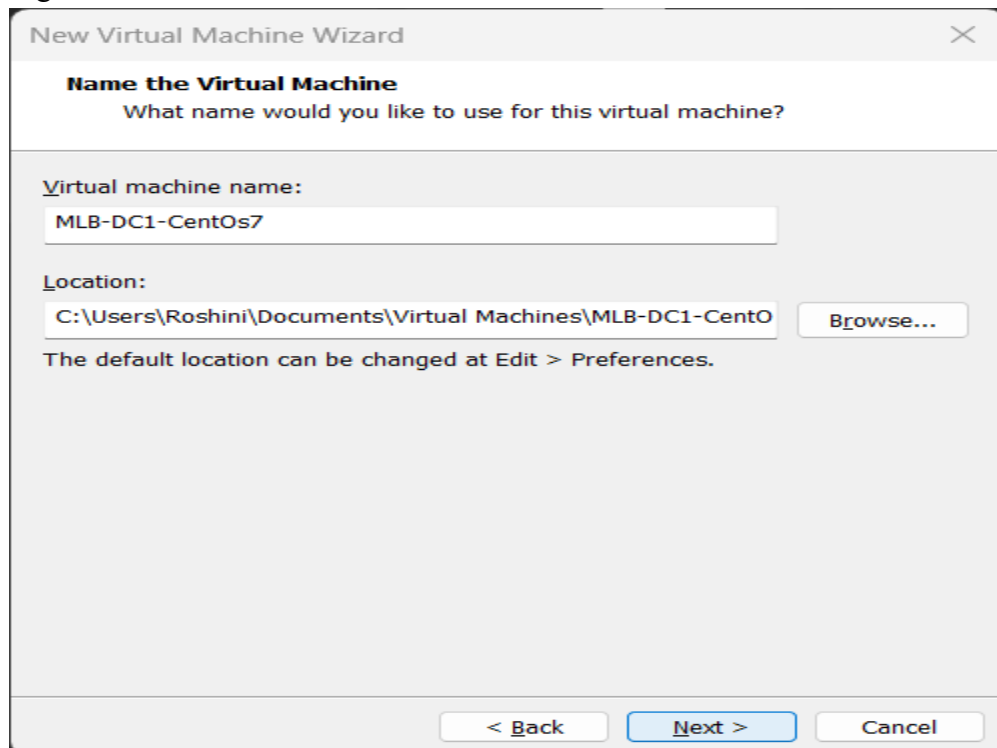
11. Choose 'I will install the operating system later' and click 'Next'.



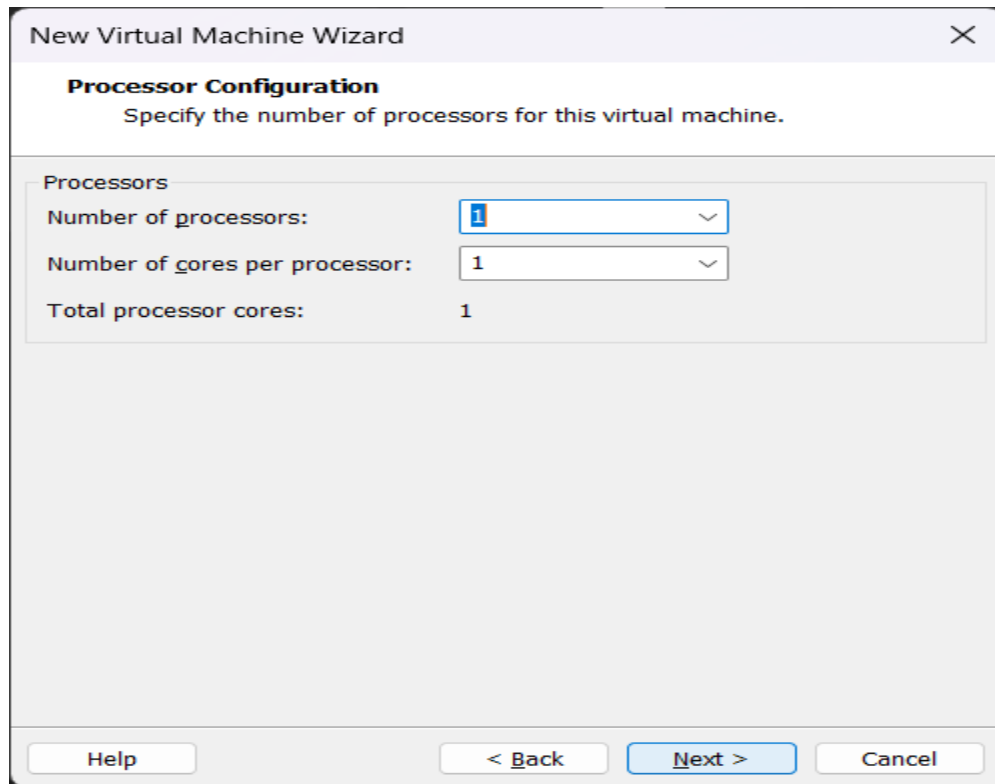
12. Choose Linux as the guest operating system and CentOS 7 (64-bit) as the version, then click 'Next'.



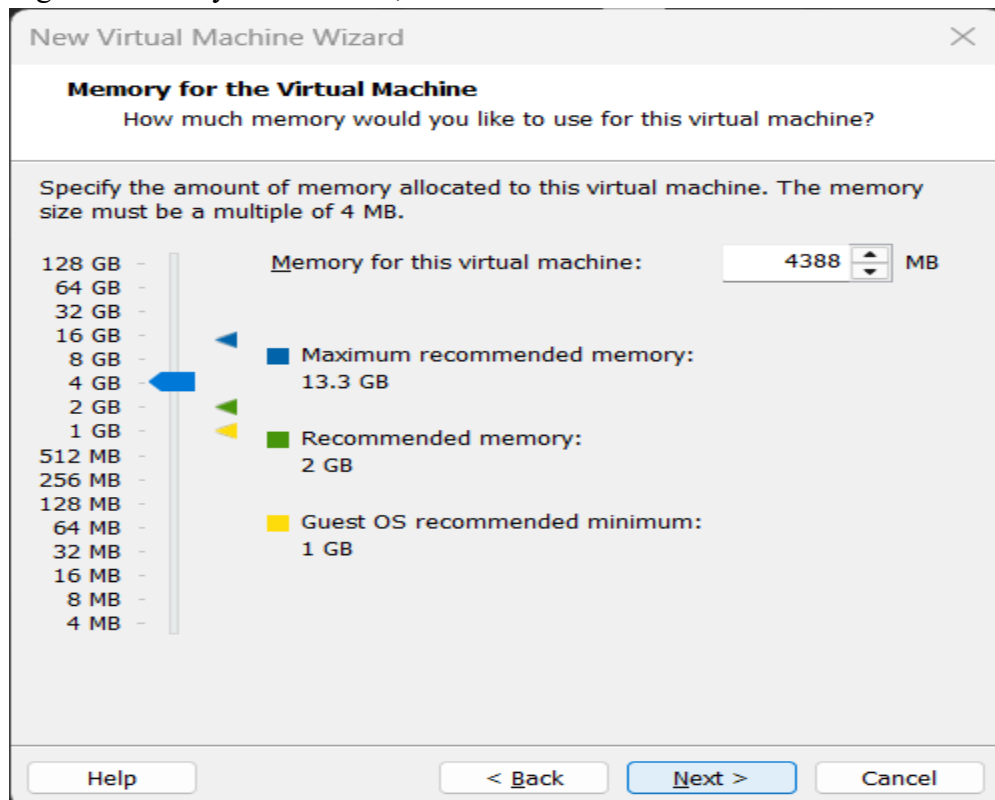
13. Change the virtual machine name to MLB-DC1-CentOS7, then click Next.



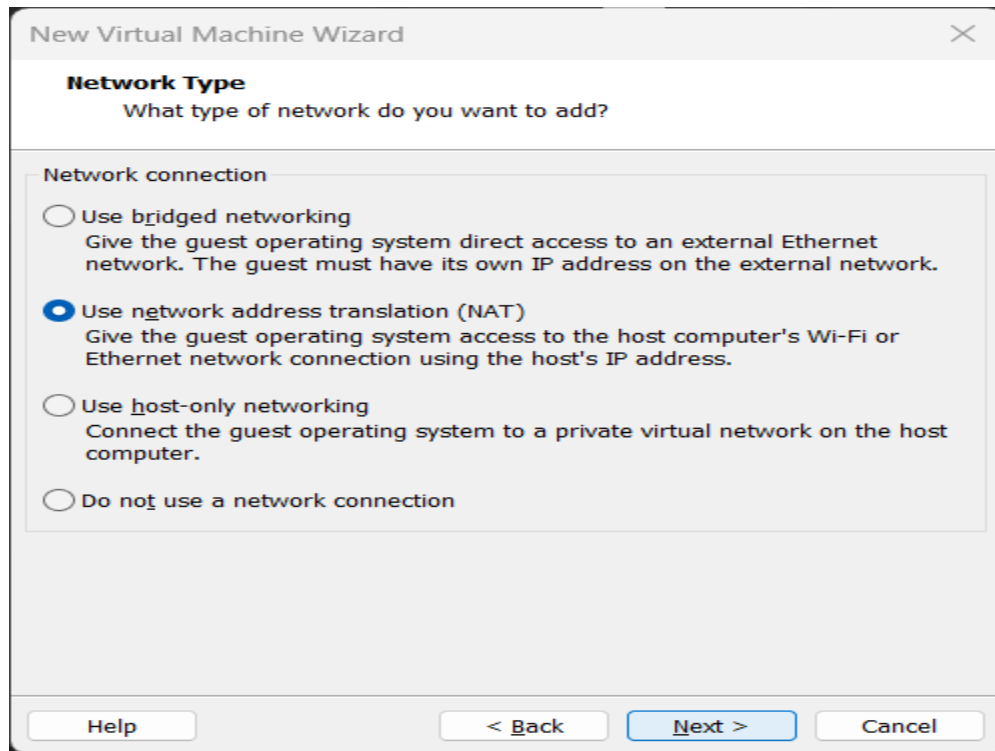
14. Click 'Next'.



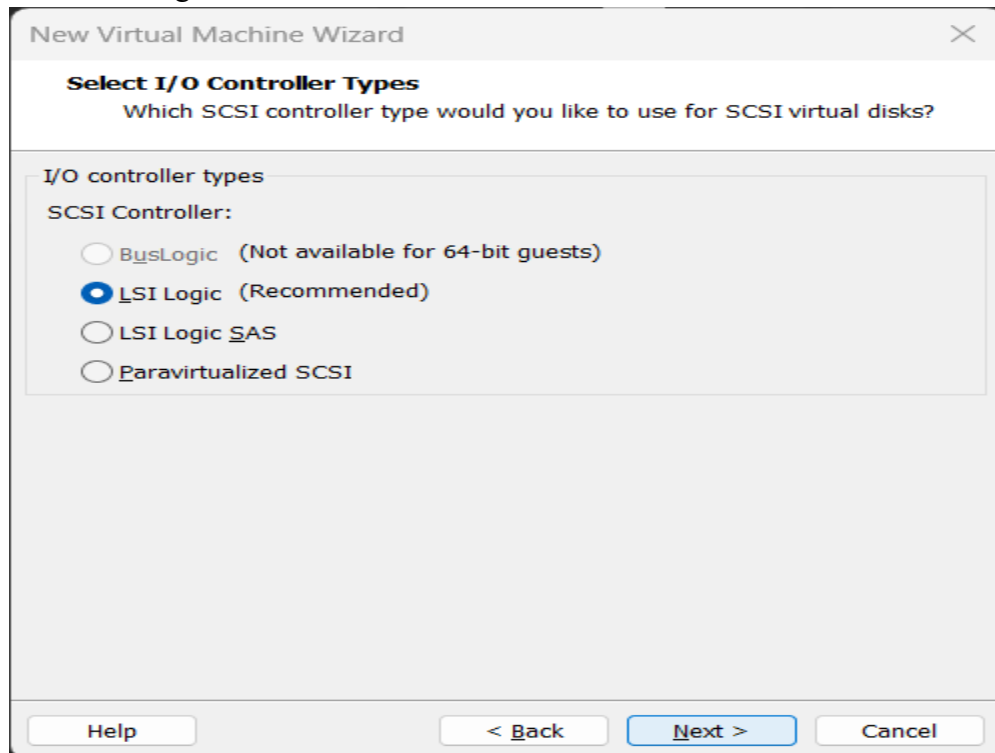
15. Change the memory size to 4 GB, then click Next.



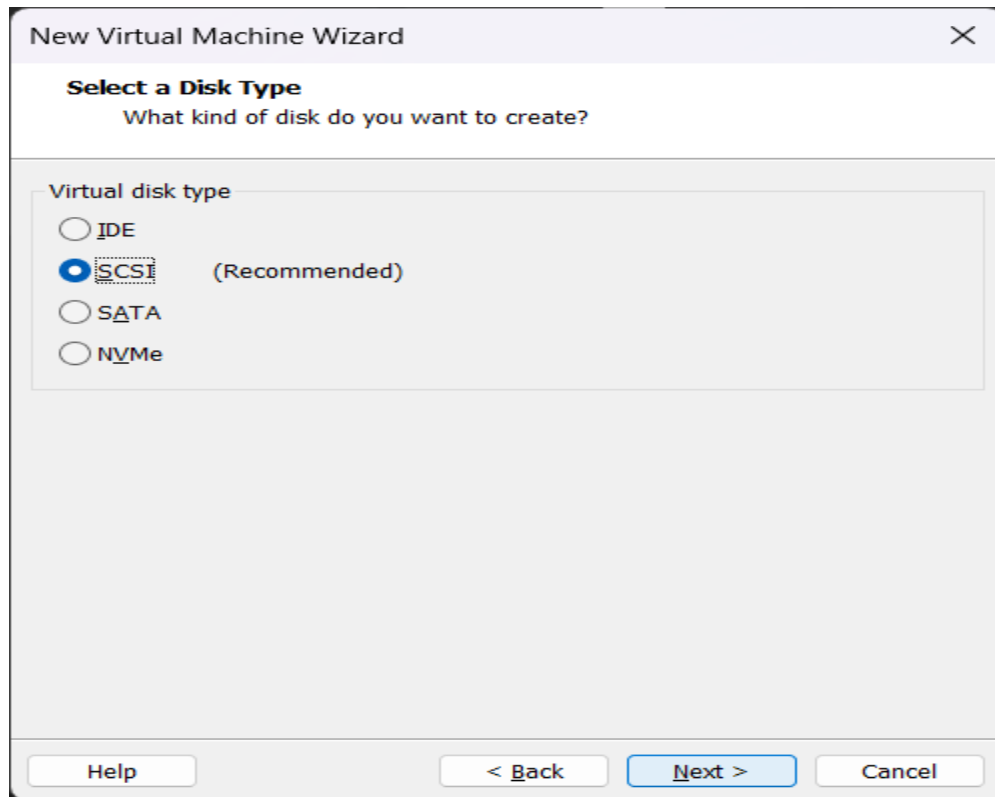
16. Choose 'Use Network Address Translation' and click 'Next'.



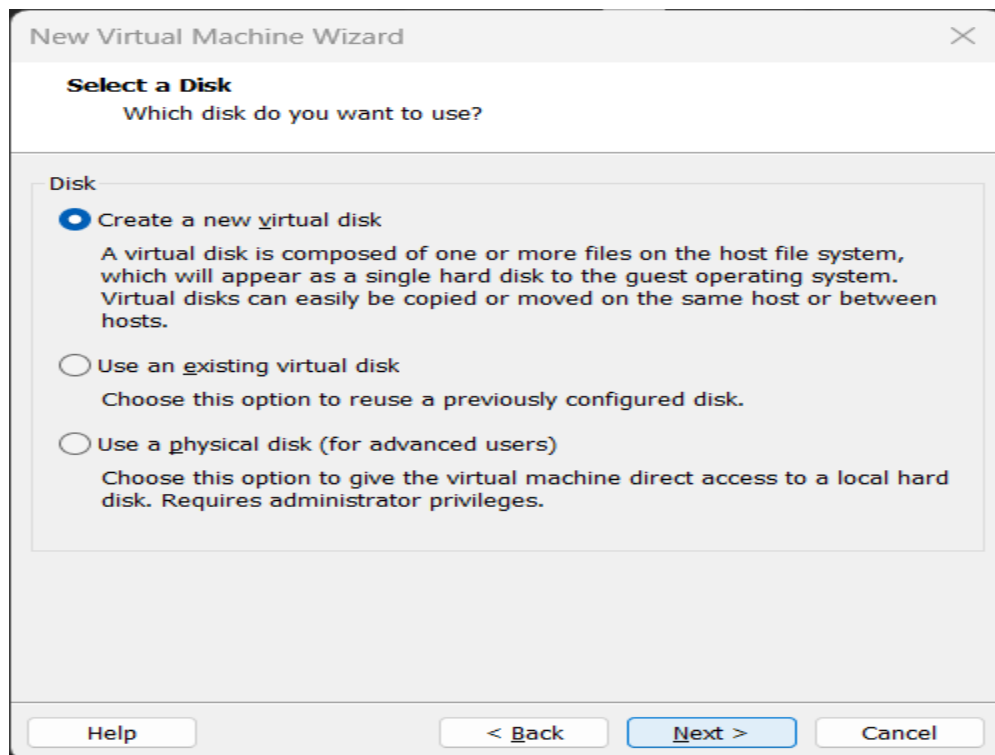
17. Choose 'LSI Logic' and click 'Next'.



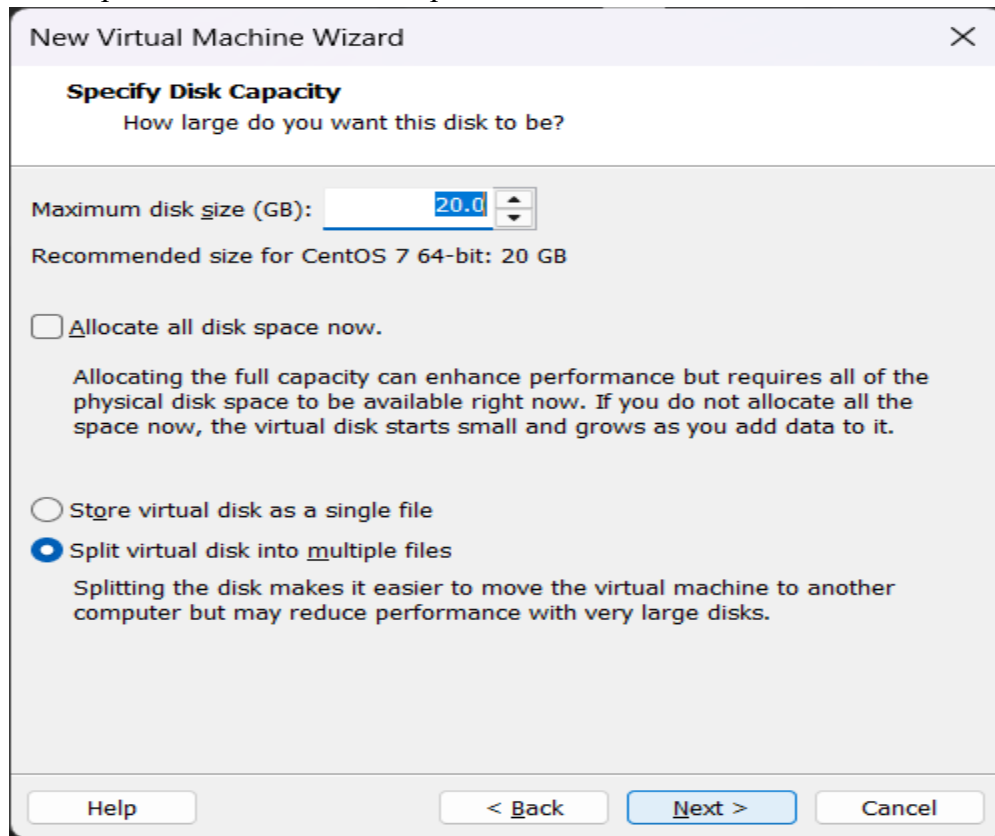
18. Choose 'SCSI' and click 'Next'.



19. Choose 'Create a new virtual disk' and click 'Next'.



20. Choose 'Split virtual disk into multiple files' and click 'Next'.



The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Specify Disk Capacity' step. The window title is 'New Virtual Machine Wizard' with a close button (X) in the top right corner. The main heading is 'Specify Disk Capacity' with the subtext 'How large do you want this disk to be?'. Below this, there is a text input field for 'Maximum disk size (GB):' with the value '20.0' entered. To the right of the input field is a small spinner control. Below the input field, it says 'Recommended size for CentOS 7 64-bit: 20 GB'. There are two radio button options: 'Allocate all disk space now.' (which is unselected) and 'Split virtual disk into multiple files' (which is selected). Below the second option, there is explanatory text: 'Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.' At the bottom of the window, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

Maximum disk size (GB):

Recommended size for CentOS 7 64-bit: 20 GB

☐ Allocate all disk space now.

Allocating the full capacity can enhance performance but requires all of the physical disk space to be available right now. If you do not allocate all the space now, the virtual disk starts small and grows as you add data to it.

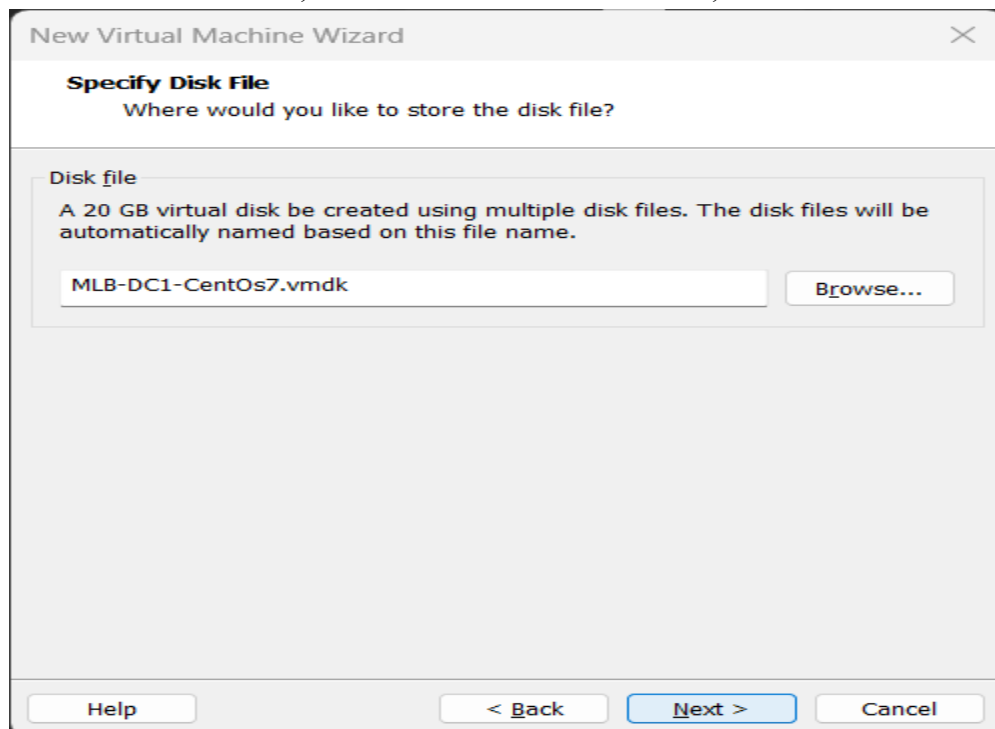
☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Help < Back Next > Cancel

21. Click the 'Browse' button, choose the NDM-Server folder, and click 'Save'.



The screenshot shows the 'New Virtual Machine Wizard' window, specifically the 'Specify Disk File' step. The window title is 'New Virtual Machine Wizard' with a close button (X) in the top right corner. The main heading is 'Specify Disk File' with the subtext 'Where would you like to store the disk file?'. Below this, there is a text input field for 'Disk file' containing the text 'MLB-DC1-CentOs7.vmdk'. To the right of the input field is a 'Browse...' button. Below the input field, there is explanatory text: 'A 20 GB virtual disk be created using multiple disk files. The disk files will be automatically named based on this file name.' At the bottom of the window, there are four buttons: 'Help', '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a blue border.

New Virtual Machine Wizard

Specify Disk File
Where would you like to store the disk file?

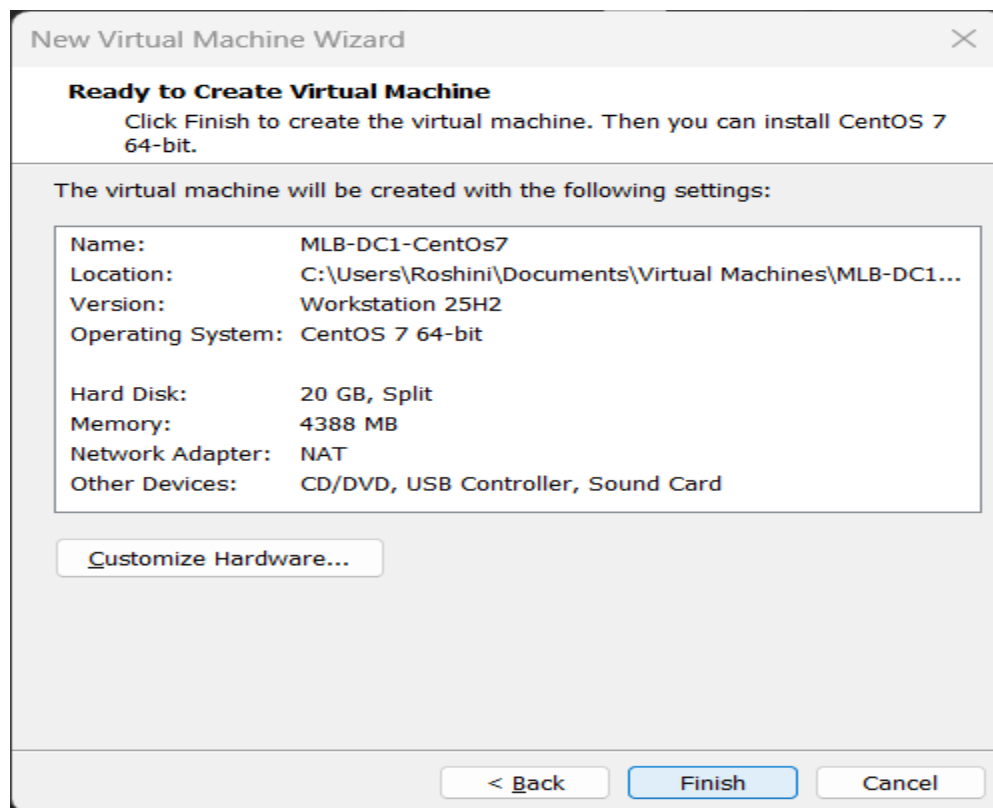
Disk file

A 20 GB virtual disk be created using multiple disk files. The disk files will be automatically named based on this file name.

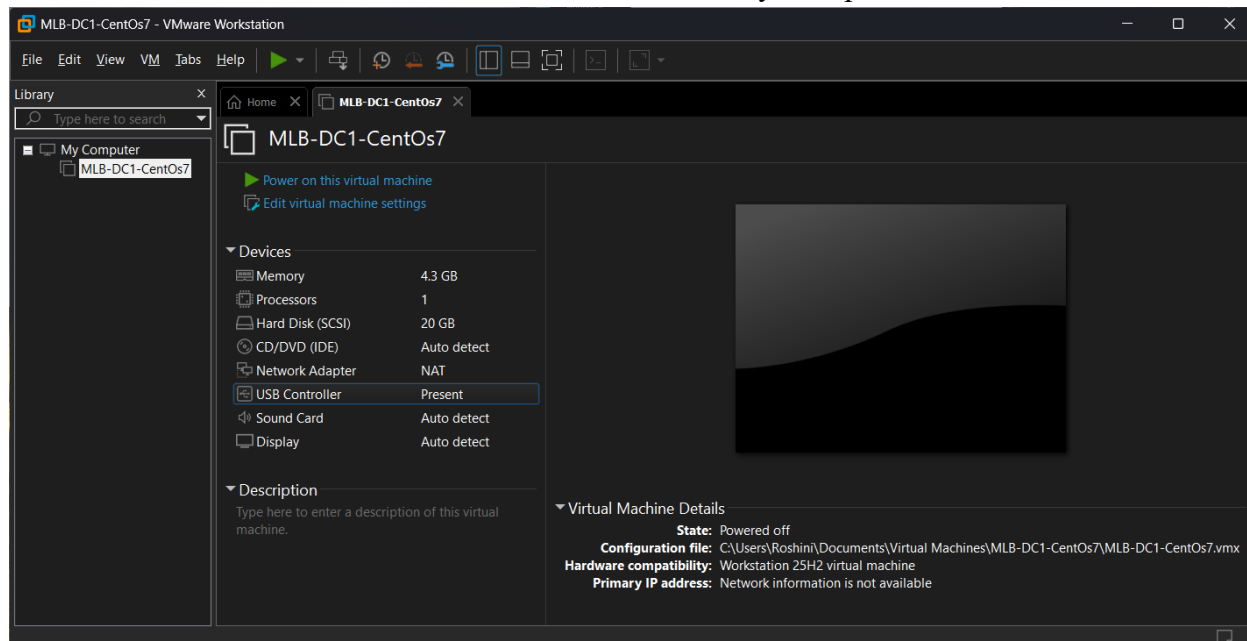
Browse...

Help < Back Next > Cancel

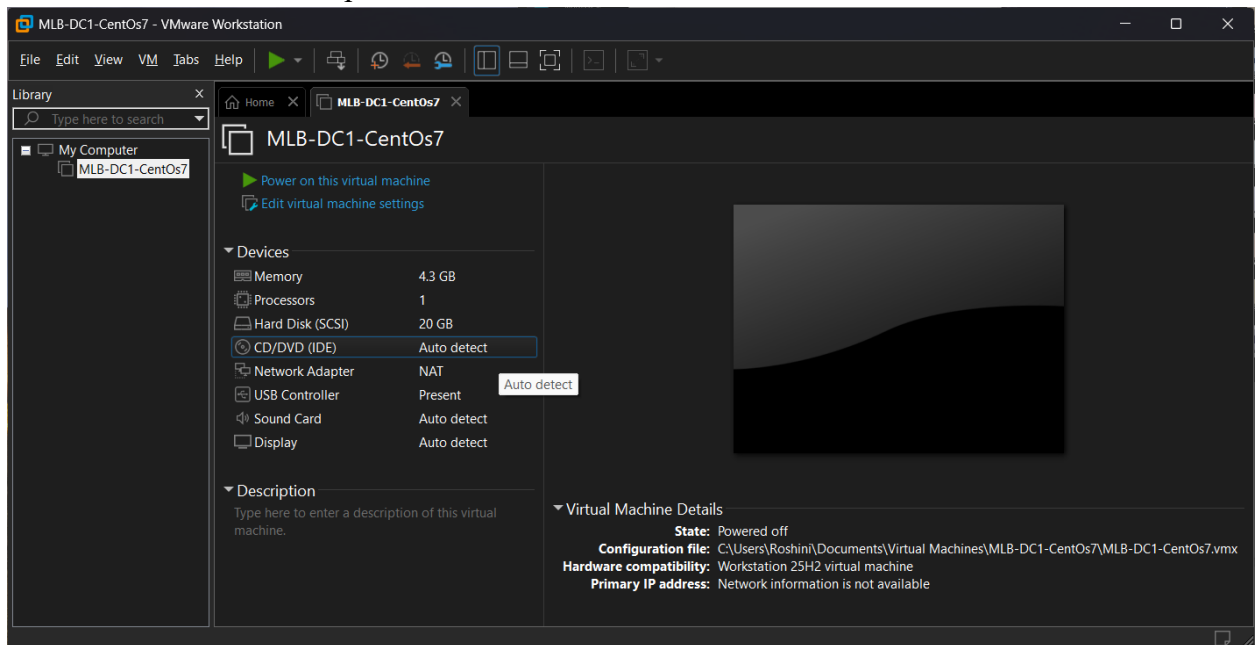
22. Then click the 'Finish' button.



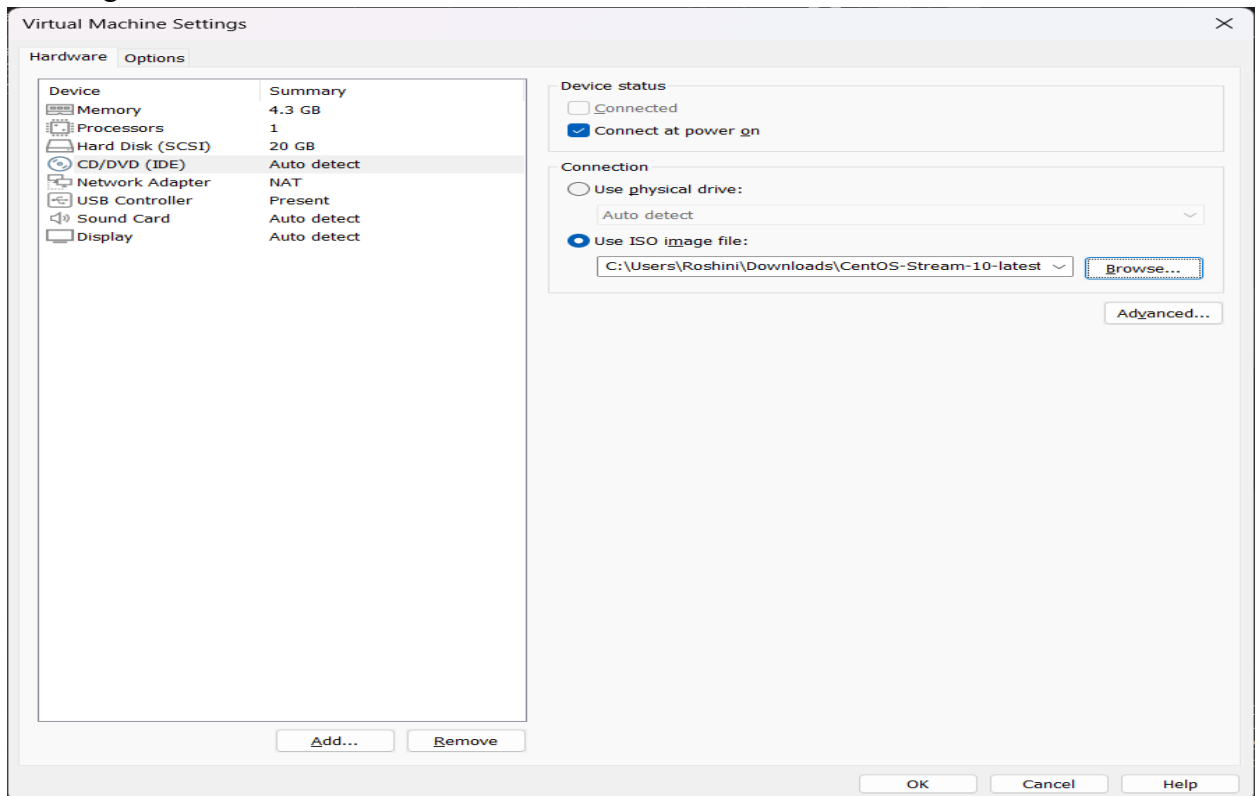
23. After this, the virtual machine will be visible under 'My Computer' in VMware.



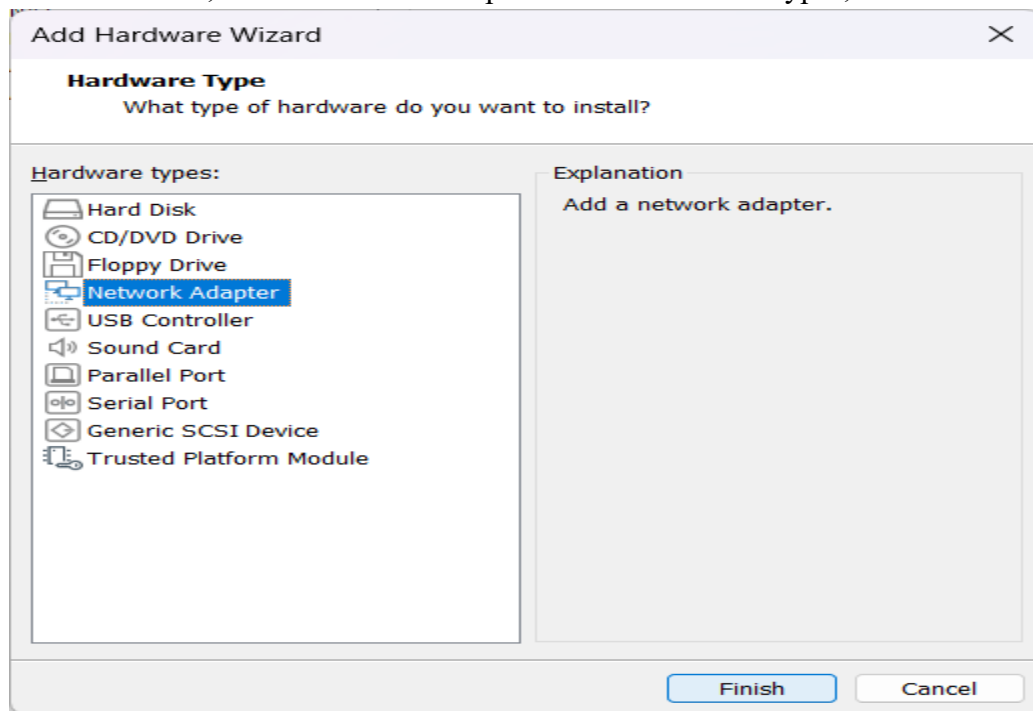
24. Click the 'CD/DVD' option.



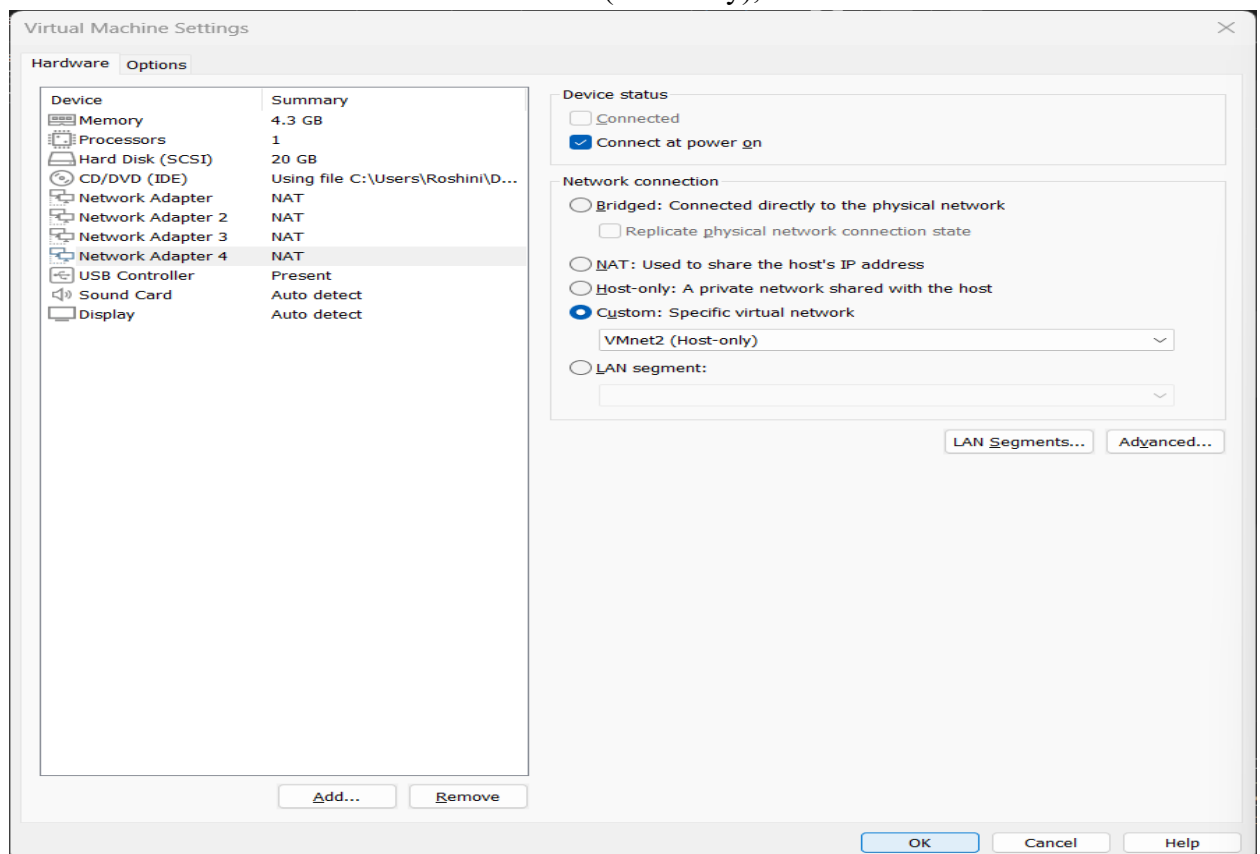
25. Choose the option 'Use ISO image file', then click 'Browse' to select the CentOS image.



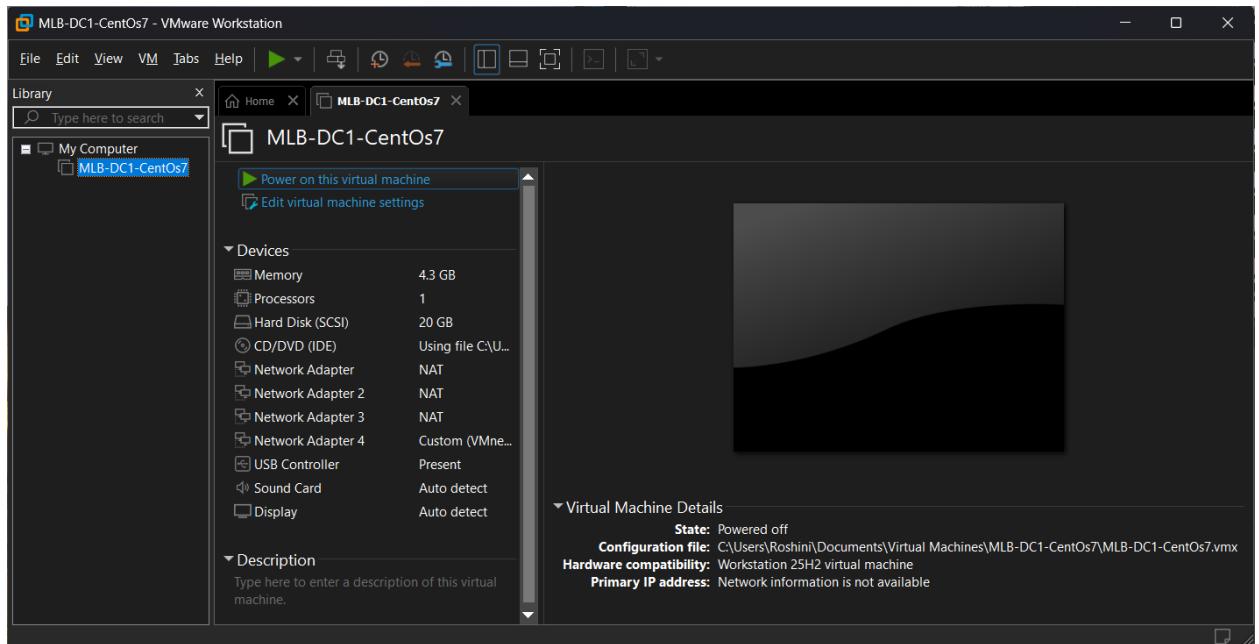
26. Then click 'Add', select 'Network Adapter' under Hardware Types, and click 'Finish'.



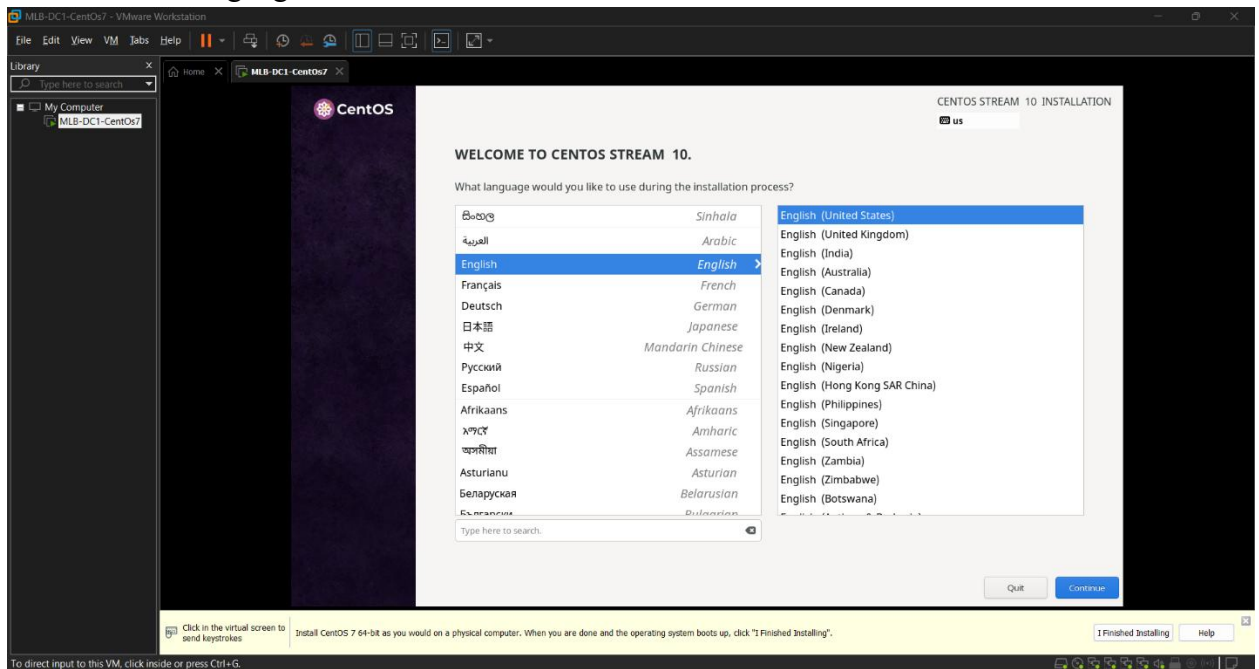
27. Then select 'Custom' and choose VMnet2 (Host-only), then click 'OK'



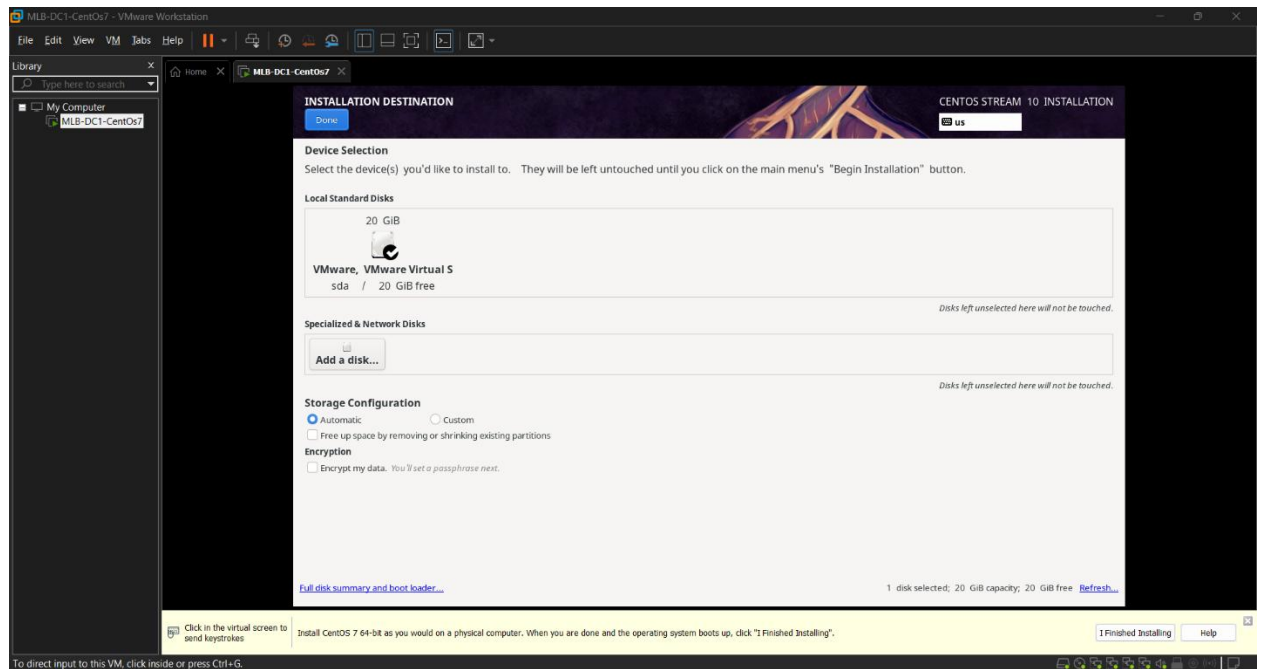
28. Click 'Power on this Virtual Machine'.



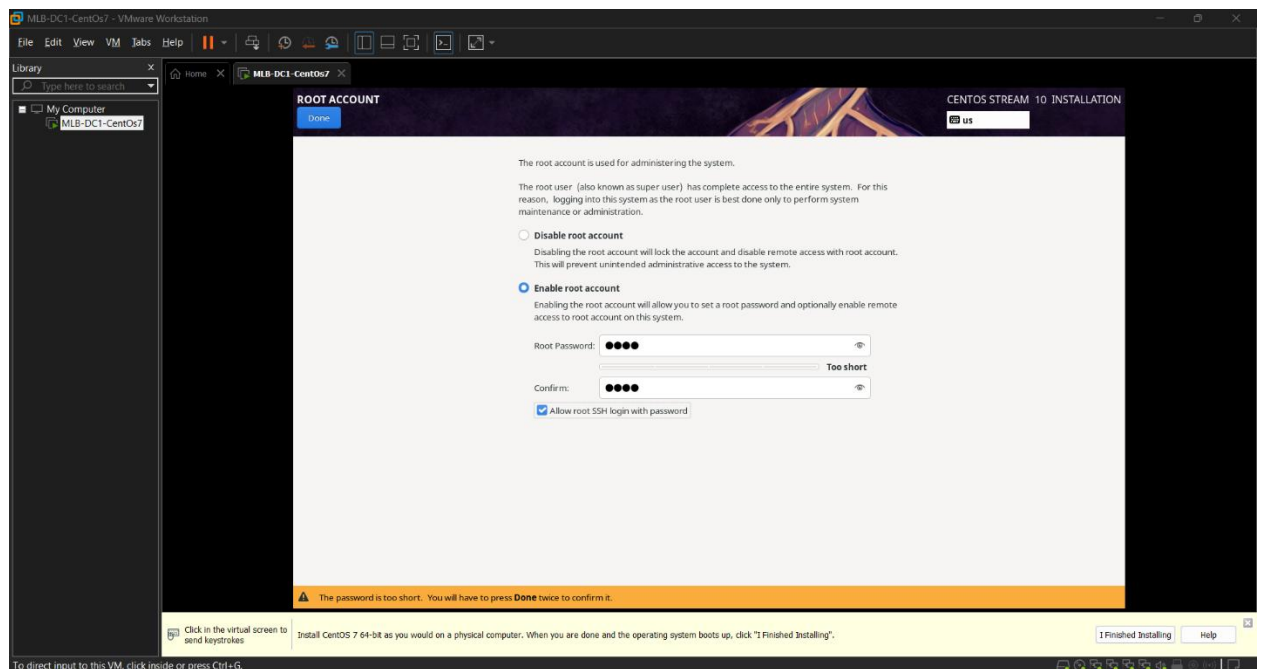
29. Select the language, then click 'continue'.



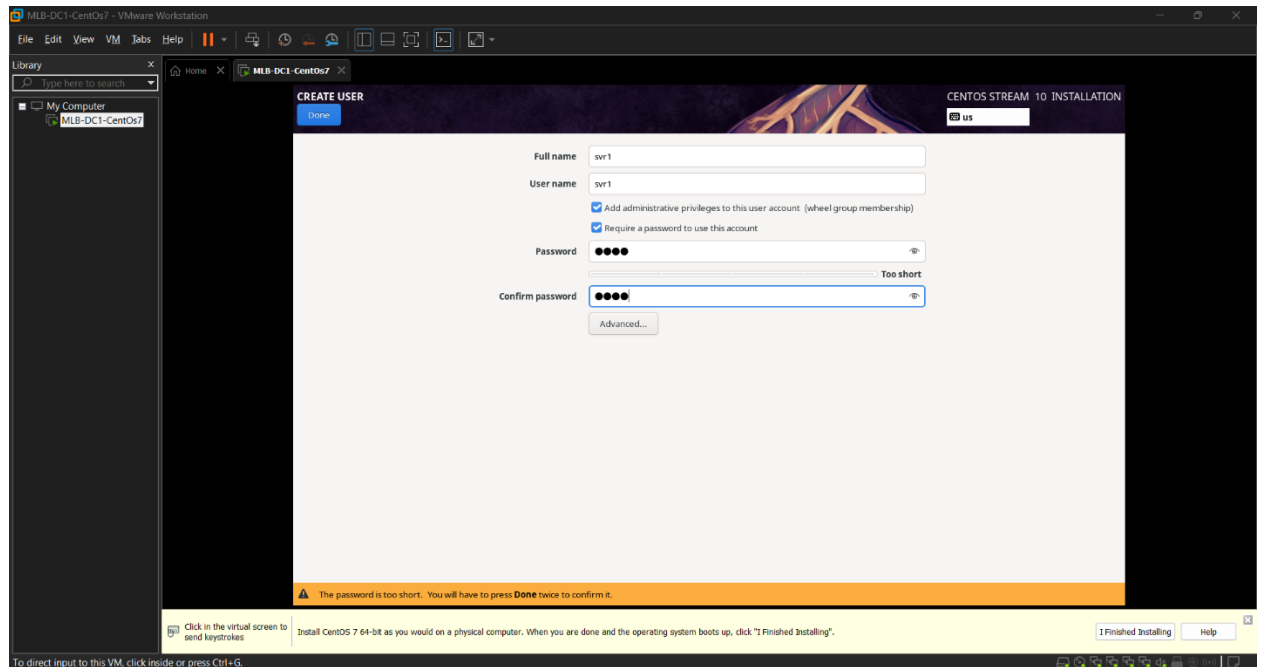
30. Go to the installation destination and click 'Done'.



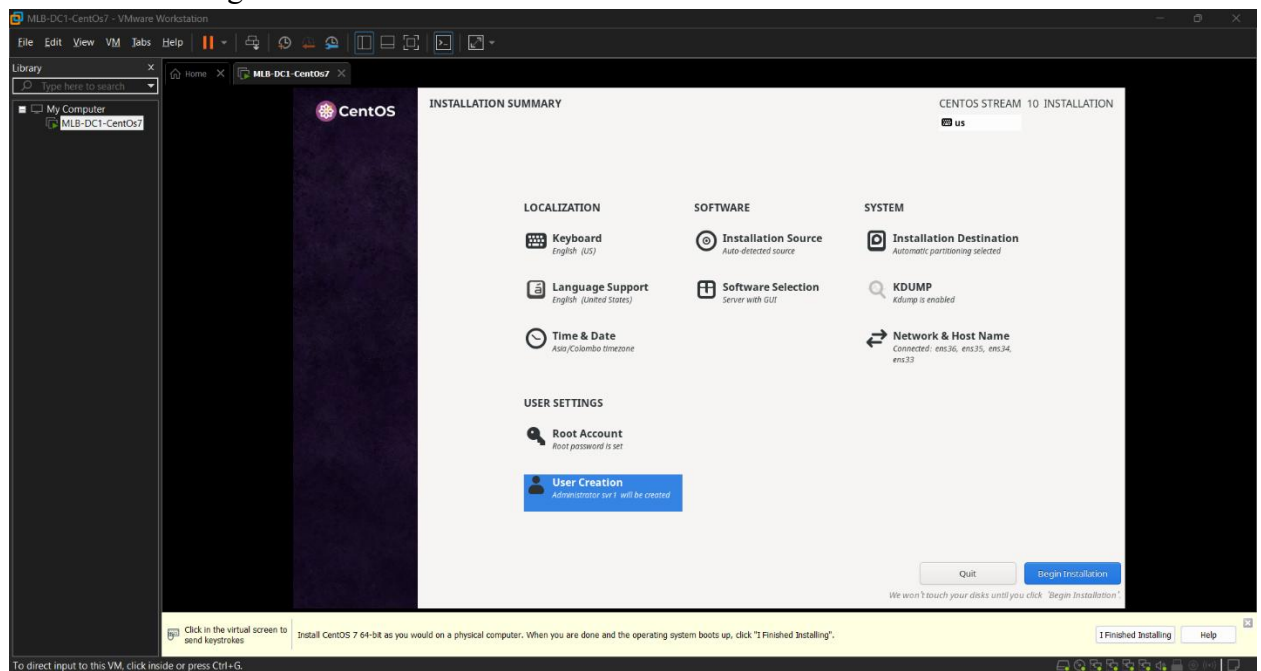
31. Go to the Root Account section, enter the password as 'root', and type 'root' again to confirm.



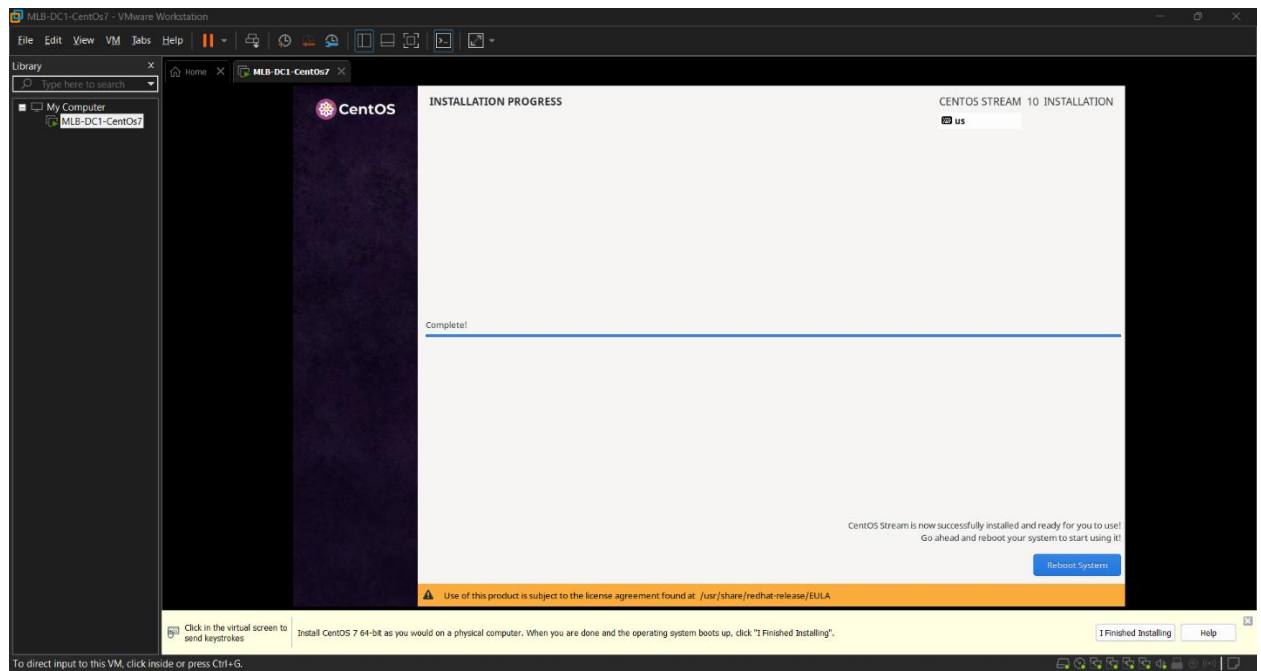
32. Go to User Creation, enter the username as 'svr1' and the password as 'svr1', then click 'Done'.



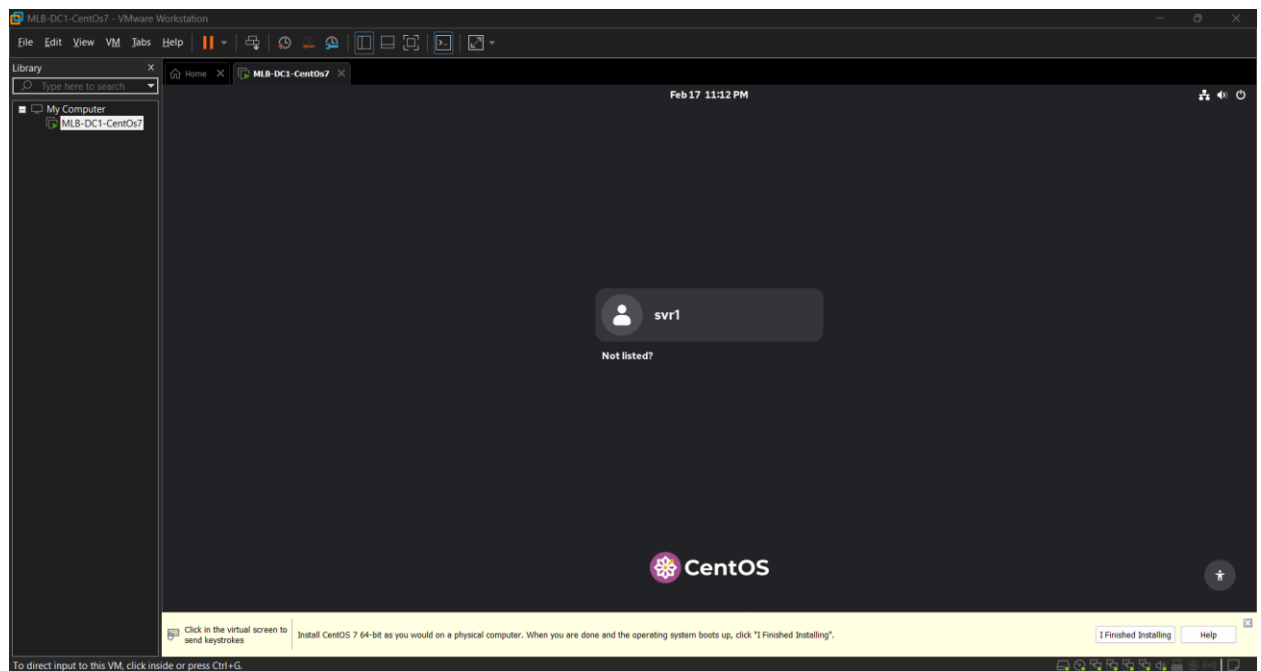
33. Click the 'Begin Installation' button.



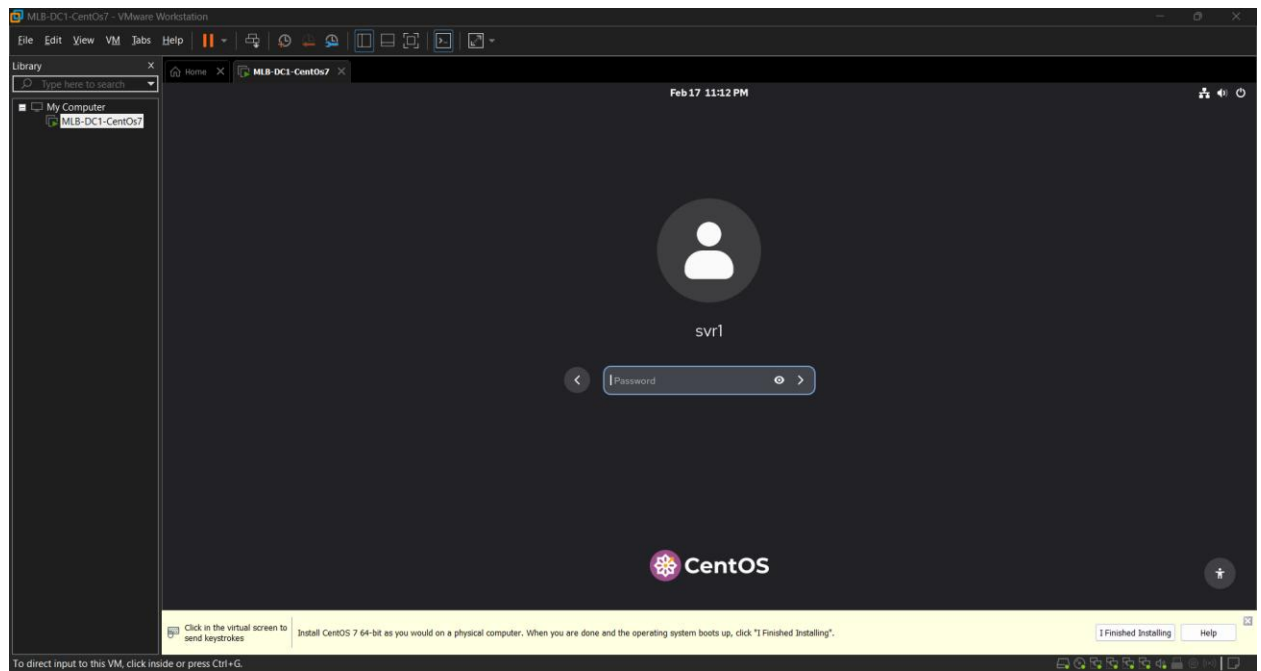
34. After the installation, click ‘Reboot System’.



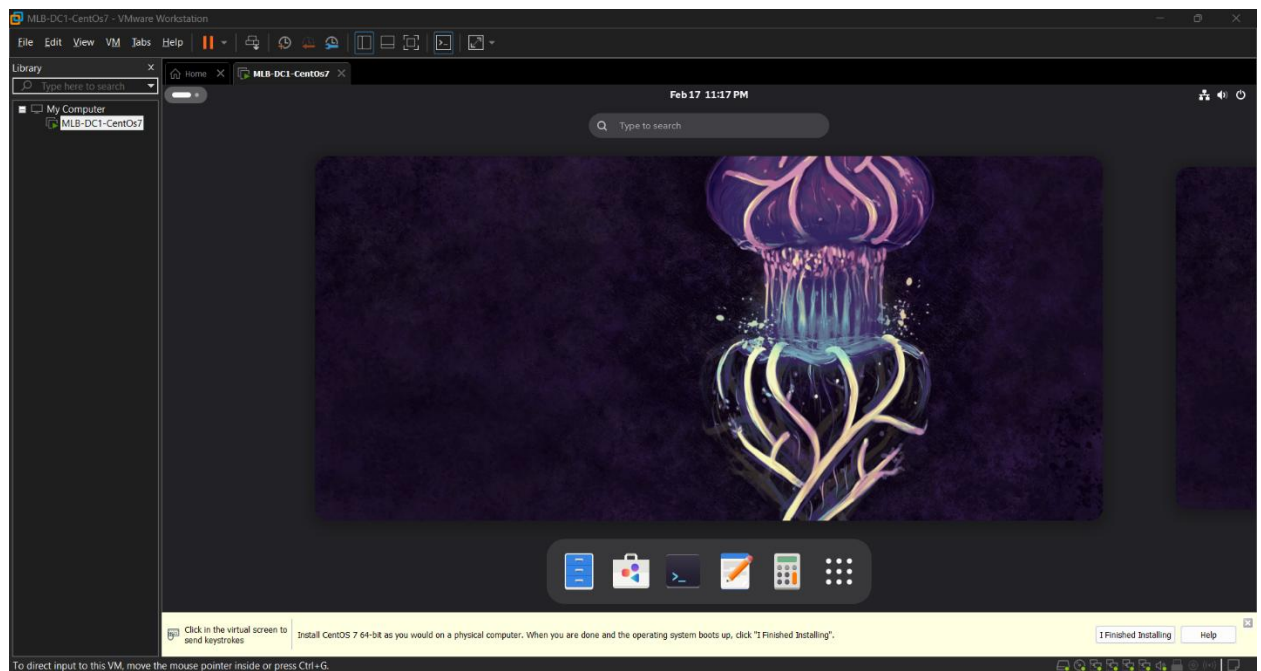
35. Select the ‘svr1’ user account.



36. Enter the account password

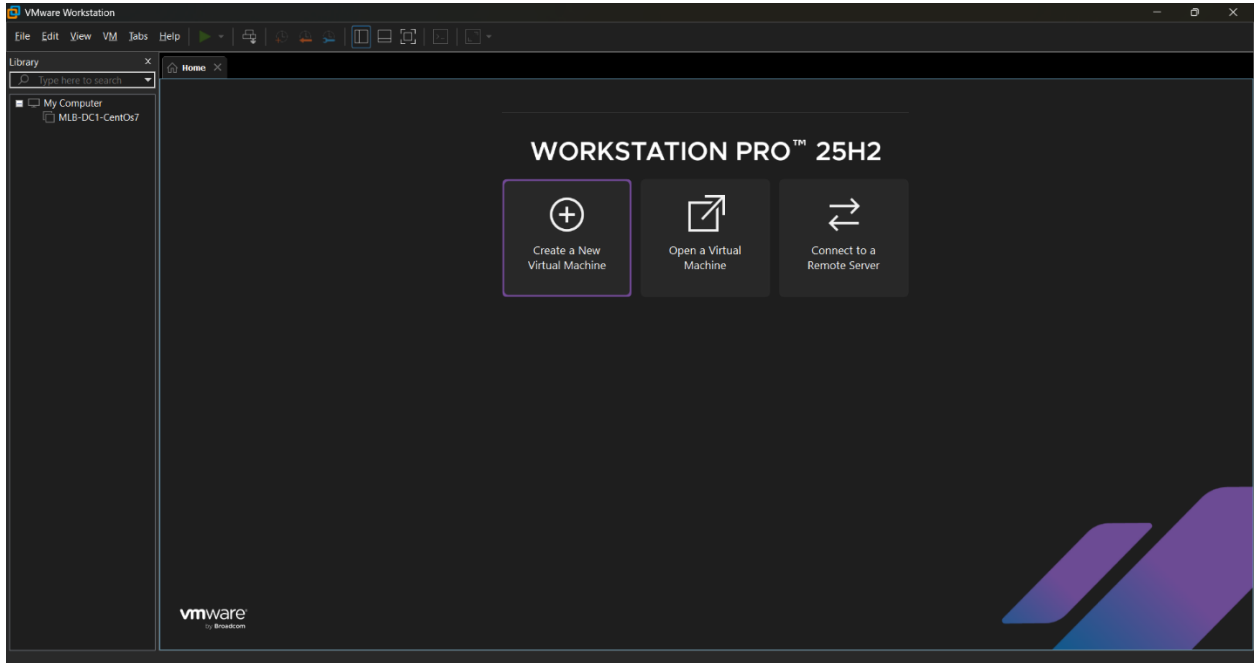


37. Finally, log in to the server account.

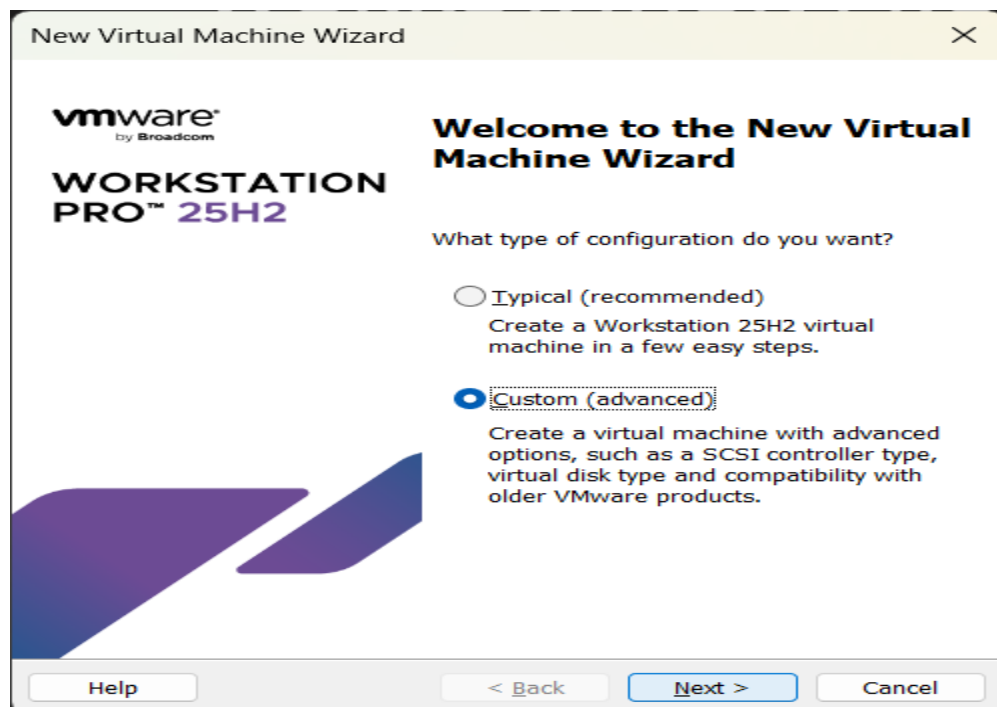


Creating the Client Specifications

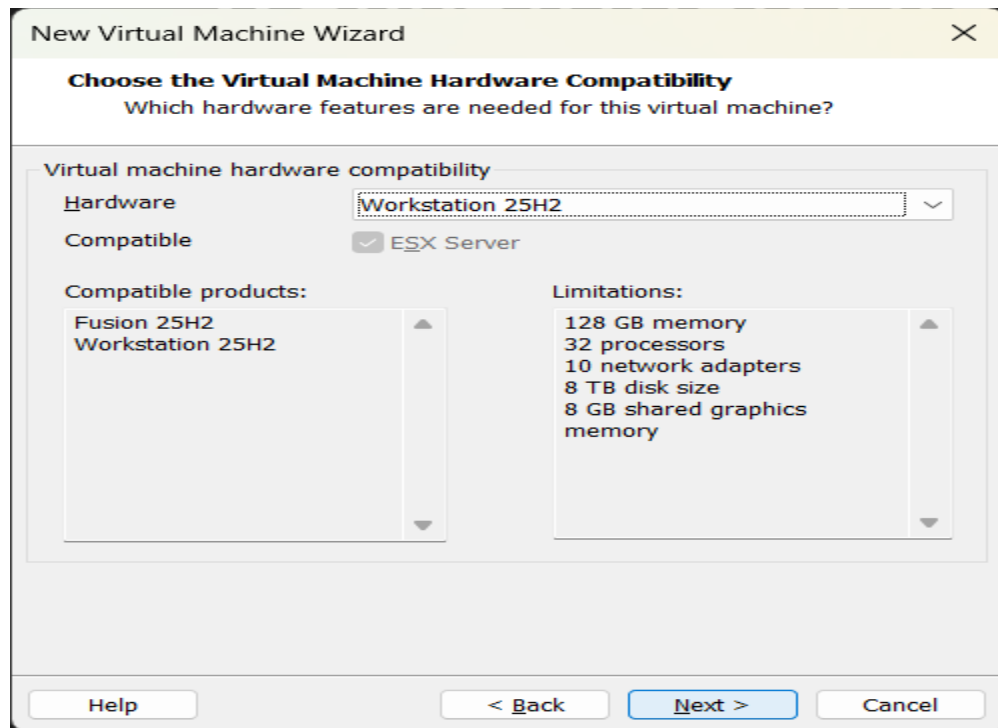
1. Open VMware and click 'Create a New Virtual Machine'.



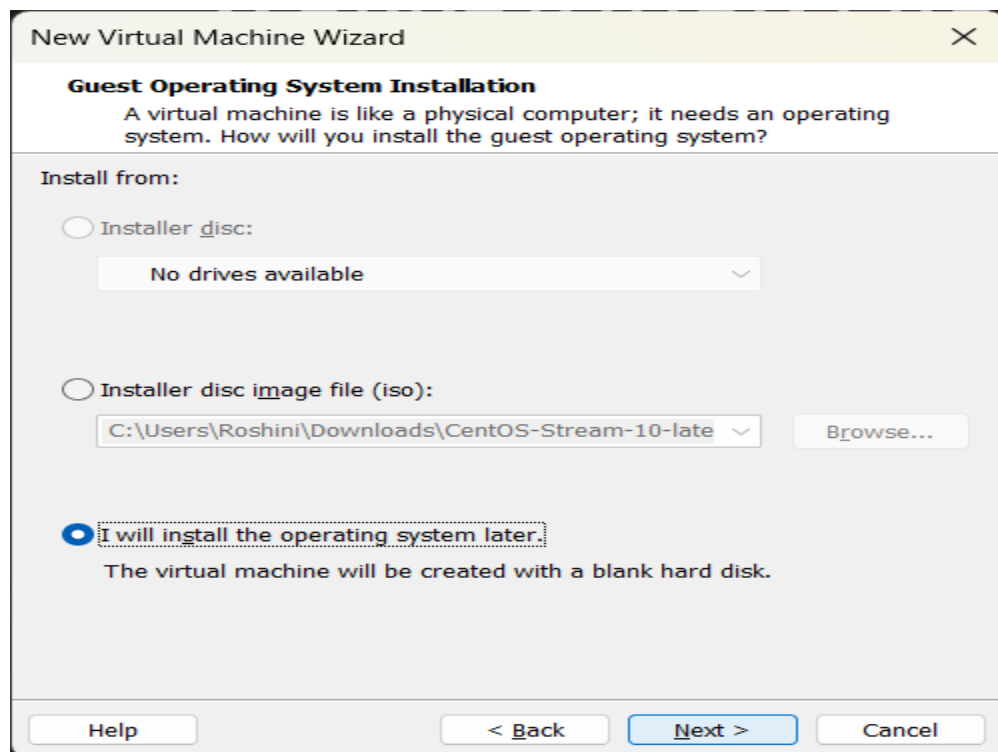
2. Choose 'Custom', then click 'Next'.



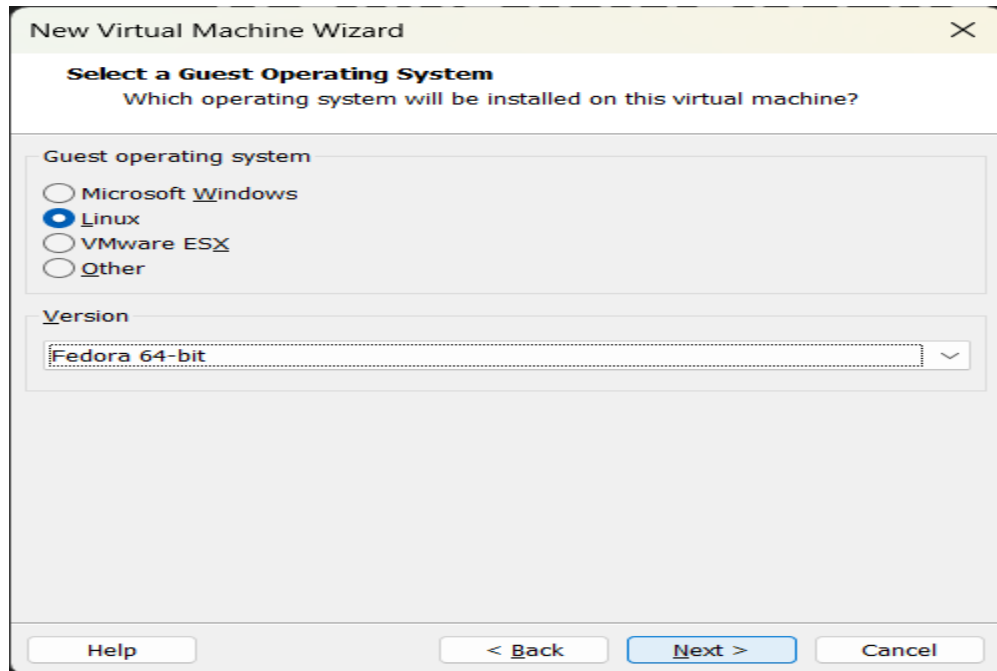
3. Click 'Next'.



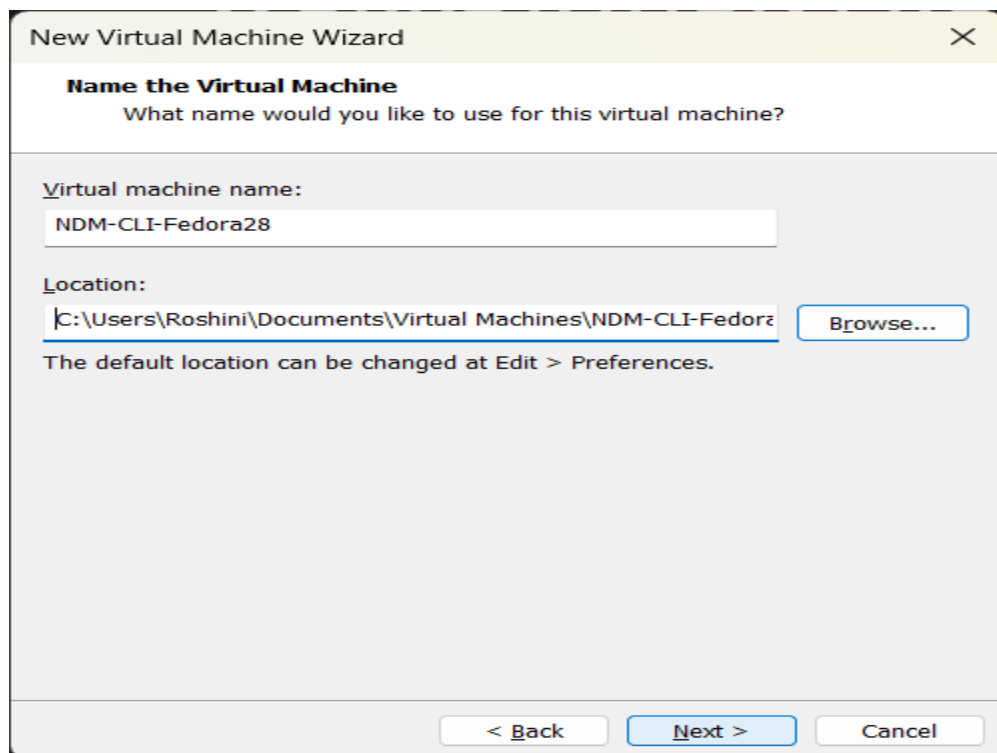
4. Choose 'I will install the operating system later' and click 'Next'.



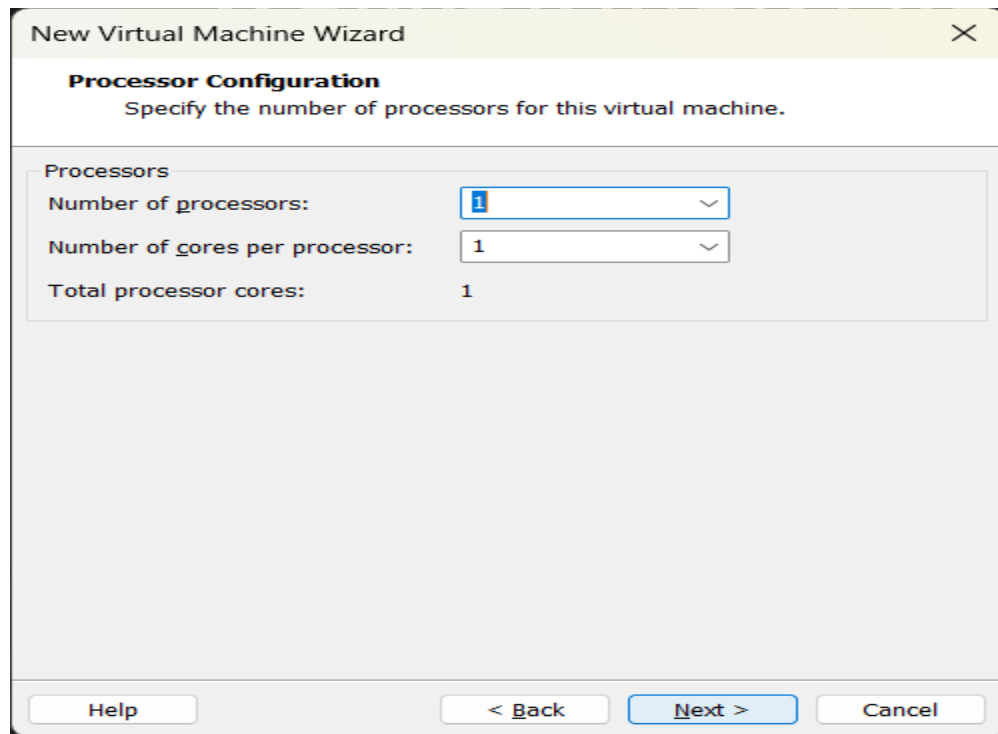
5. Choose Linux as the guest operating system and Fedora 64-bit as the version, then click Next.



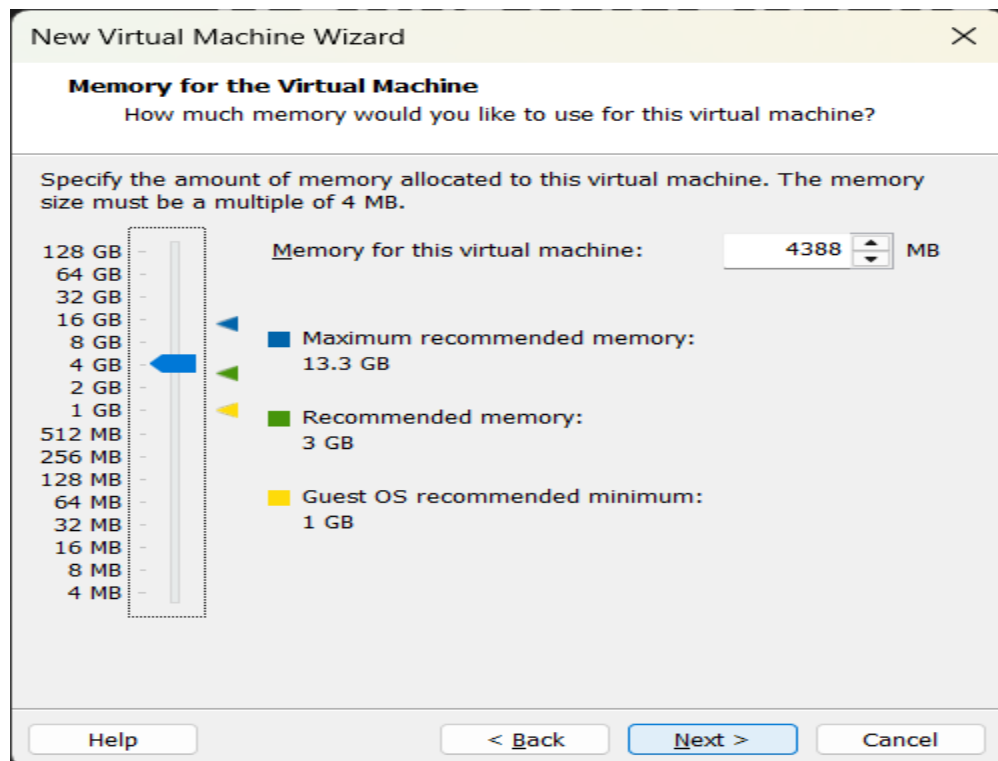
6. Change the virtual machine name to NDM-CLI-Fedora28, then click Next.



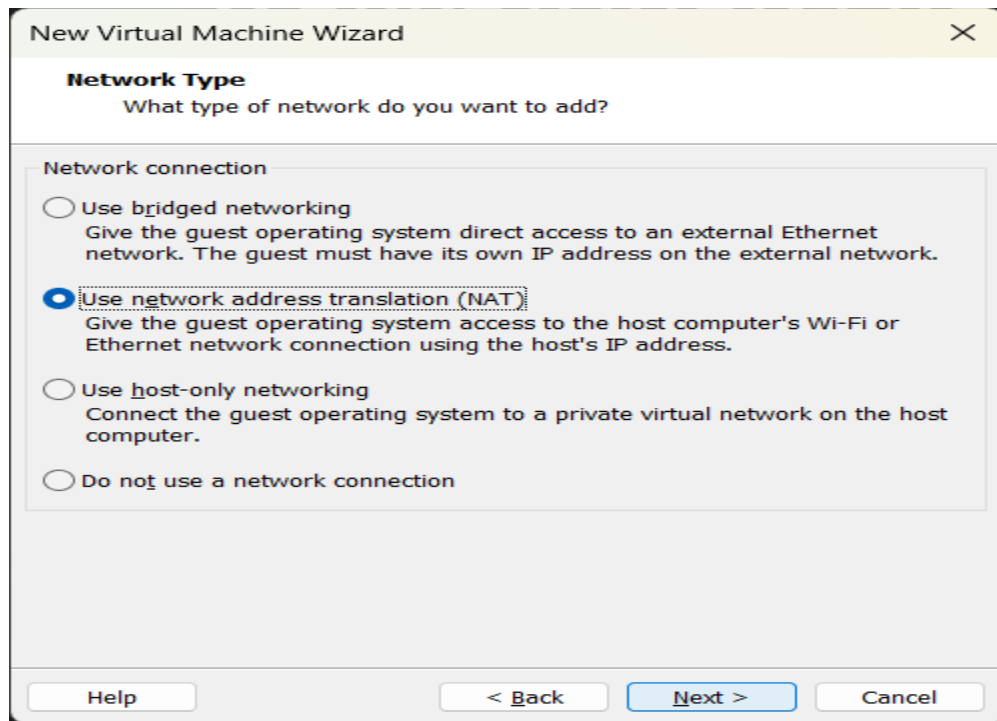
- Click 'Next'.



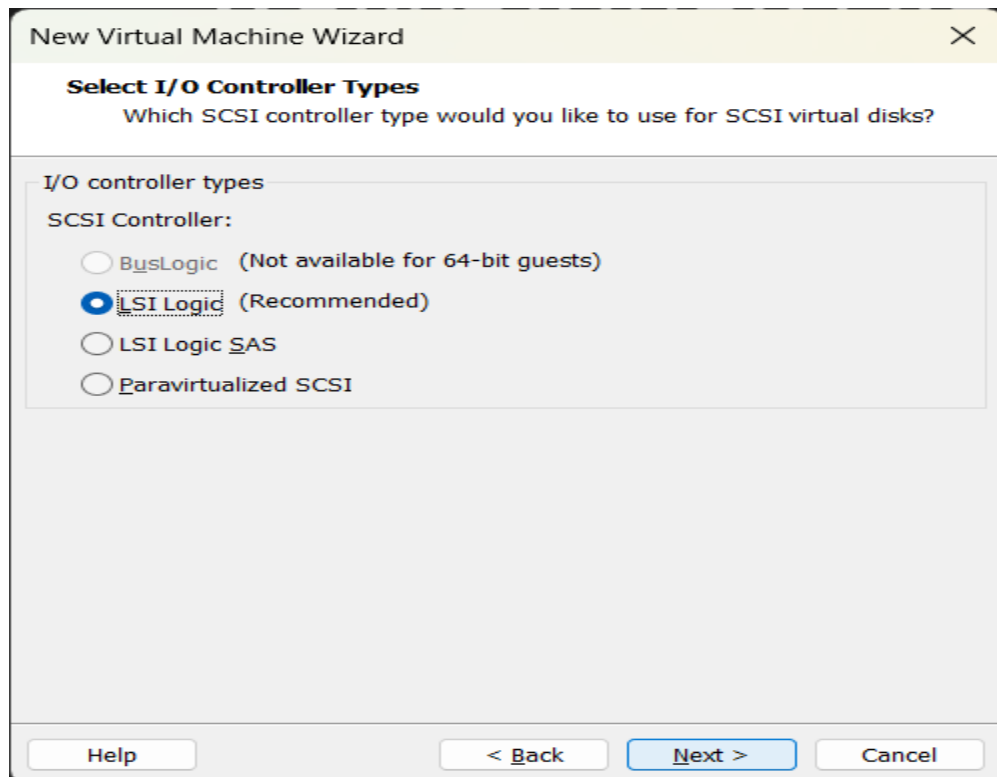
- Change the memory size to 4 GB, then click Next.



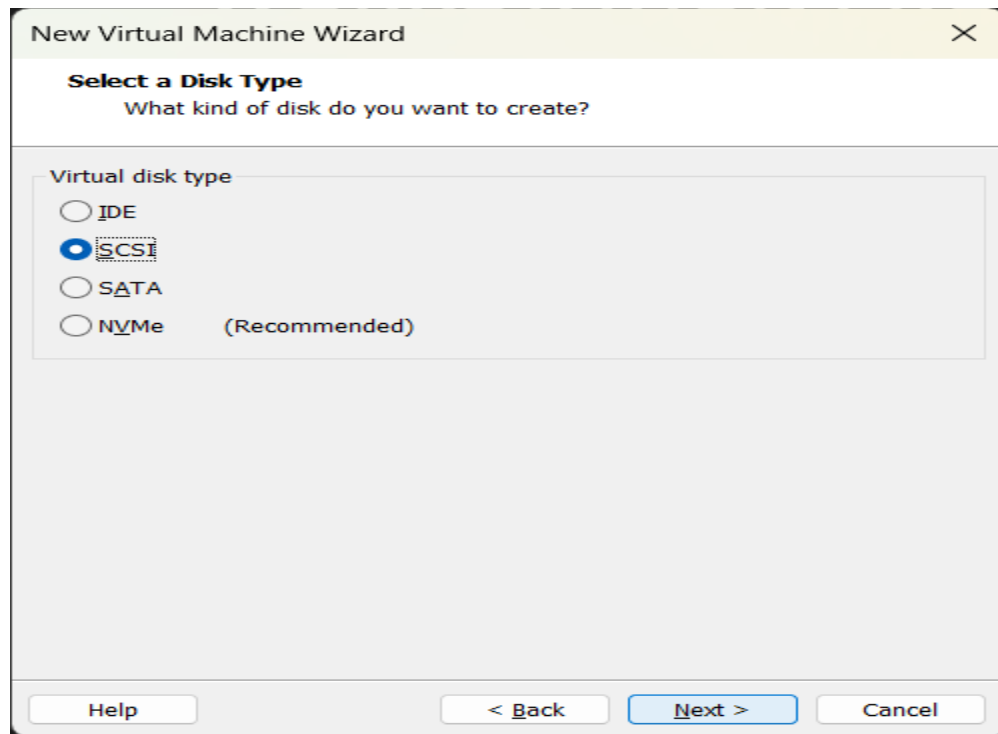
9. Choose 'Use Network Address Translation' and click 'Next'.



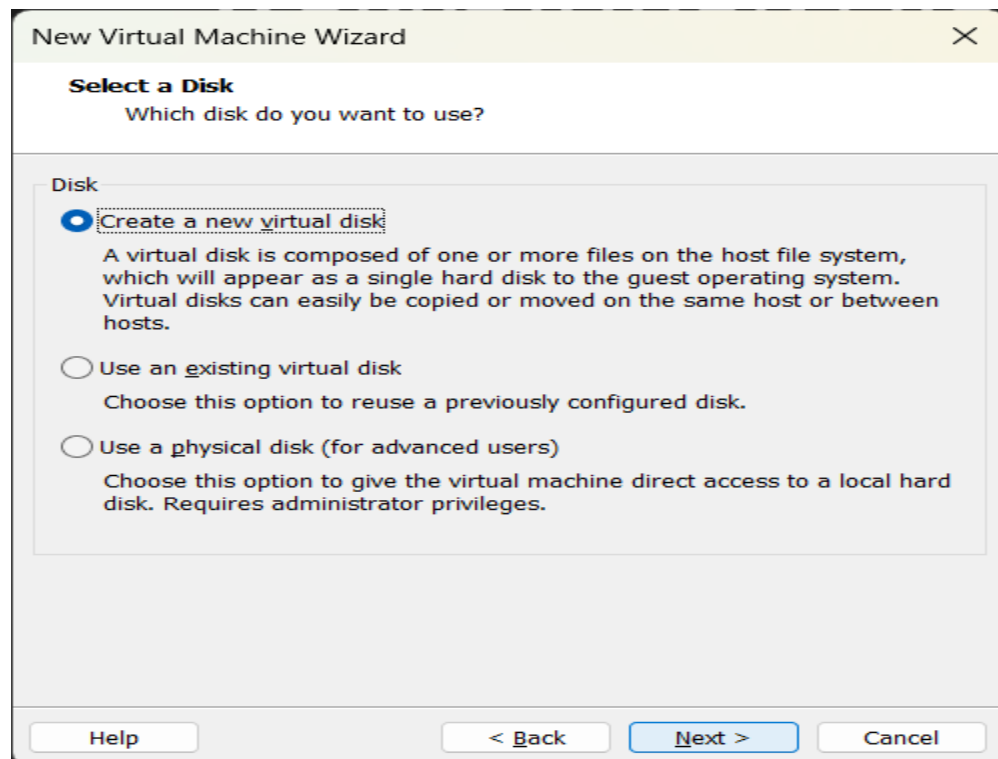
10. Choose 'LSI Logic' and click 'Next'.



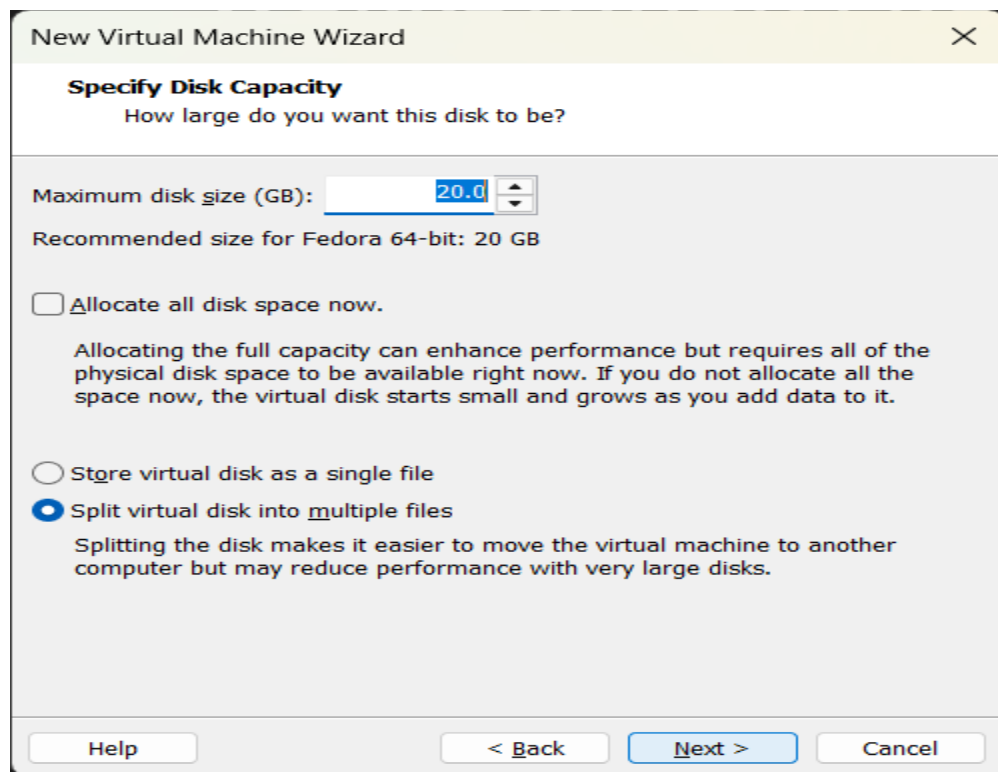
11. Choose 'SCSI' and click 'Next'.



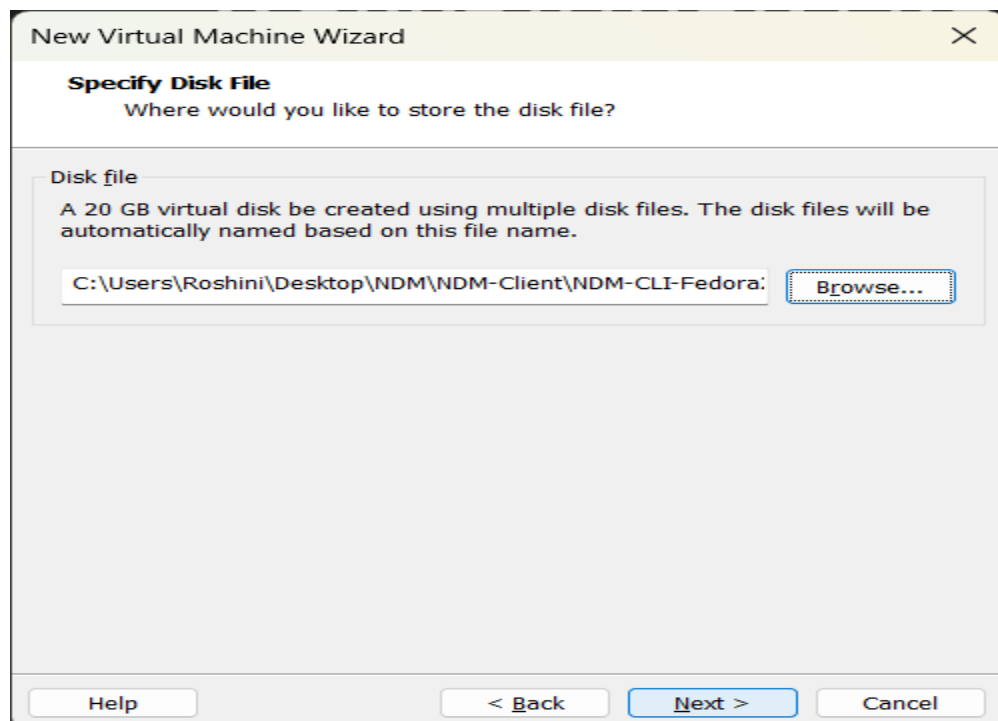
12. Choose 'Create a new virtual disk' and click 'Next'.



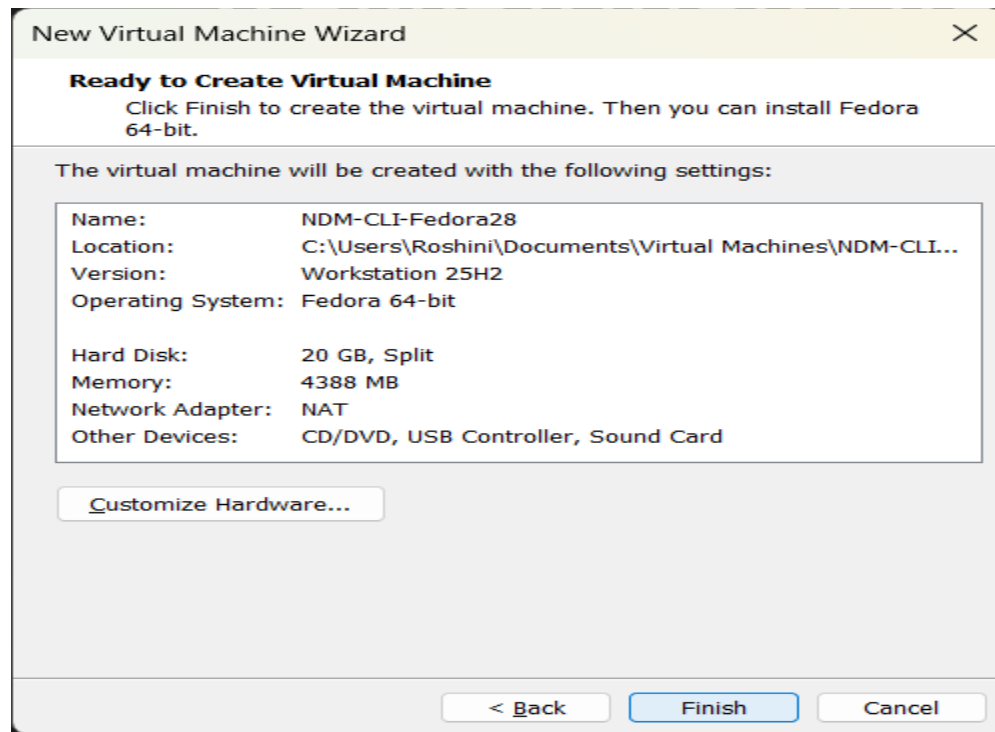
13. Choose 'Split virtual disk into multiple files' and click 'Next'.



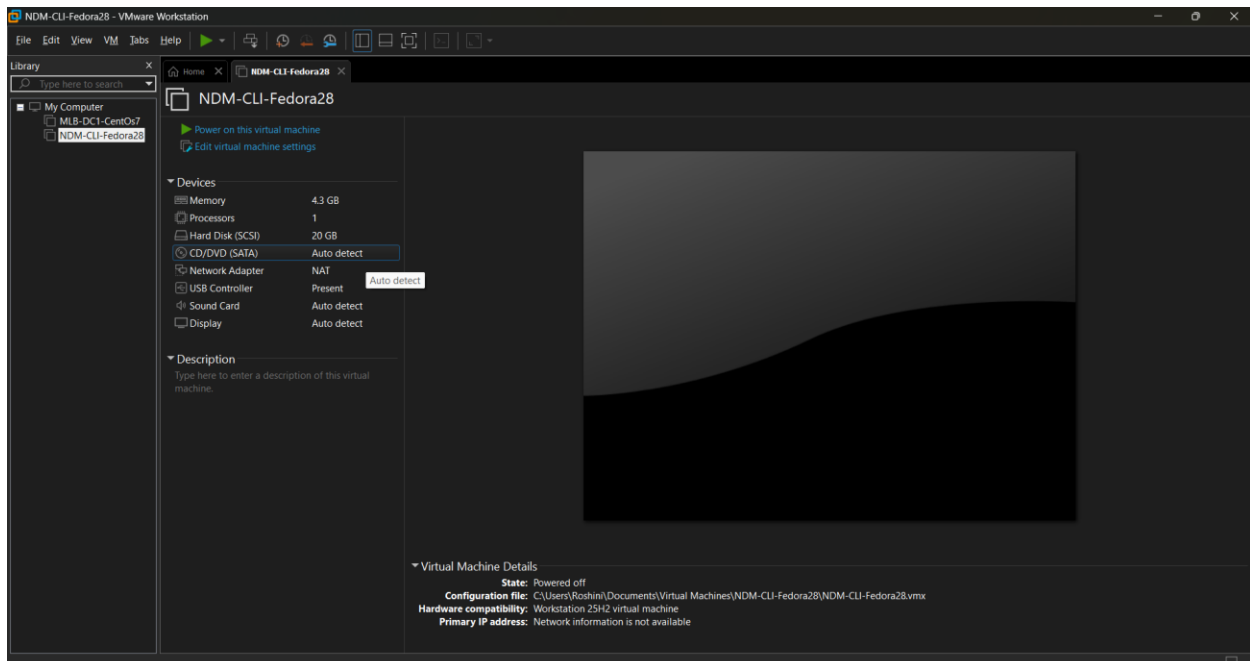
14. Click the 'Browse' button, choose the NDM-Server folder, and click 'Save'.



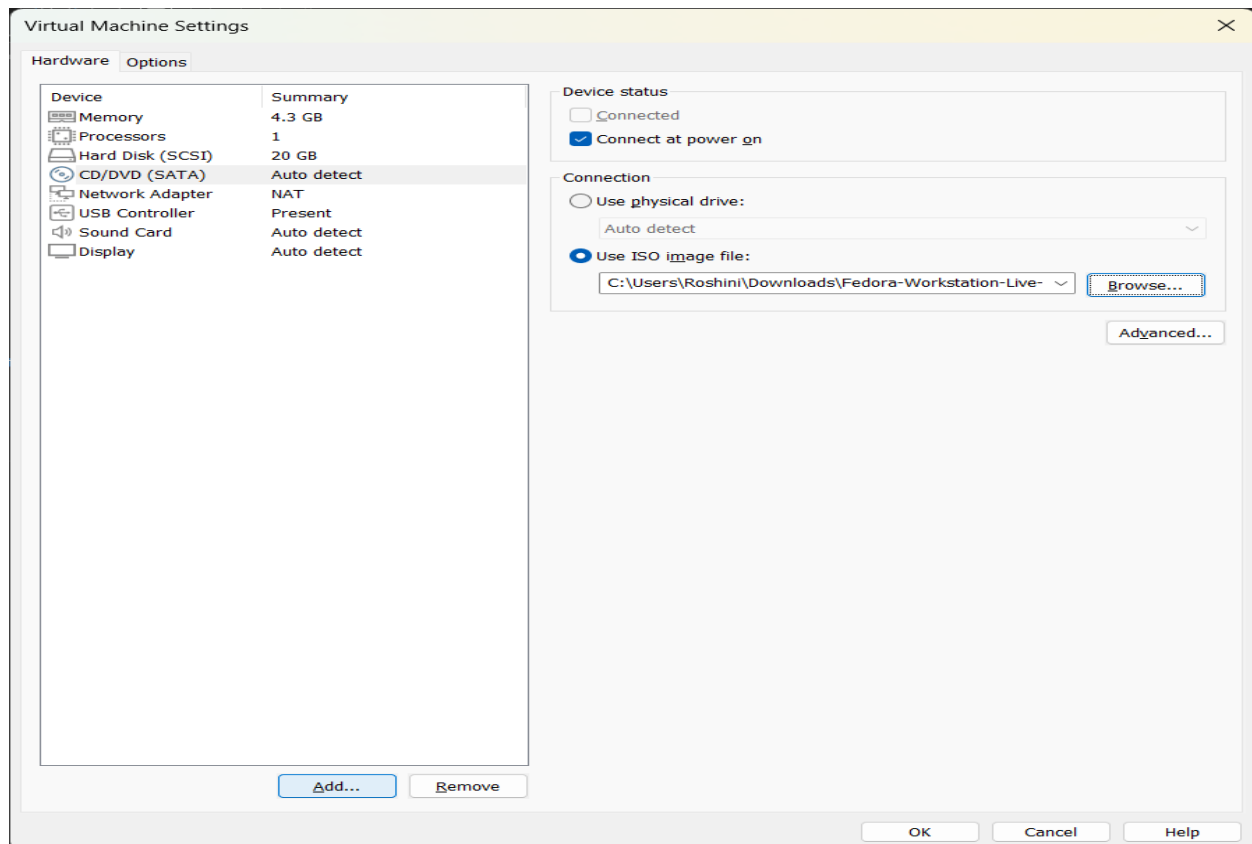
15. Then click the 'Finish' button.



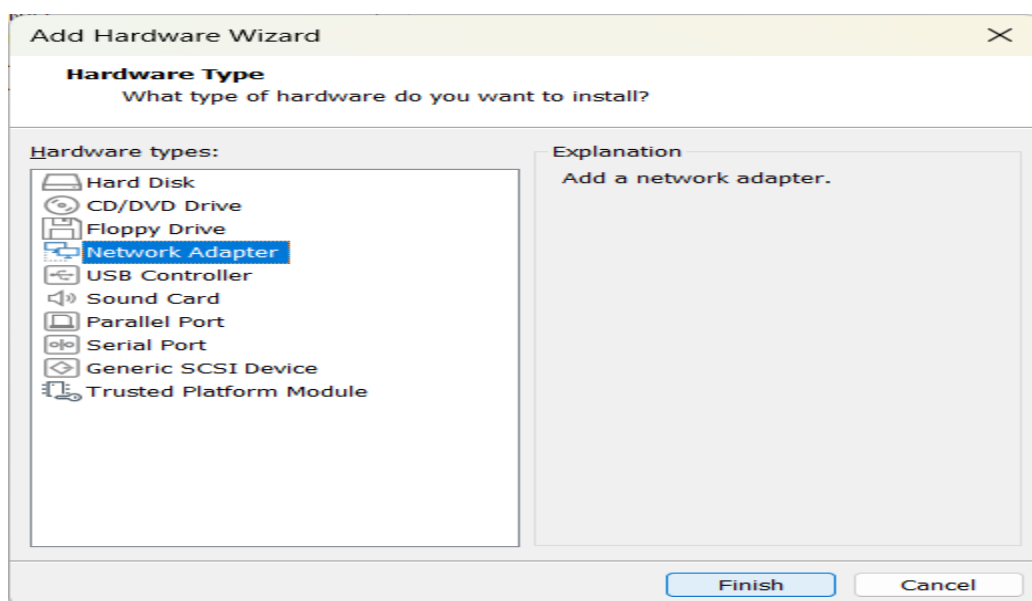
16. Click the 'CD/DVD' option.



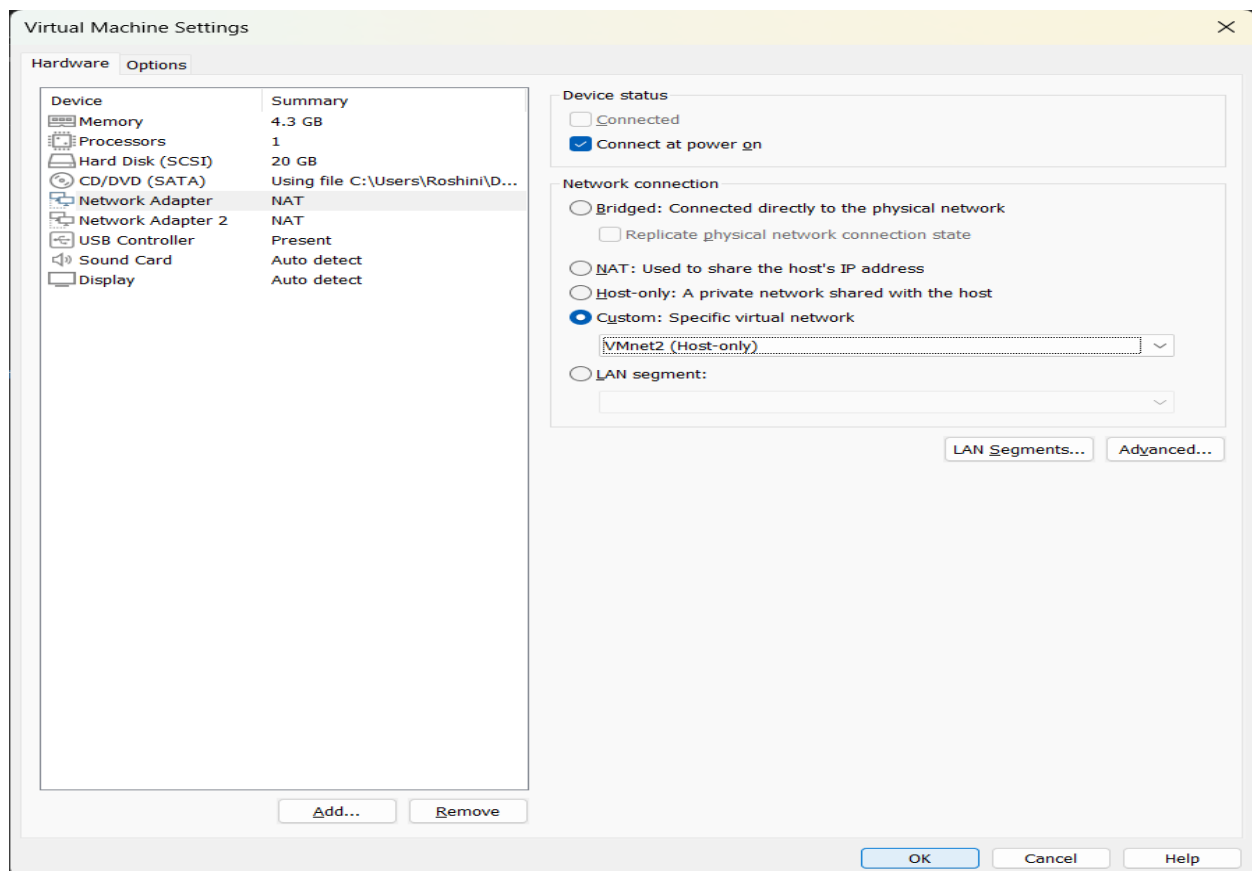
17. Choose the option 'Use ISO image file', then click 'Browse' to select the Fedora image and click 'Add'.



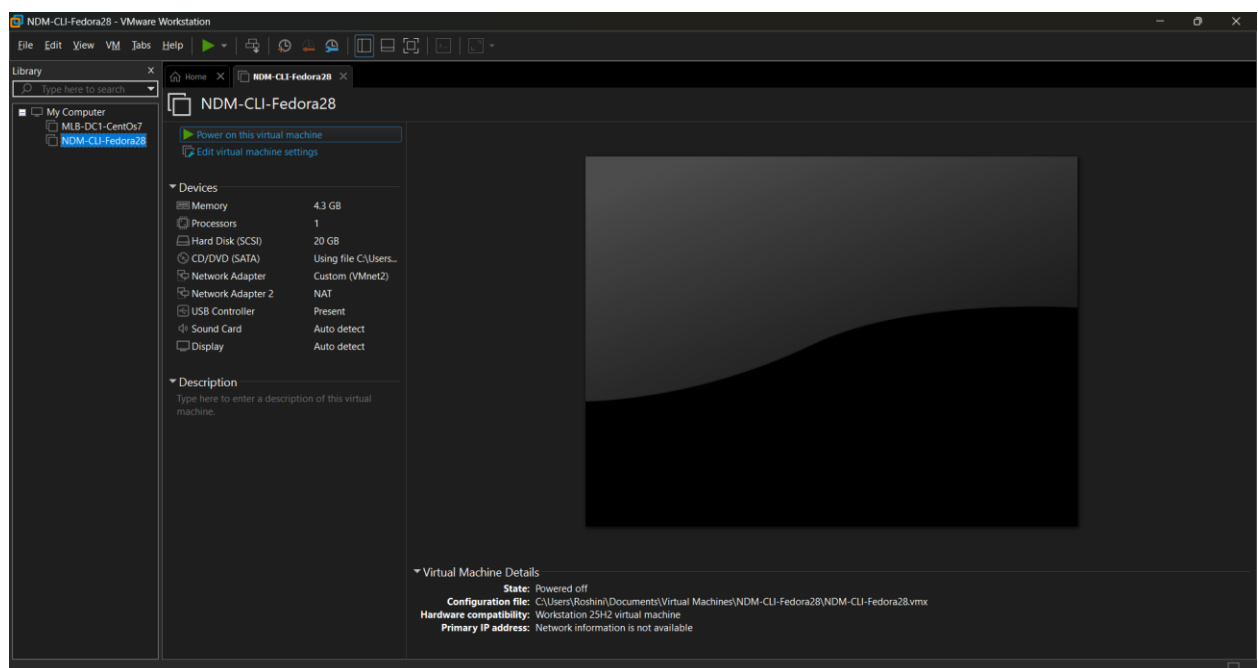
18. Select 'Network Adapter' under Hardware Types, and click 'Finish'.



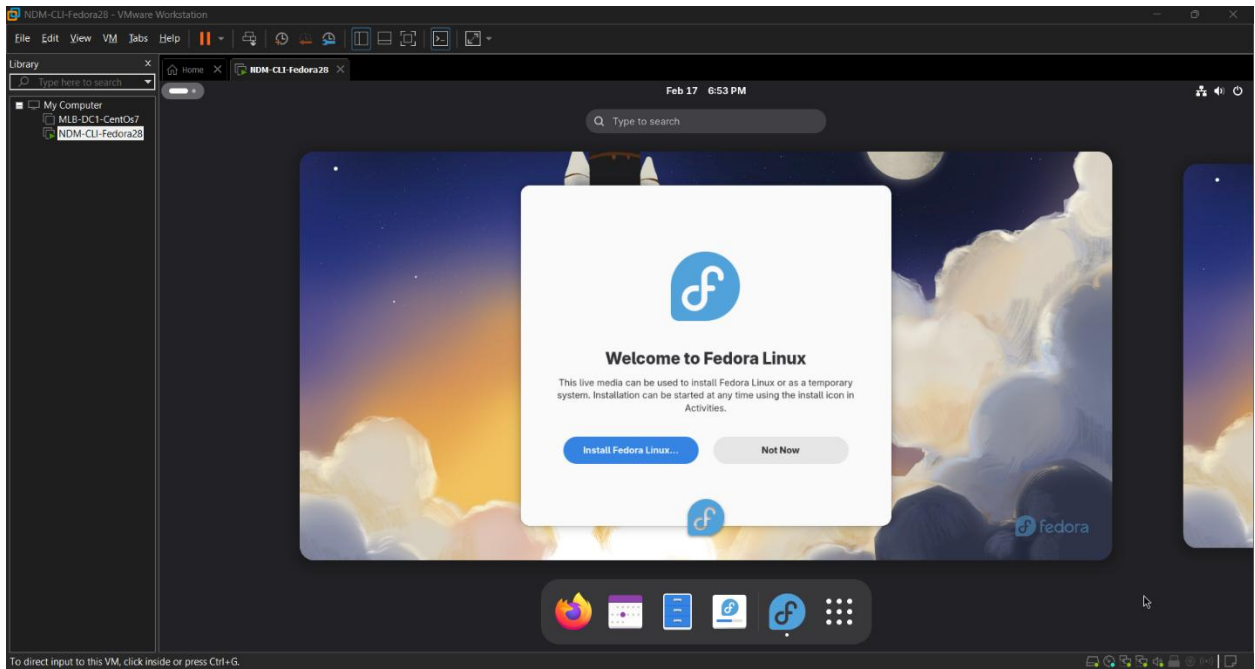
19. Then select 'Custom' and choose VMnet2 (Host-only), then click 'OK'



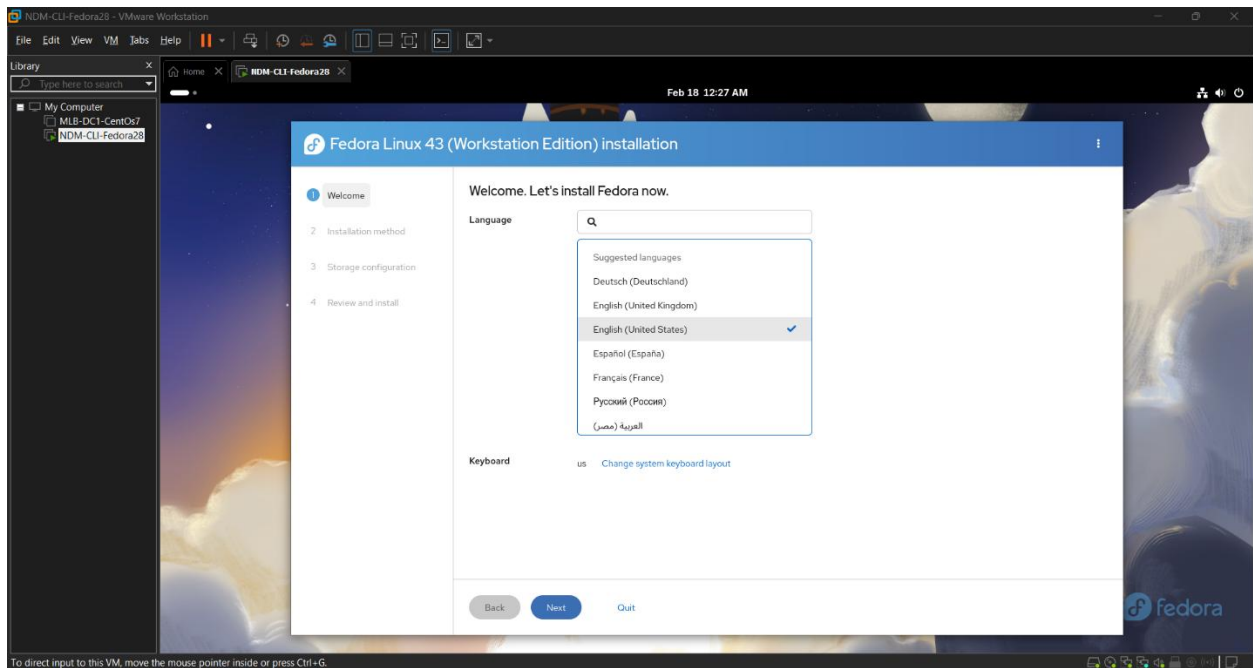
20. Click 'Power on this Virtual Machine'.



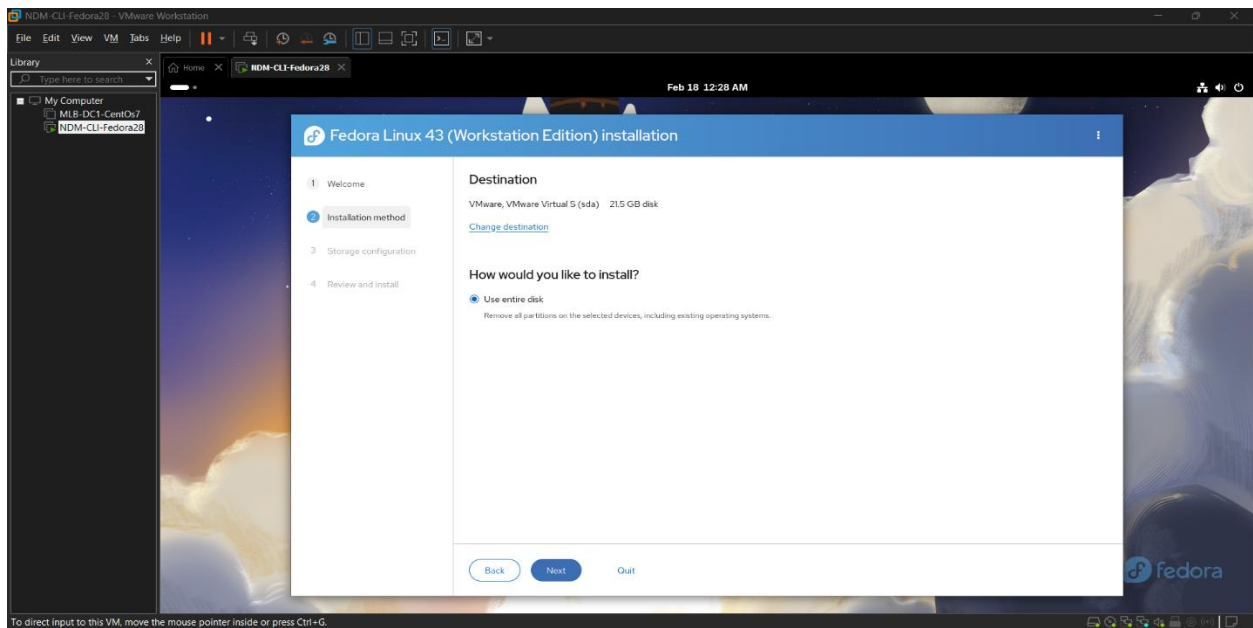
21. Click the 'Install Fedora Linux' button.



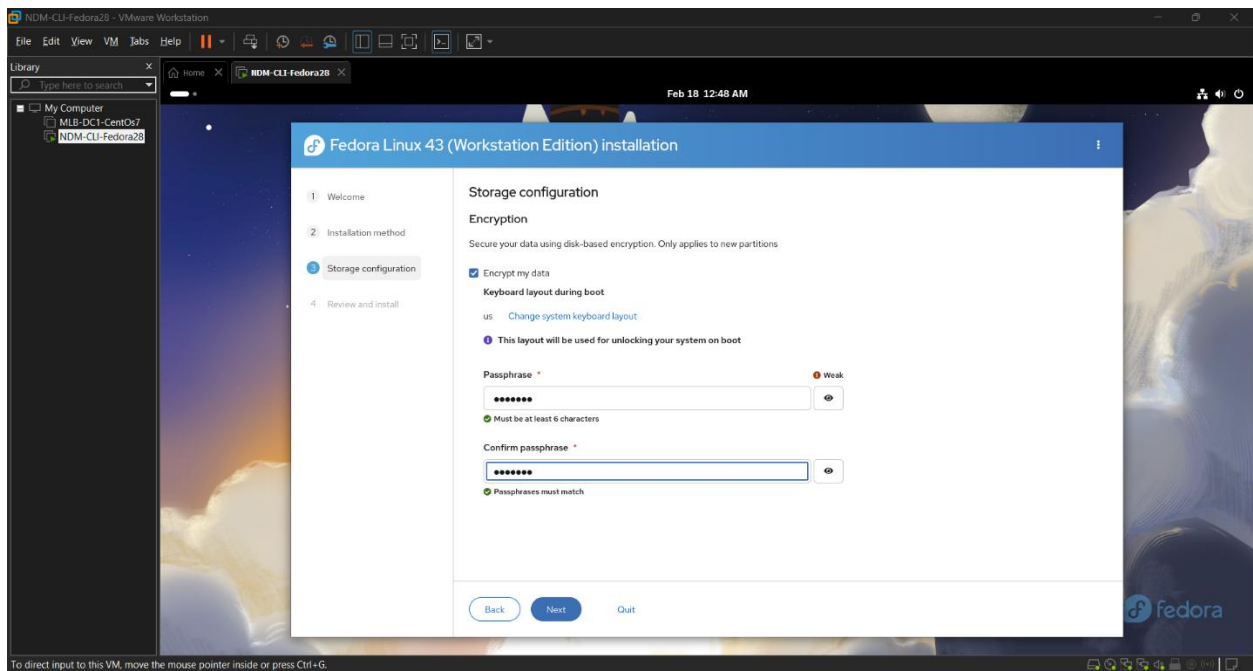
22. Select the language, then click 'Next'.



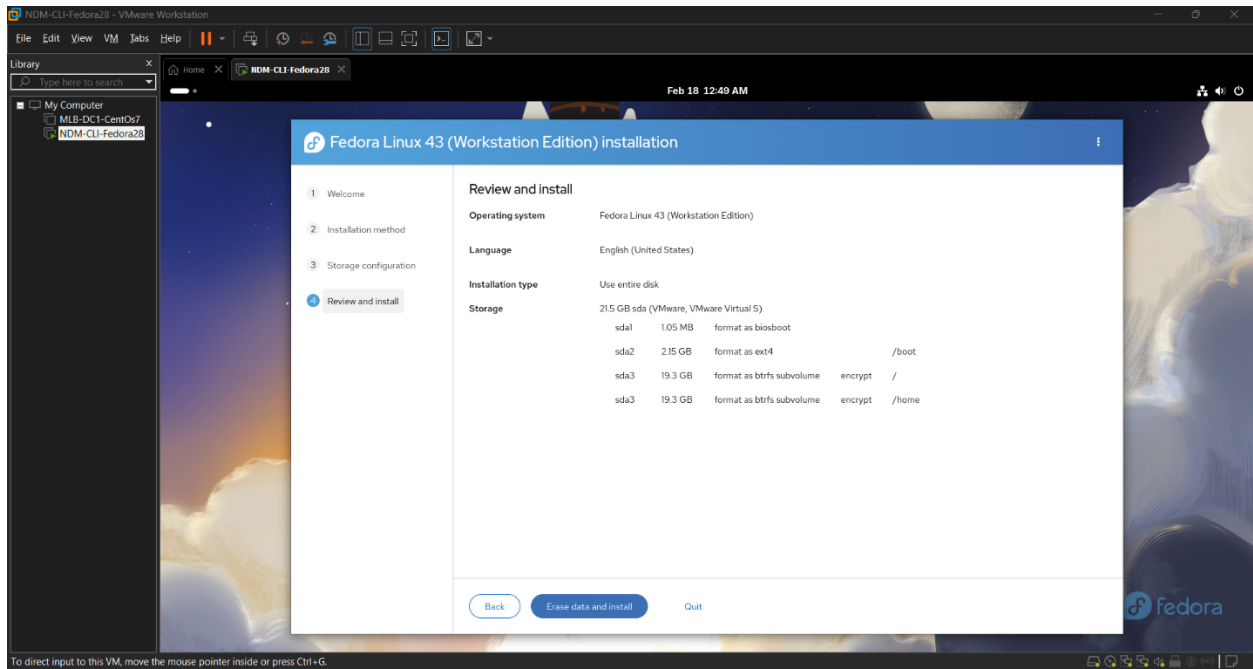
23. Click 'Next'.



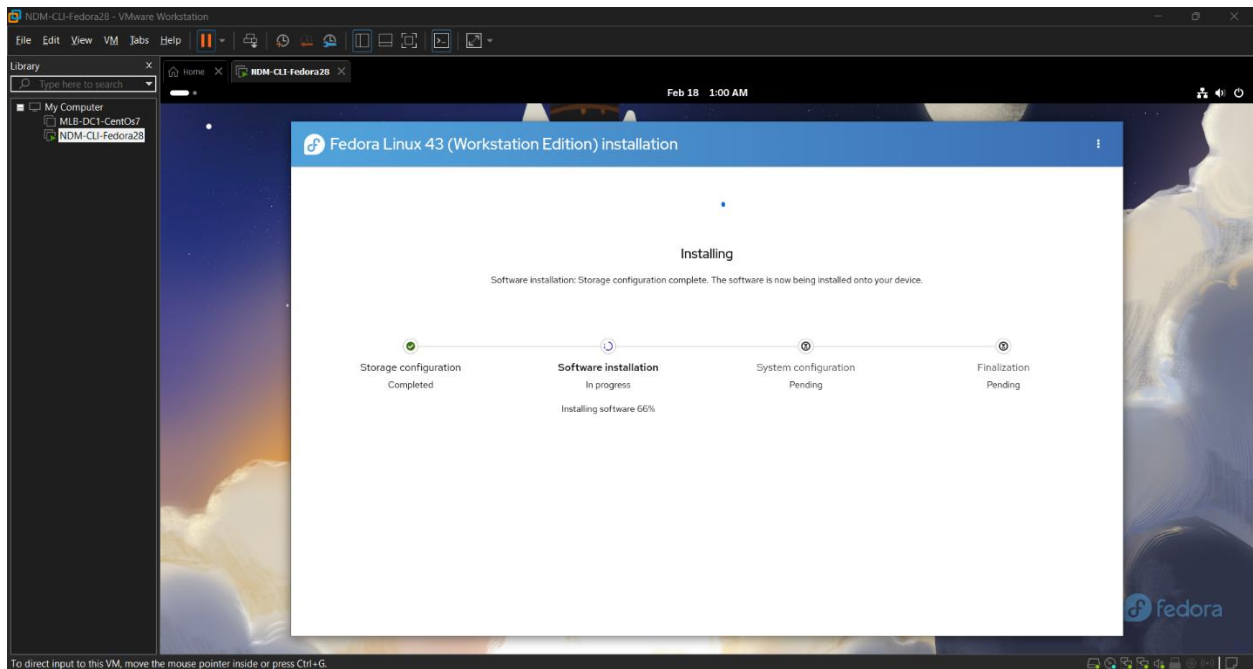
24. Type 'client1' and click 'Next'.



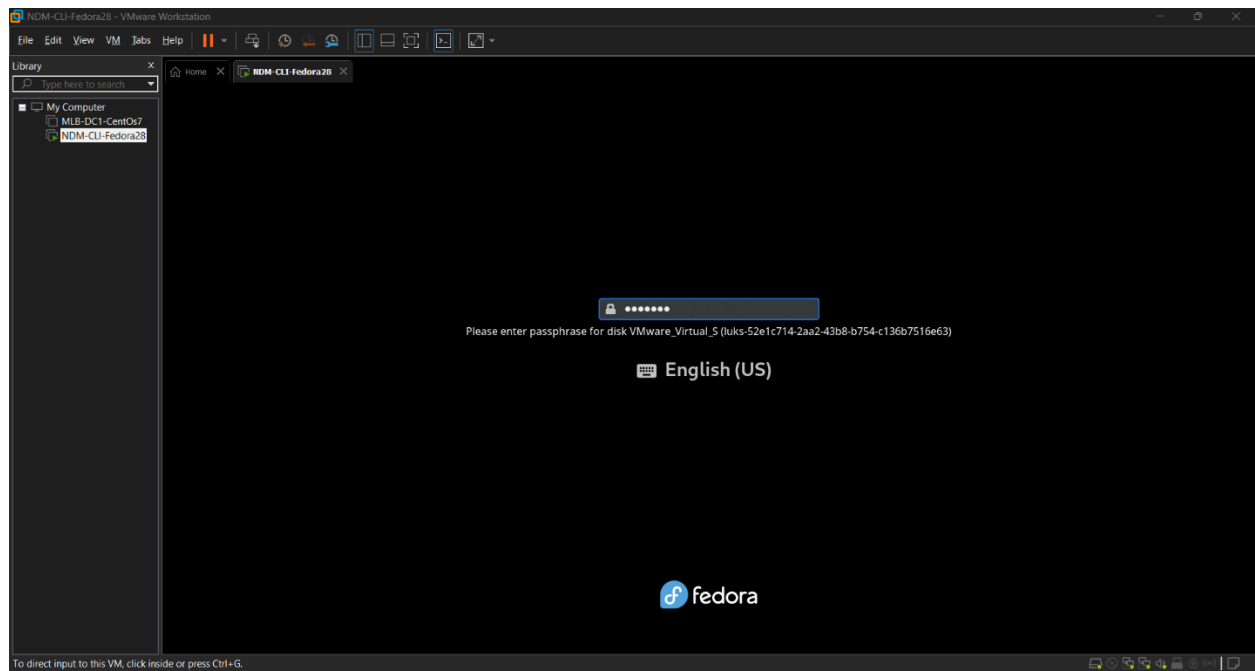
25. Click 'Erase Data and Install'.



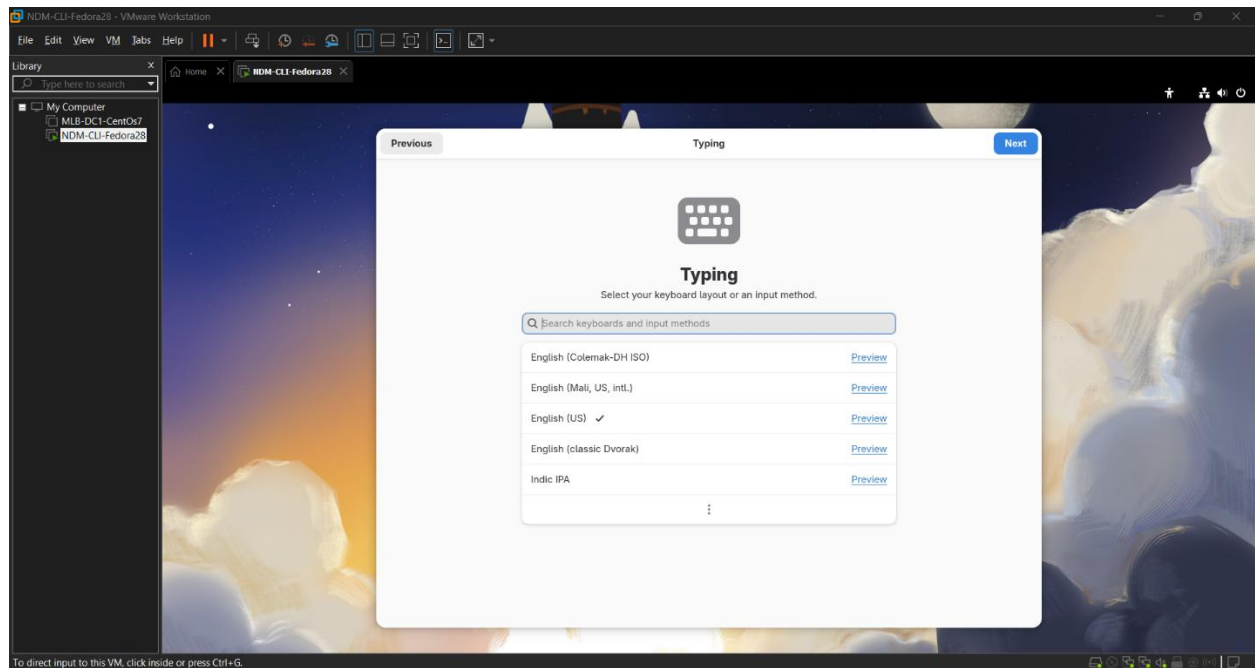
26. Wait until the installation is complete.



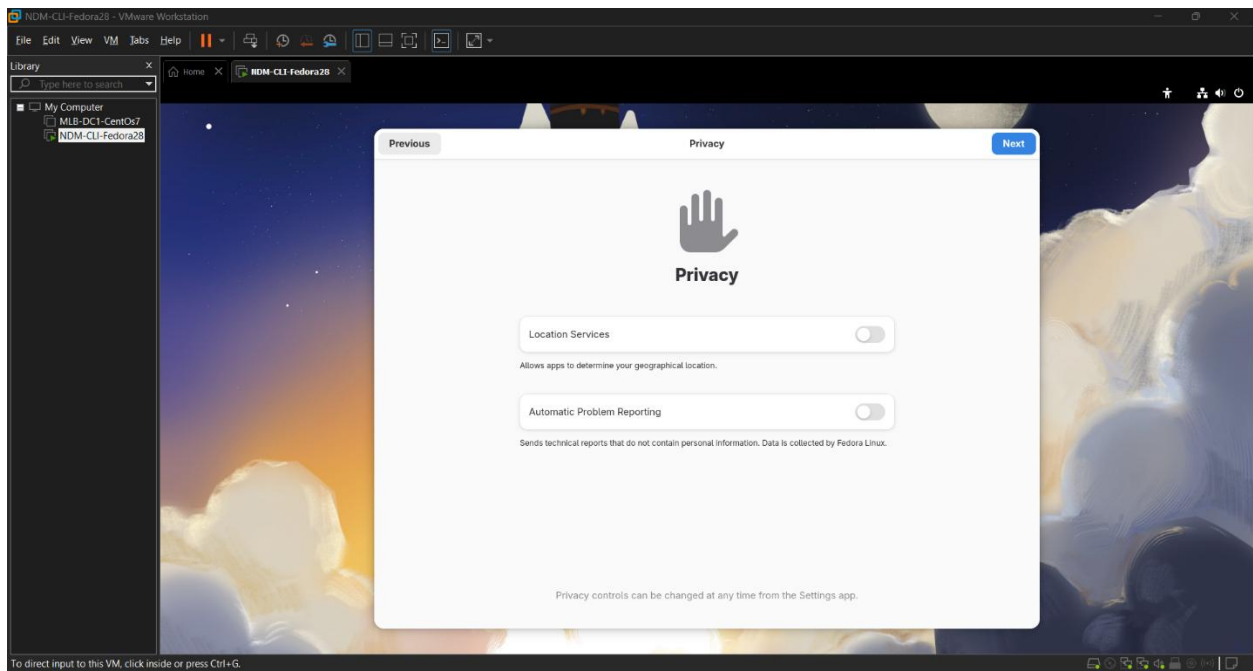
27. Restart the Fedora system.



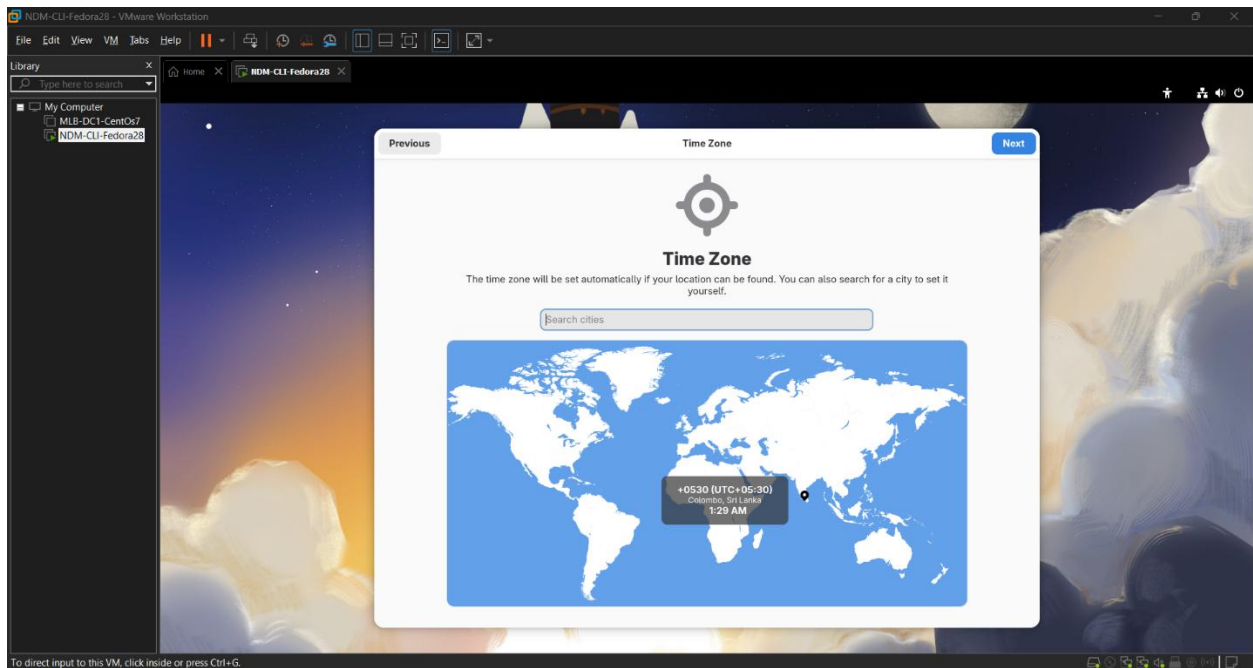
28. select the language and click next



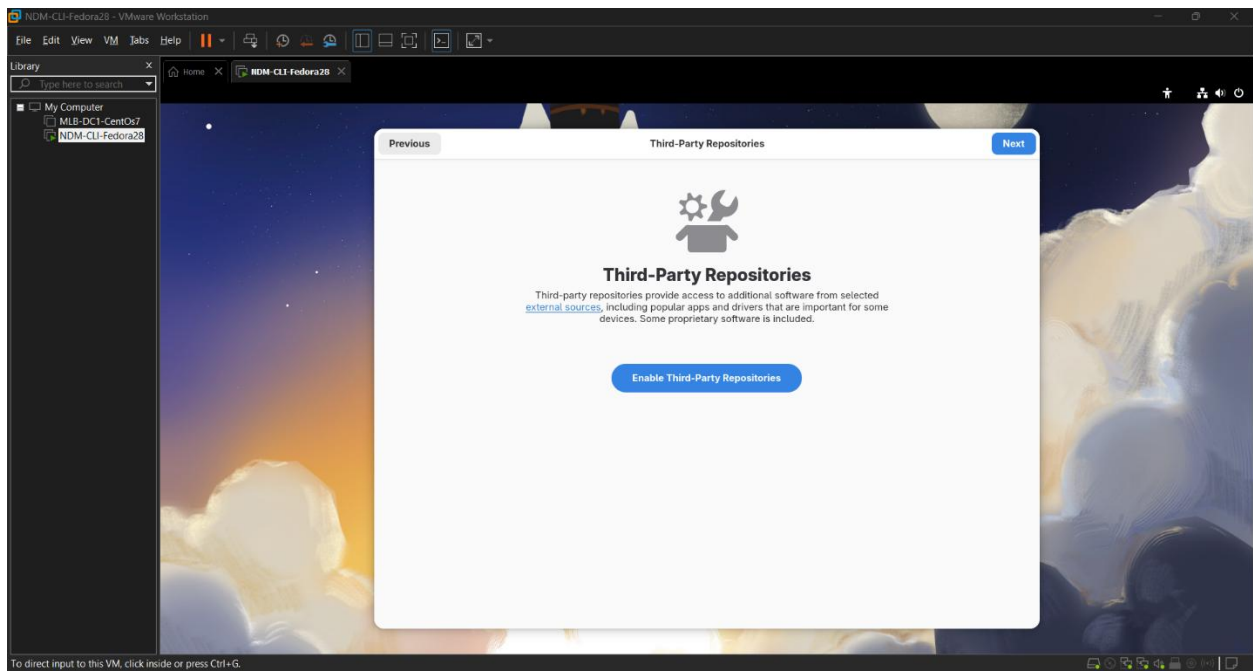
29. In the Privacy section, close the tabs, and click the 'Next' button.



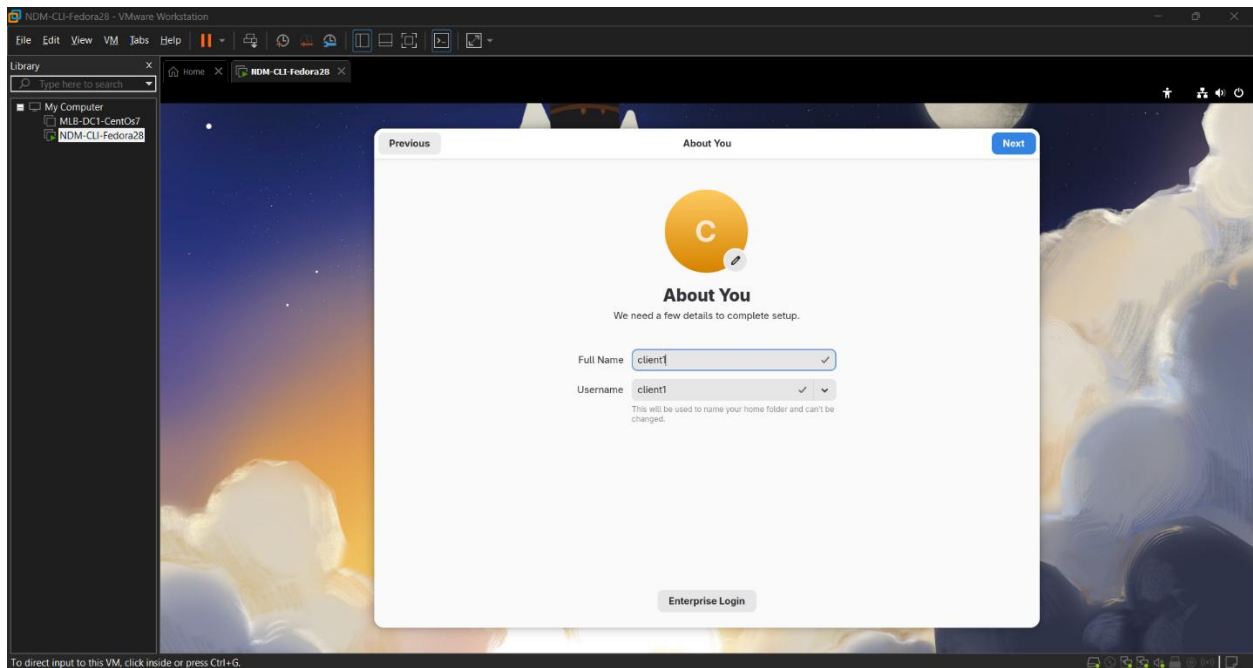
30. Choose your time zone and click the 'Next' button.



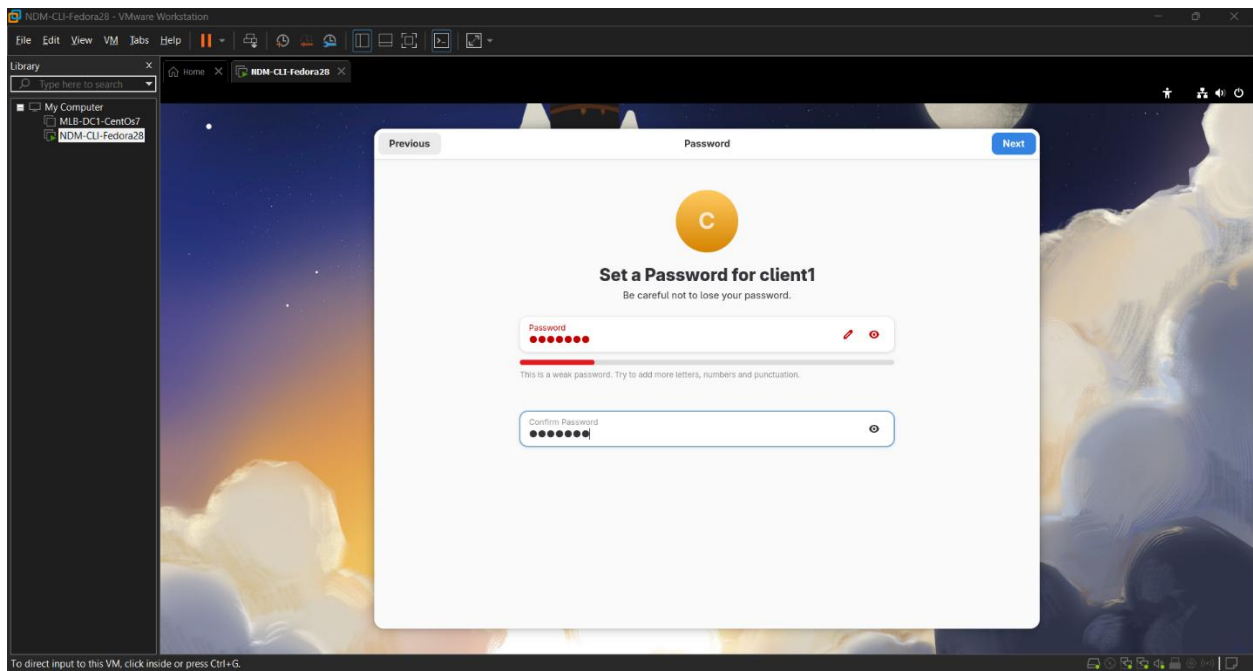
31. Click 'Next'.



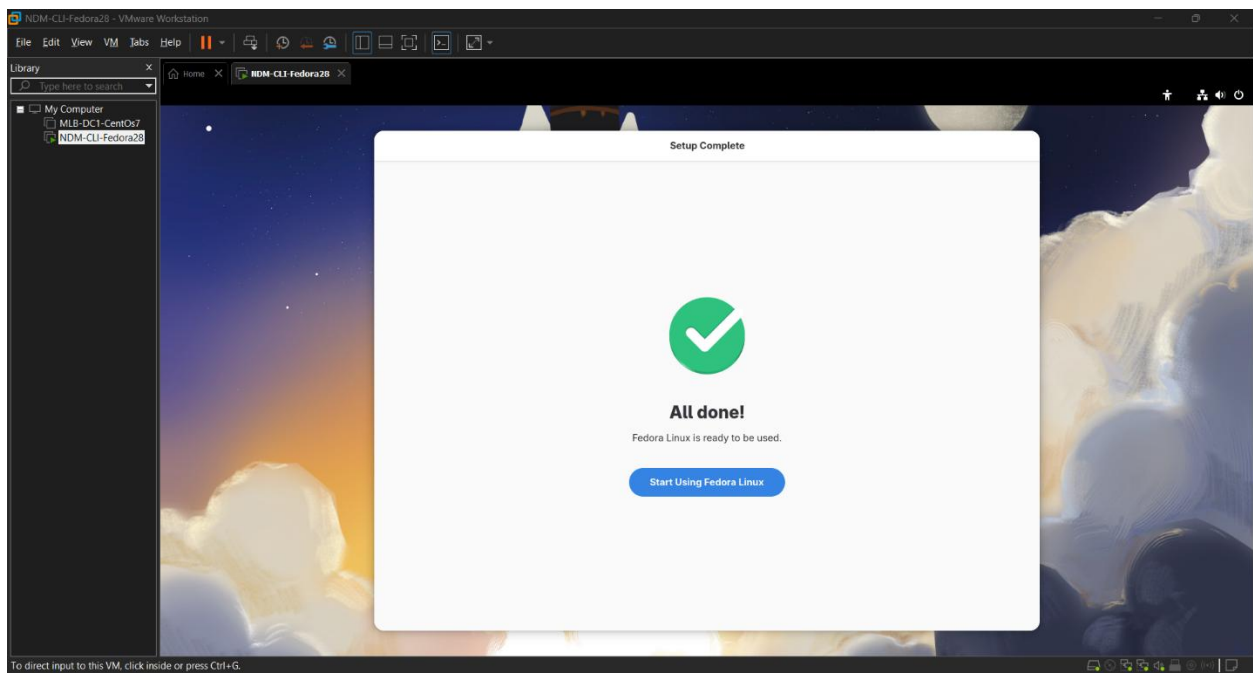
32. Type 'client1' as the name, then click 'Next'.



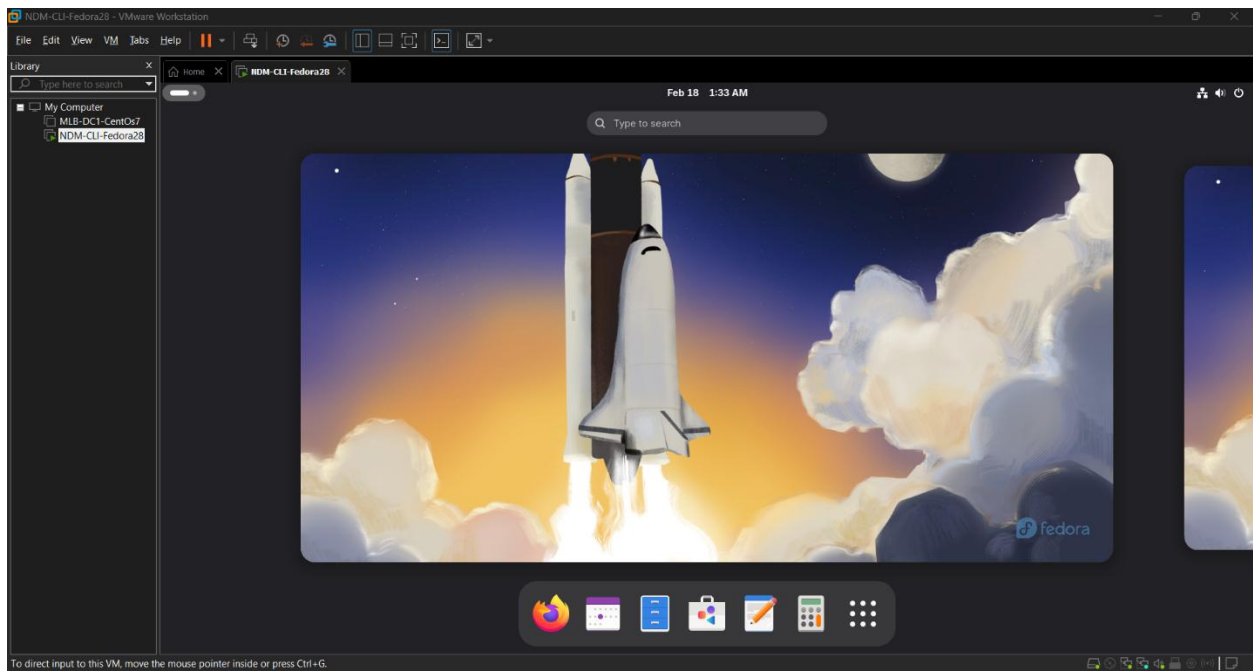
33. Set the password as 'client1'.



34. Click 'Start Using Fedora Linux.'



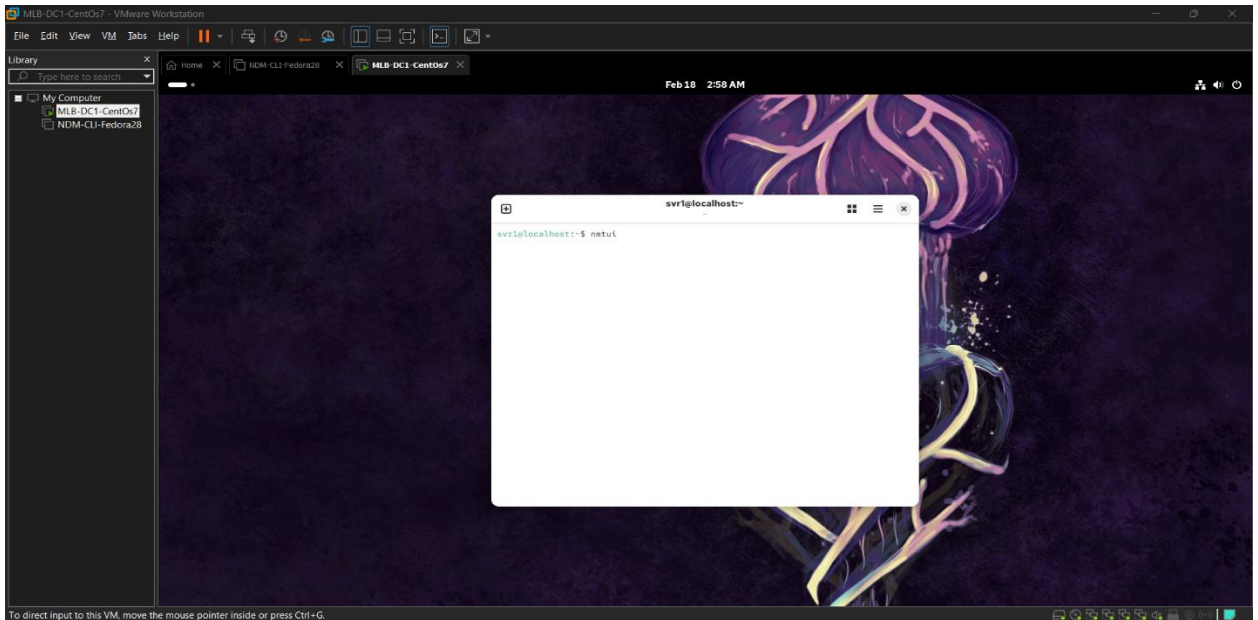
35. Finally, you are in the client account.



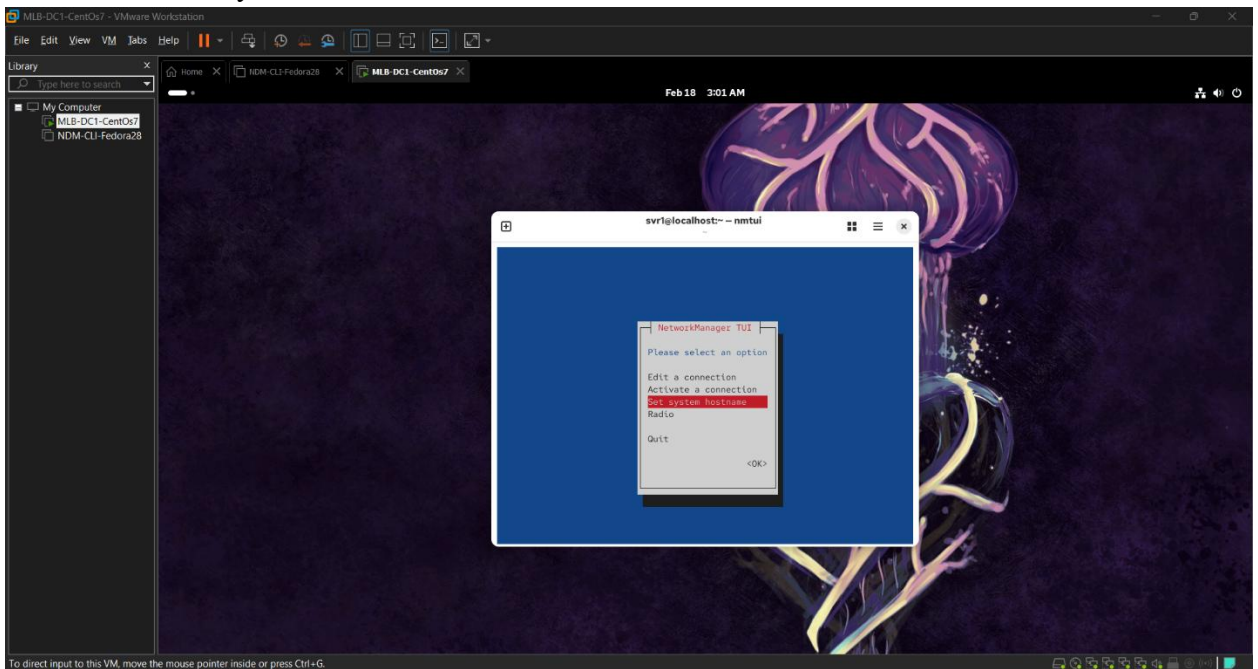
Change hostname in server and client

➤ Server

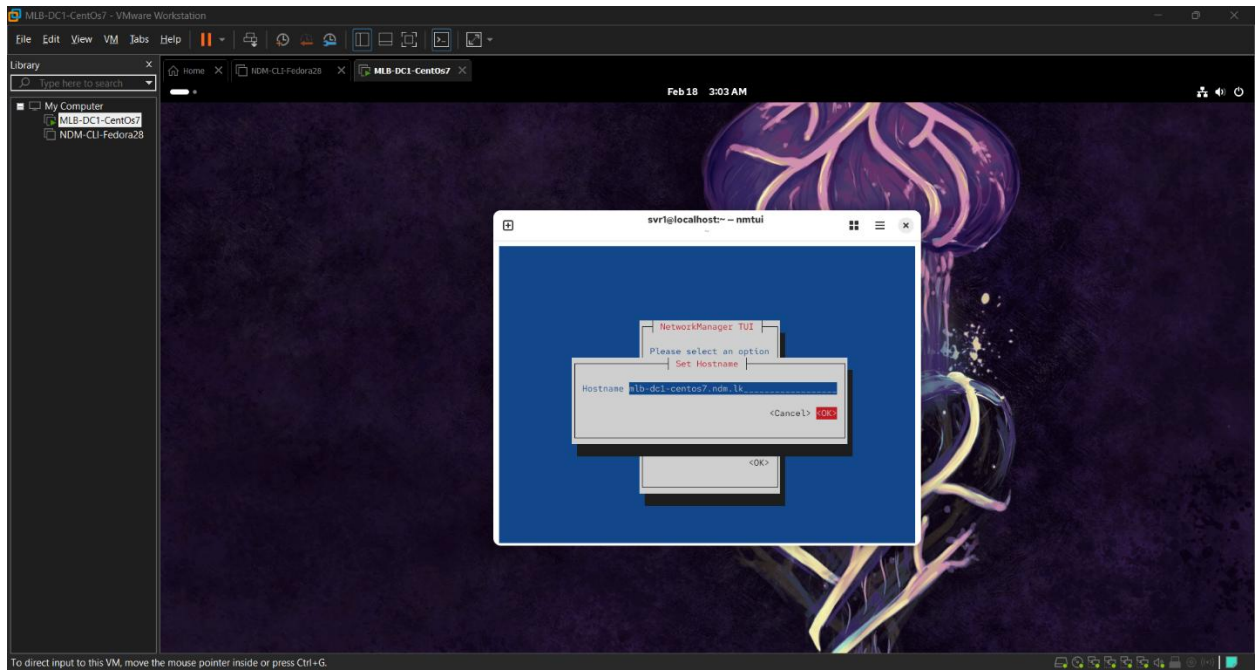
1. Power on the CentOS virtual machine.
2. Open the terminal, type nmtui, and press Enter.



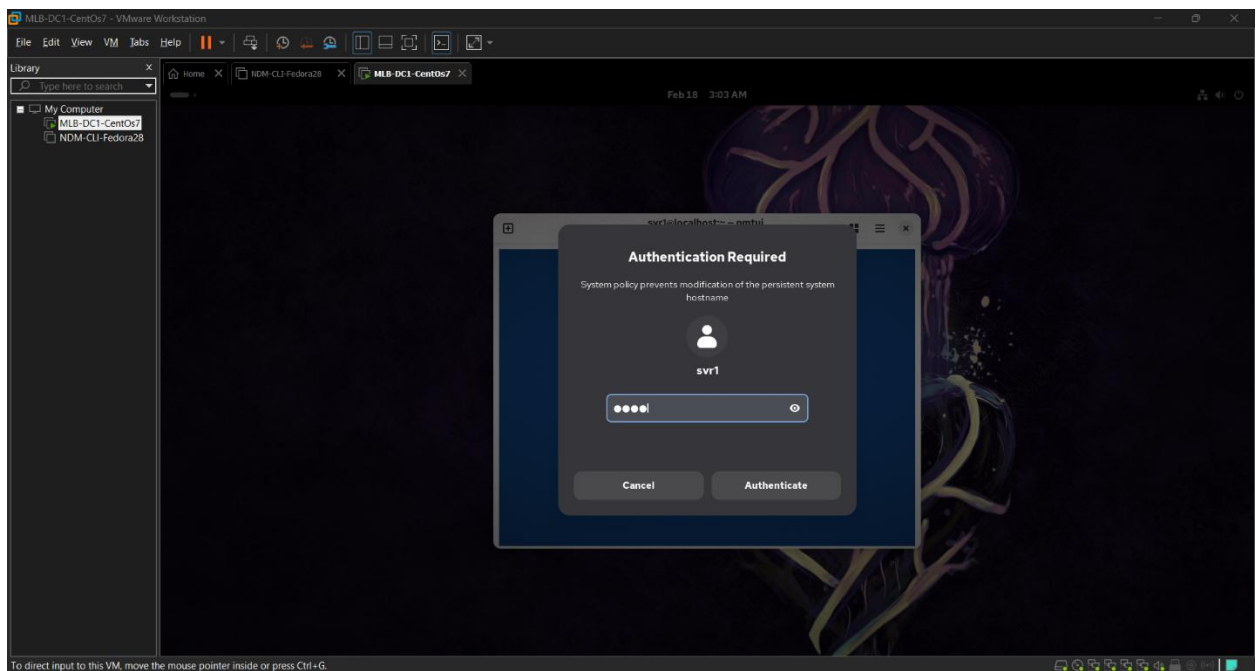
3. Then select 'Set System Hostname'.



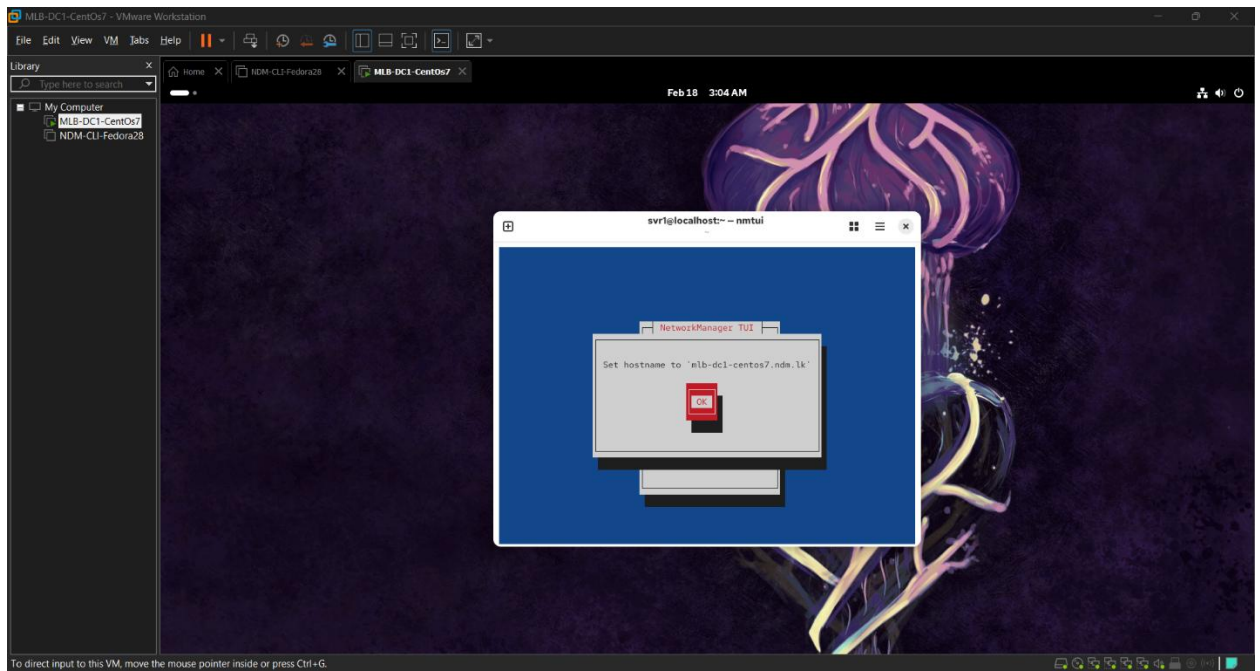
4. Change the hostname, then click 'OK'.



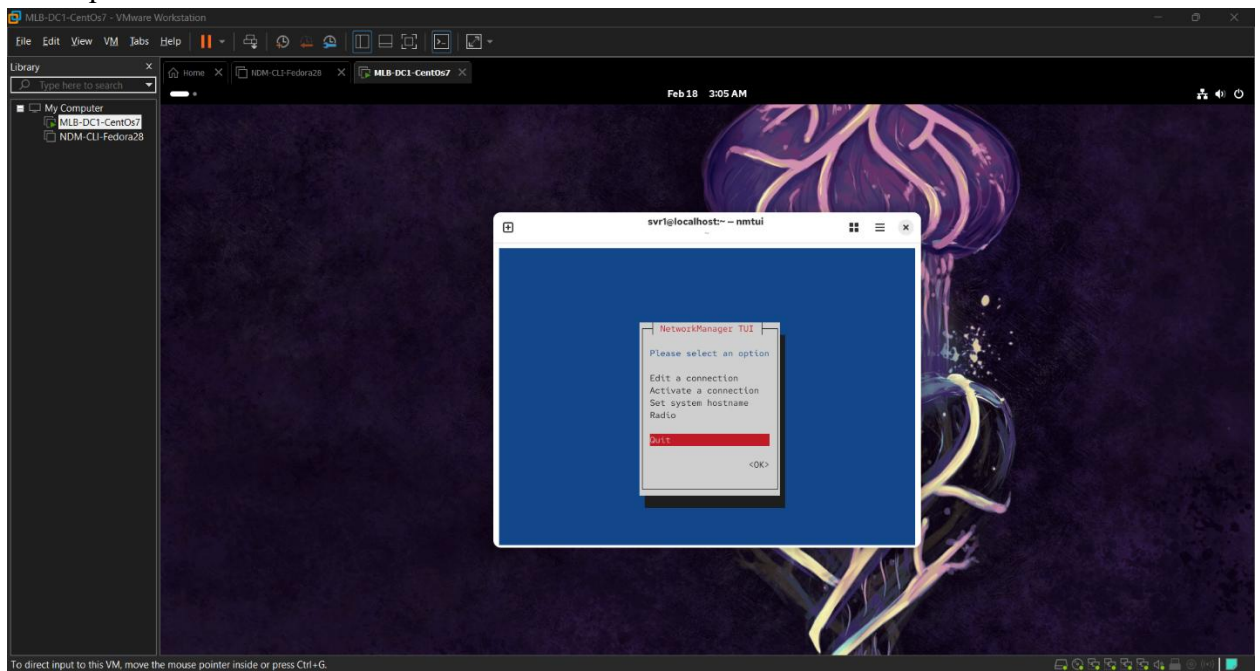
5. Provide authentication access.



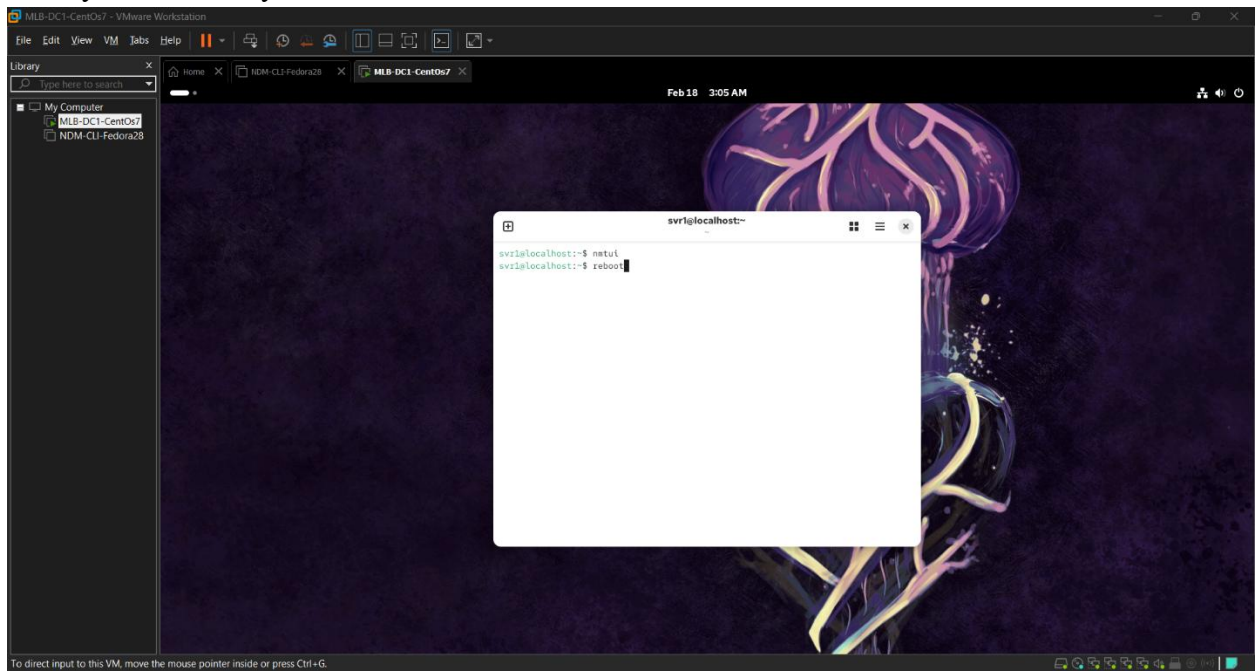
6. Then click the 'OK' button.



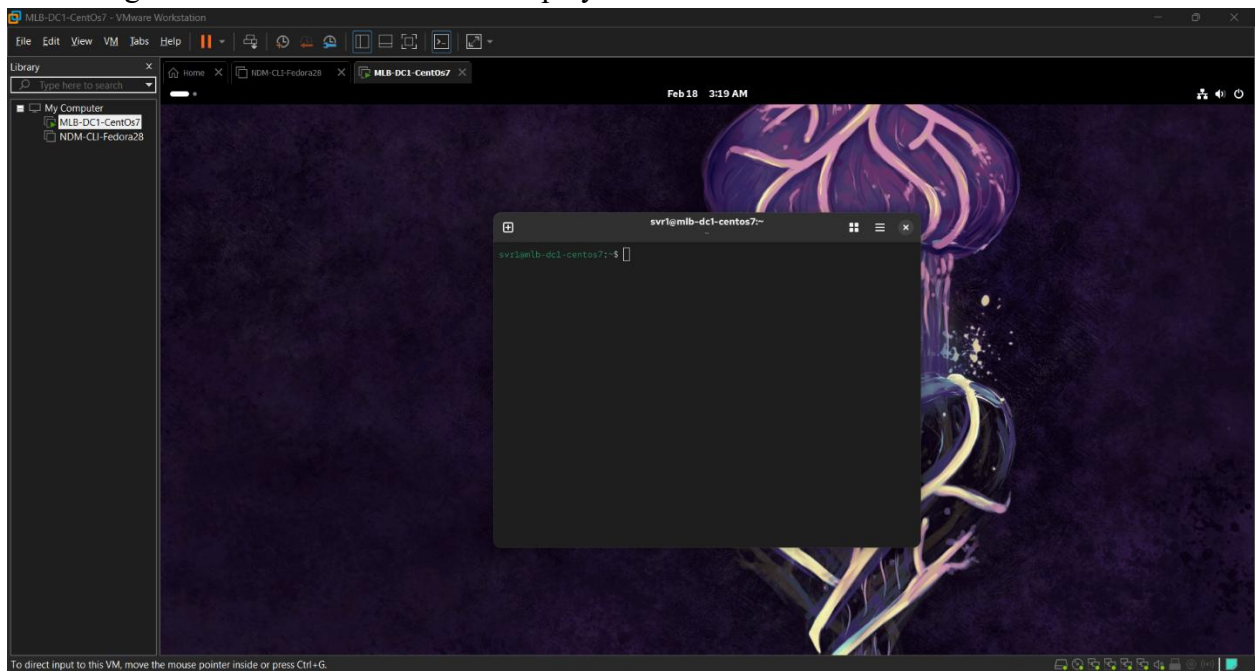
7. Then quit.



8. Finally, reboot the system.

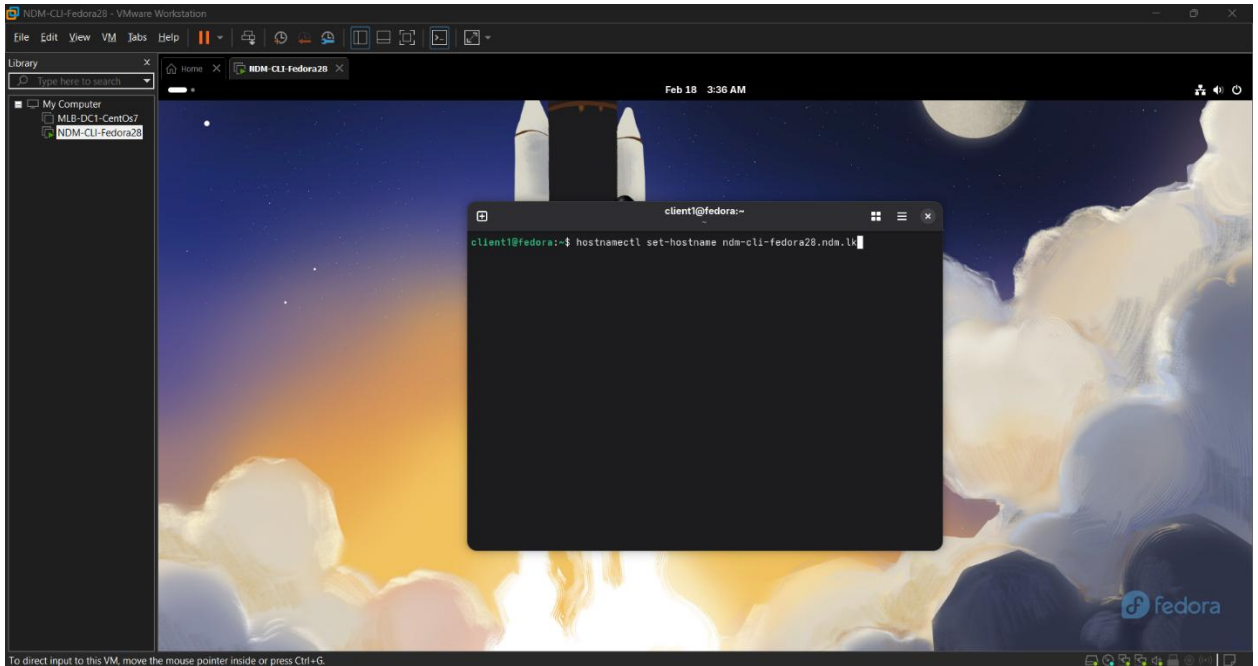


9. The changed hostname should now be displayed.

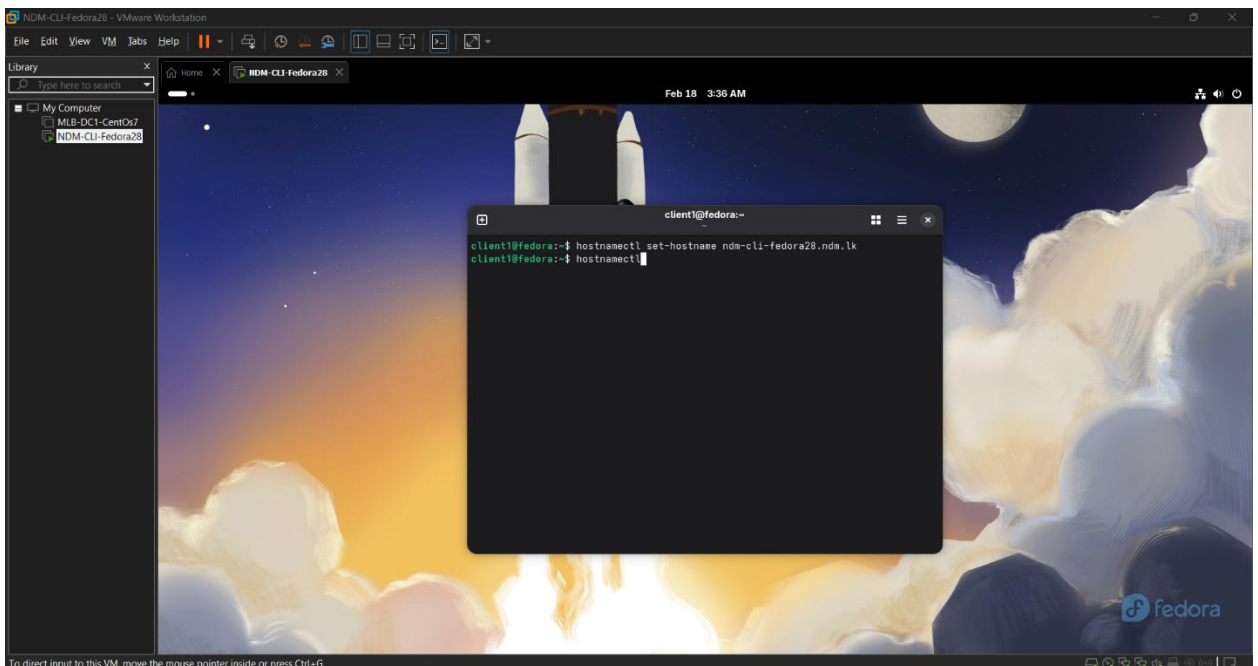


➤ Client

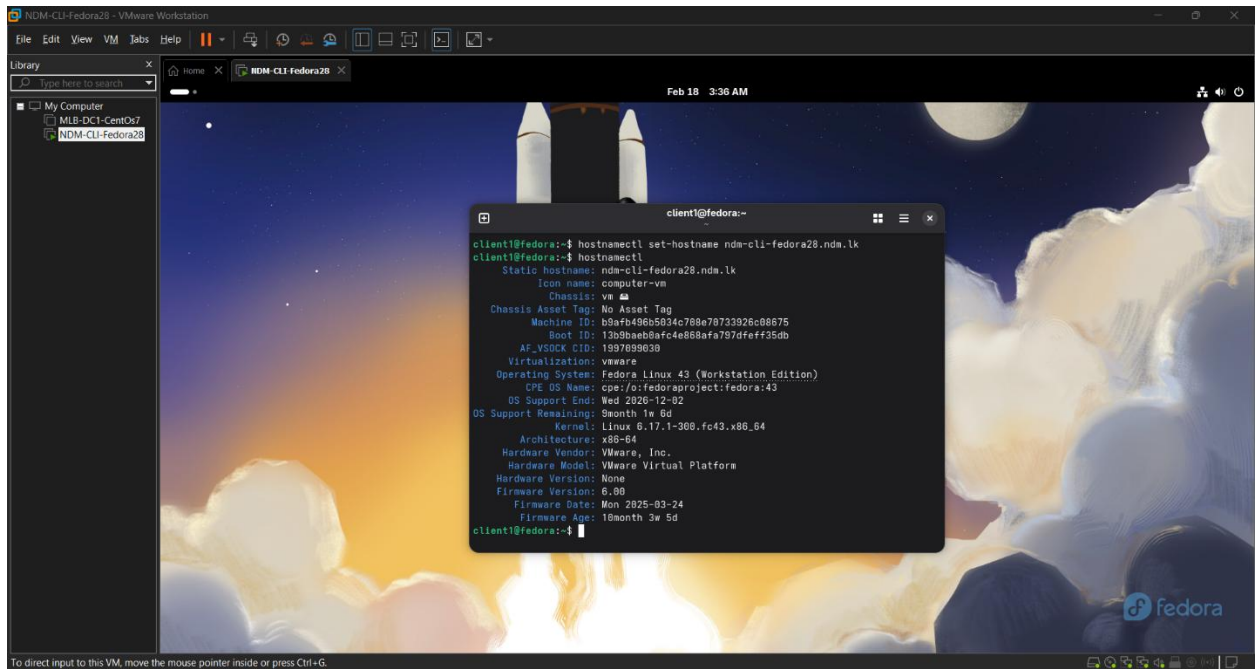
1. Power on the Fedora virtual machine.
2. Open the terminal, type `hostnamectl set-hostname ndm-cli-fedora28.ndm.lk`, and press Enter.



3. Again, type `hostnamectl` in the terminal and press Enter.



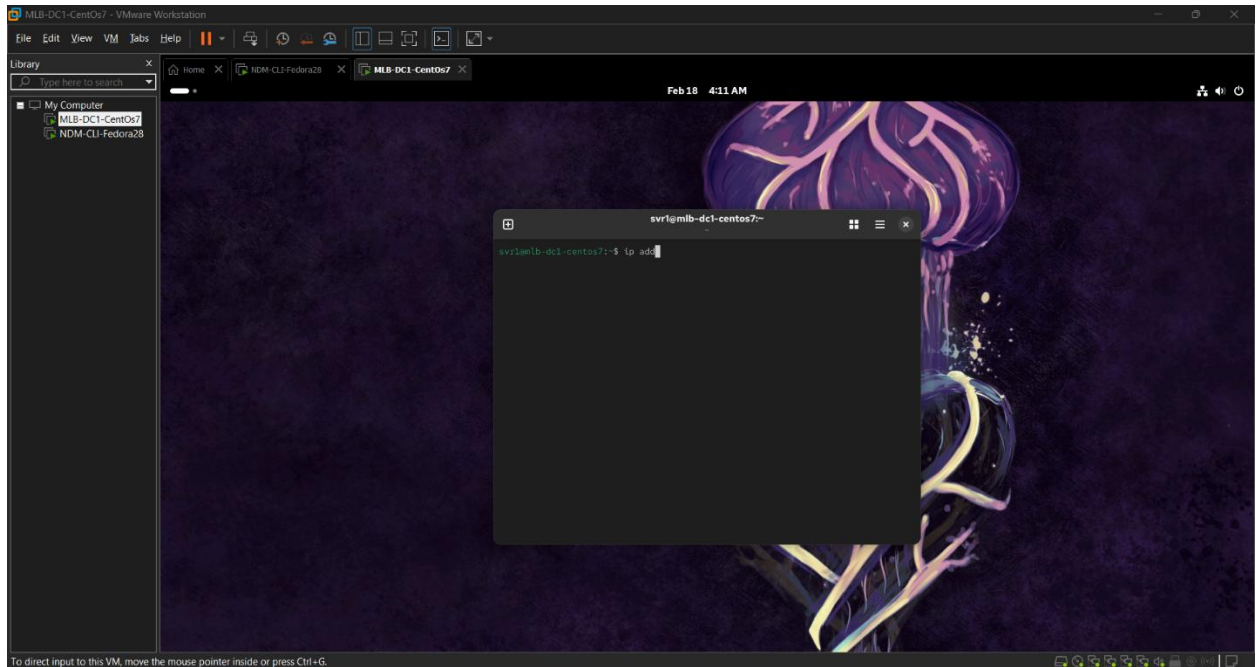
4. The changed hostname should now be displayed.



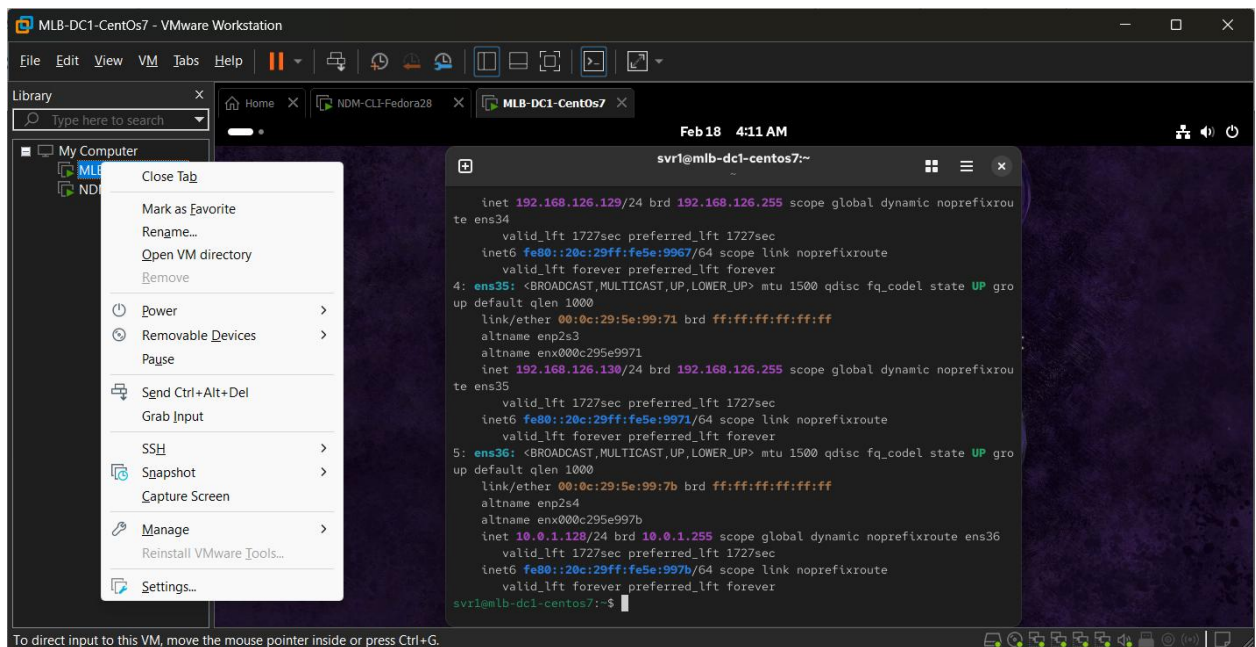
Change IP addresses

➤ Server

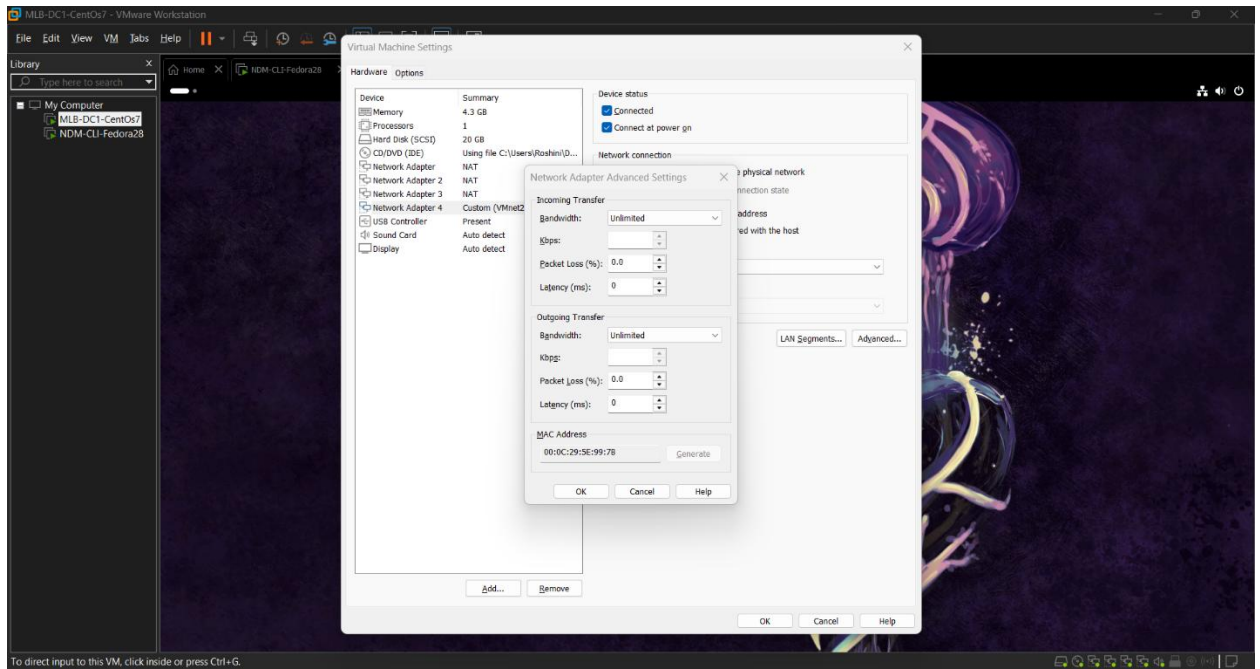
1. Power on the CentOS virtual machine.
2. Open the terminal, type `ip add`, and press Enter.



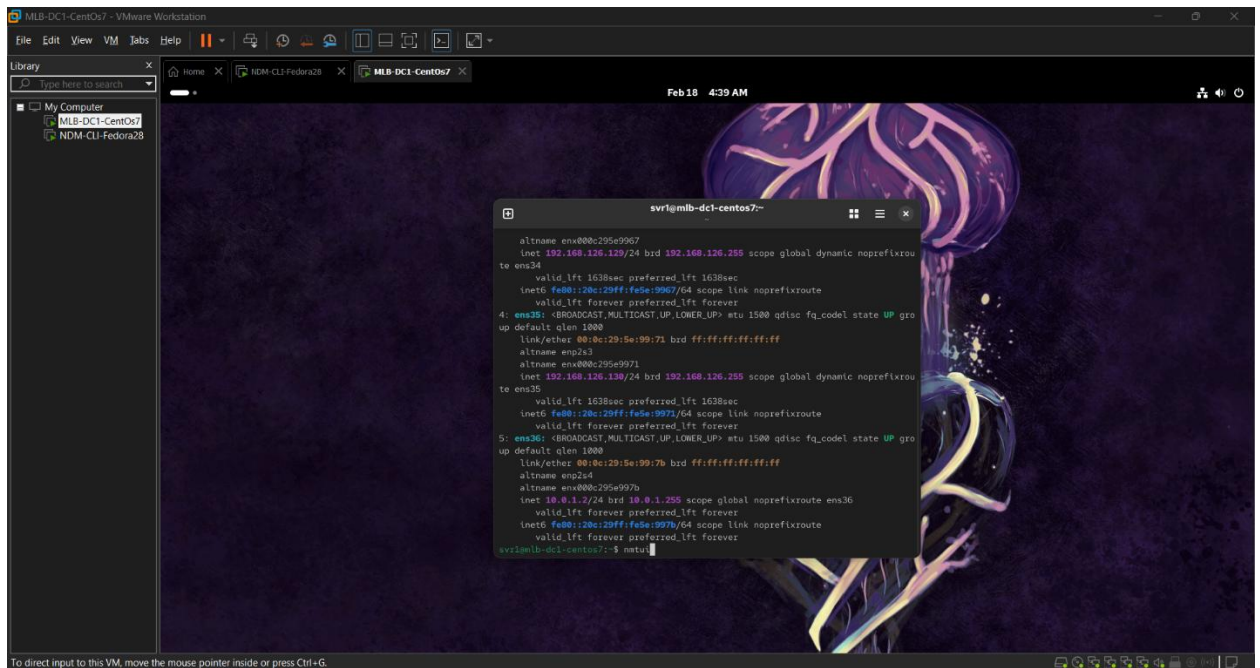
3. Right-click 'MLB-DC1-CentOS7' and go to 'Settings'.



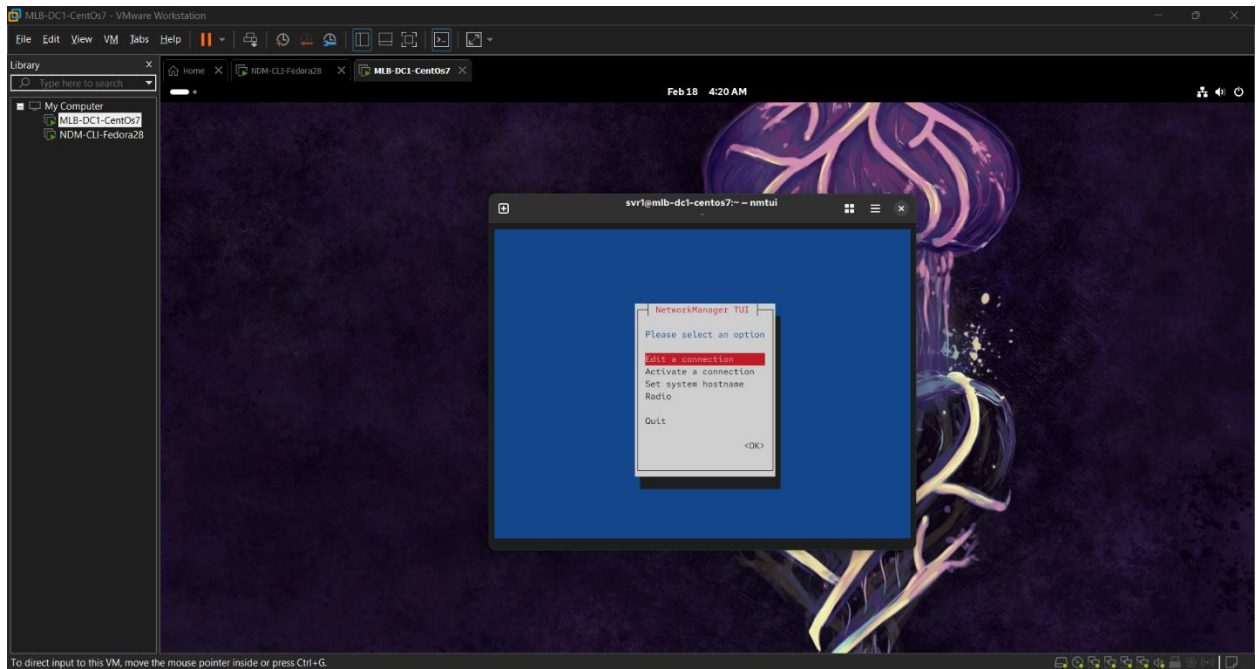
4. Go to 'Network Adapter', select 'Advanced', copy the MAC address, and click 'OK'.



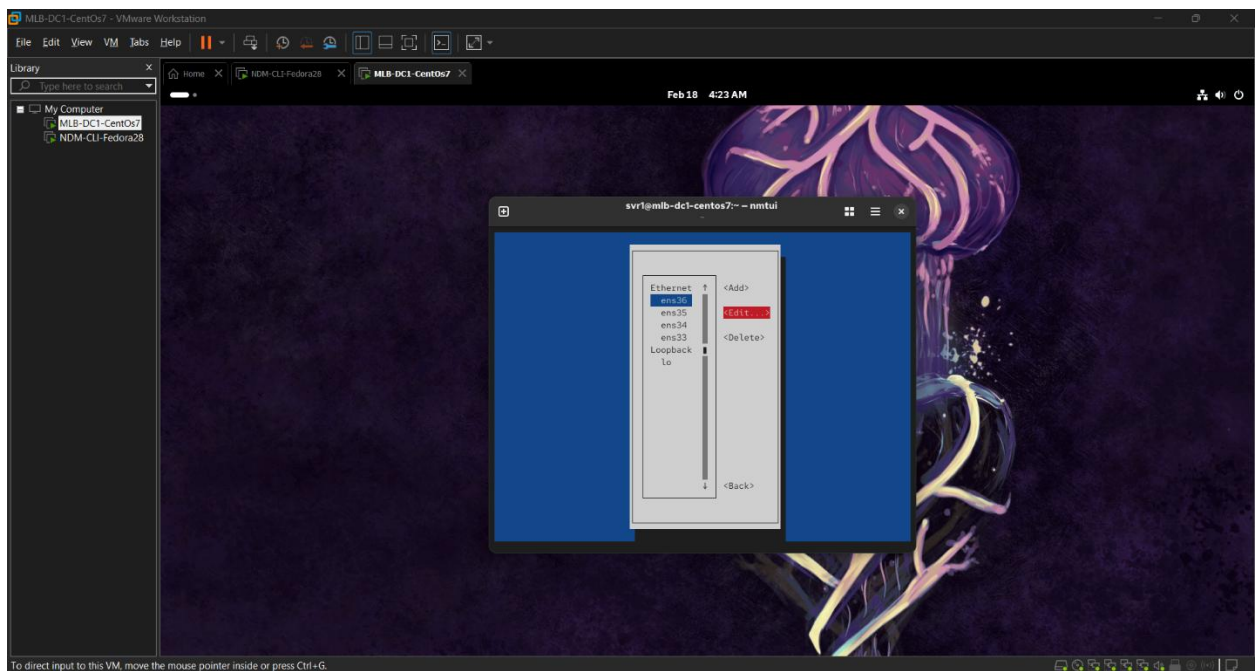
5. In the terminal, enter nmtui and press Enter.



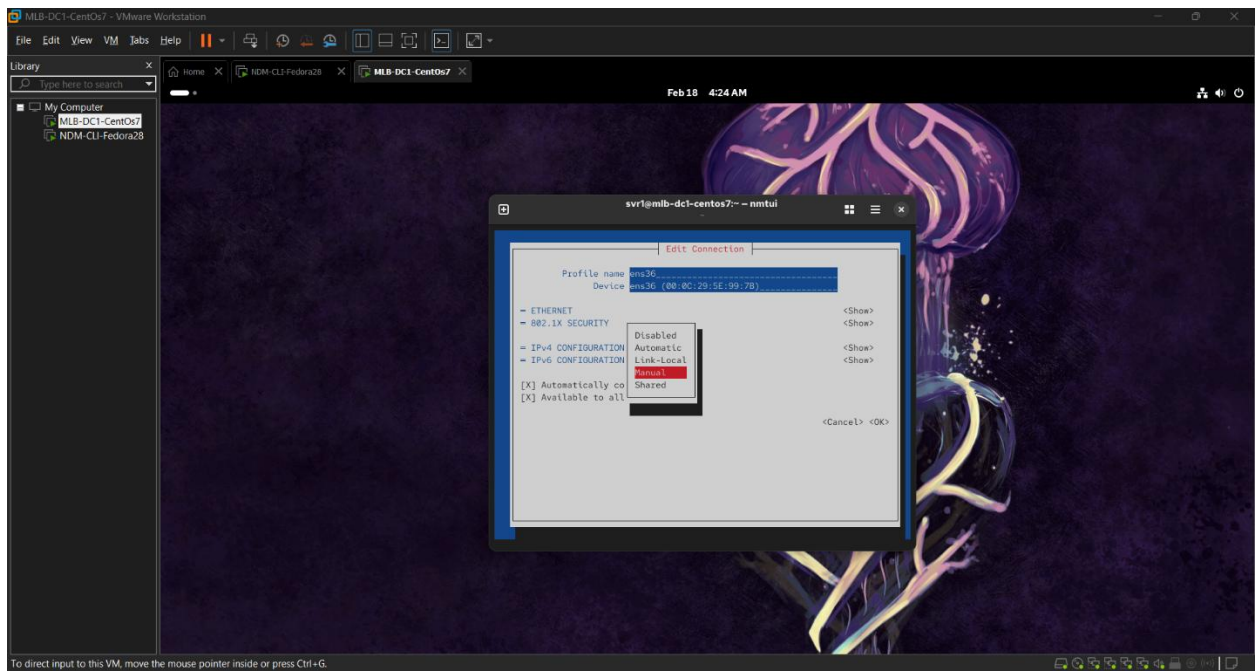
6. Then click the 'Edit Connection' button.



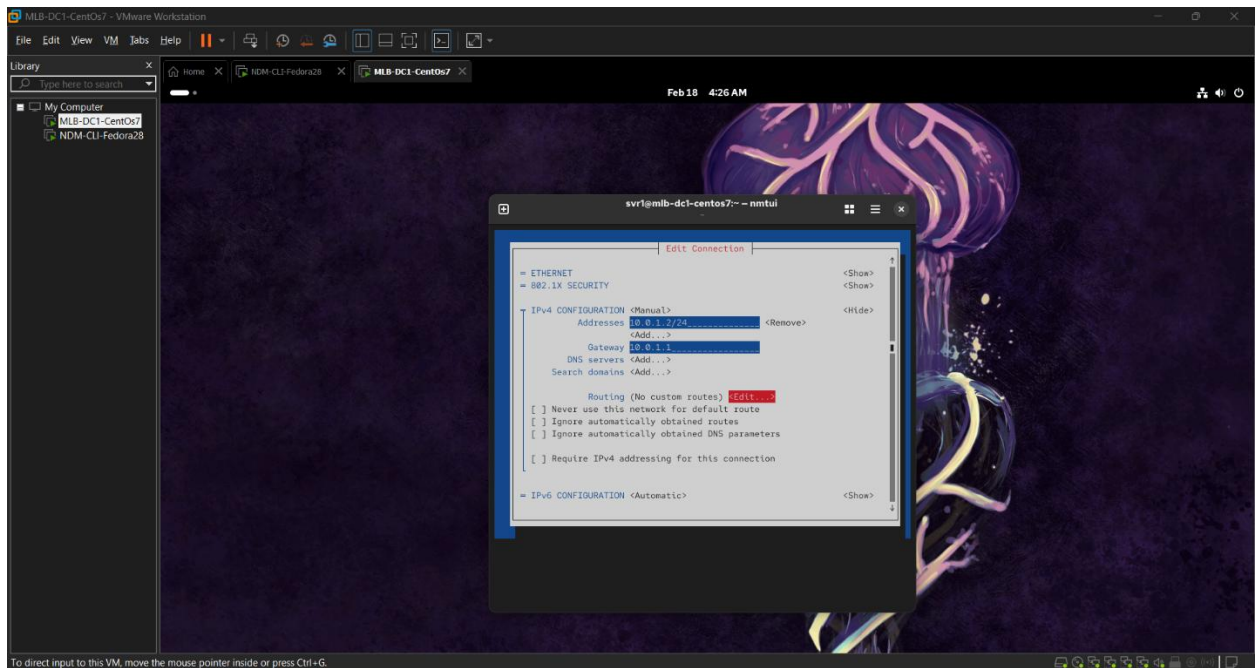
7. Choose the correct Ethernet connection and click 'Edit'.



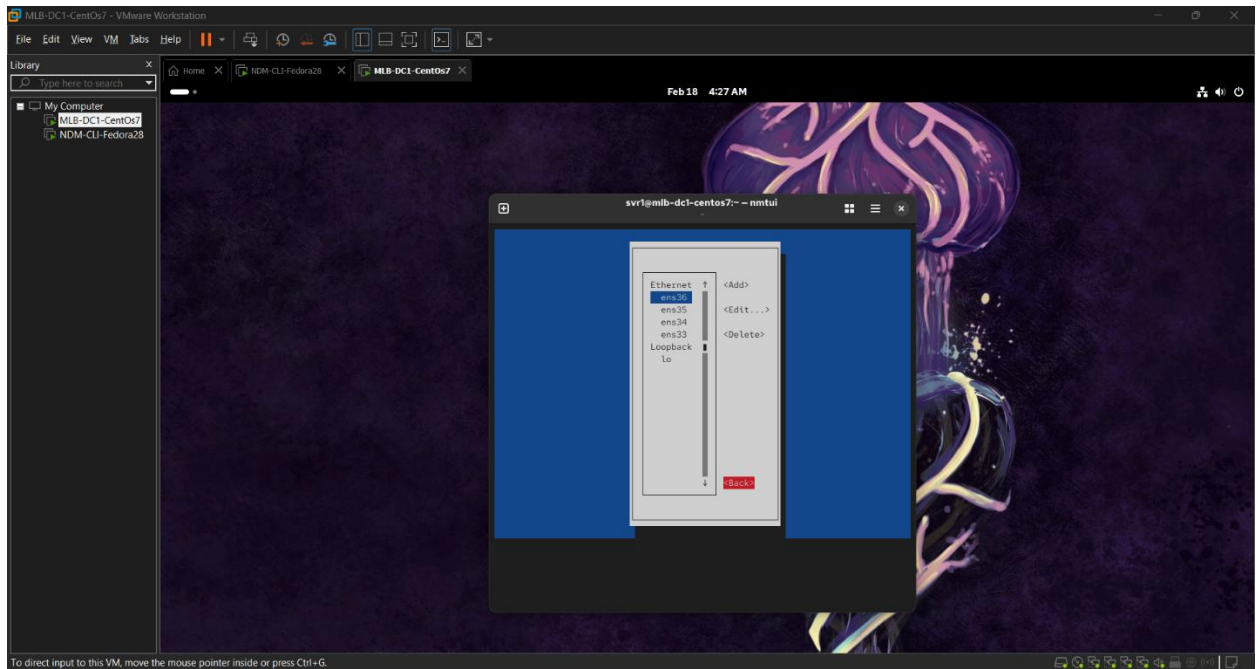
8. Change IPv4 to 'Manual', then click 'Show'.



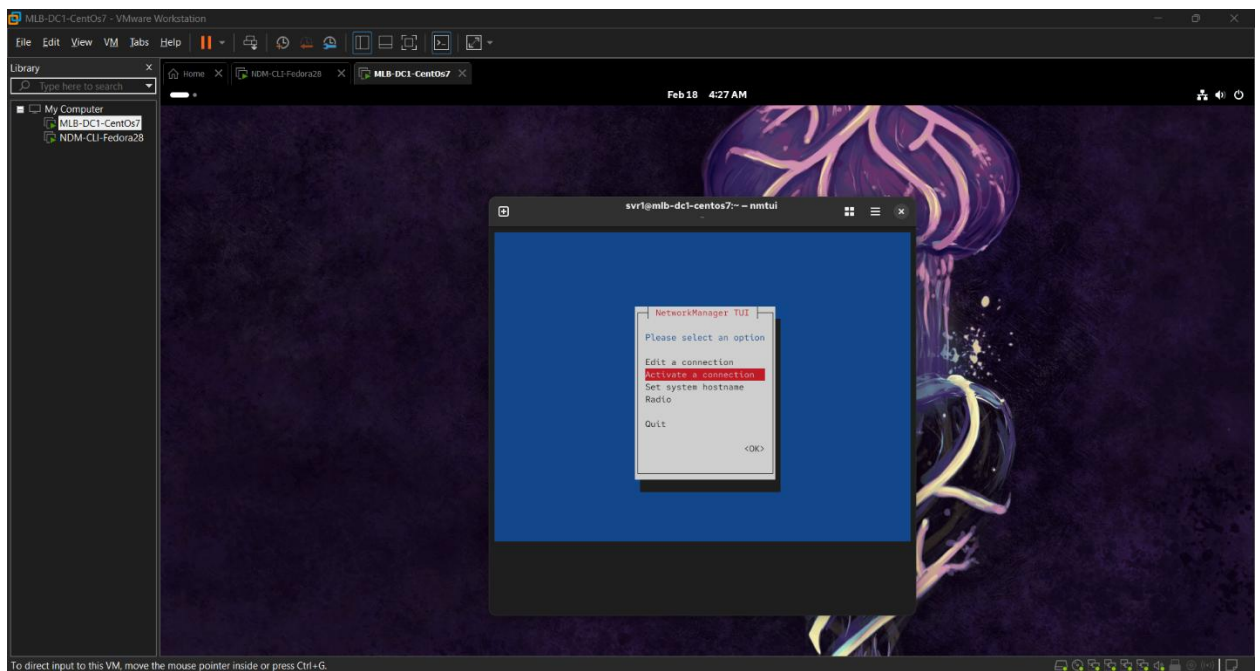
9. Enter the address and click the 'OK' button.



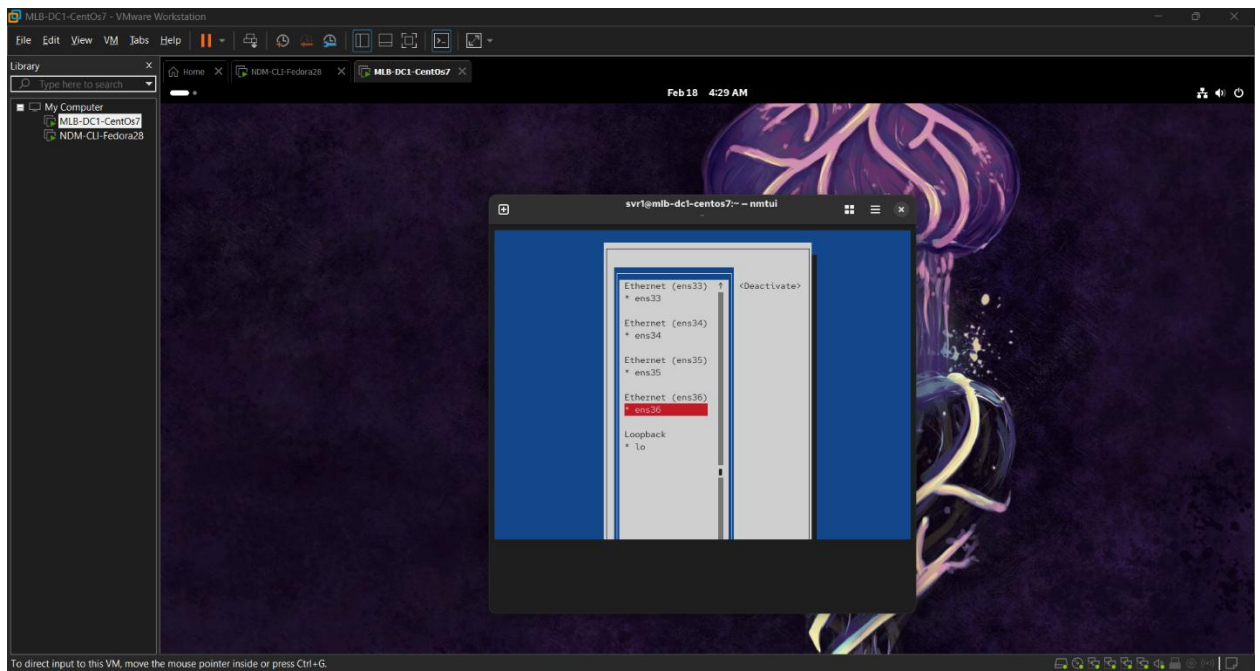
10. Then go back.



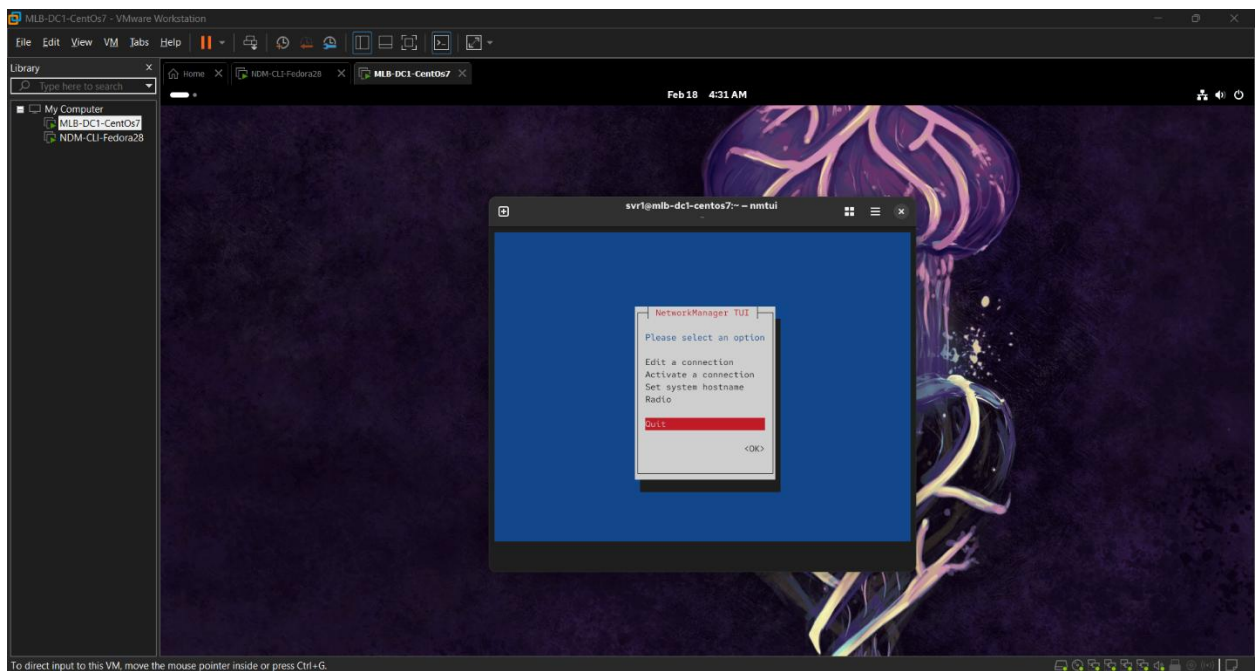
11. Open the 'Activate a Connection' option.



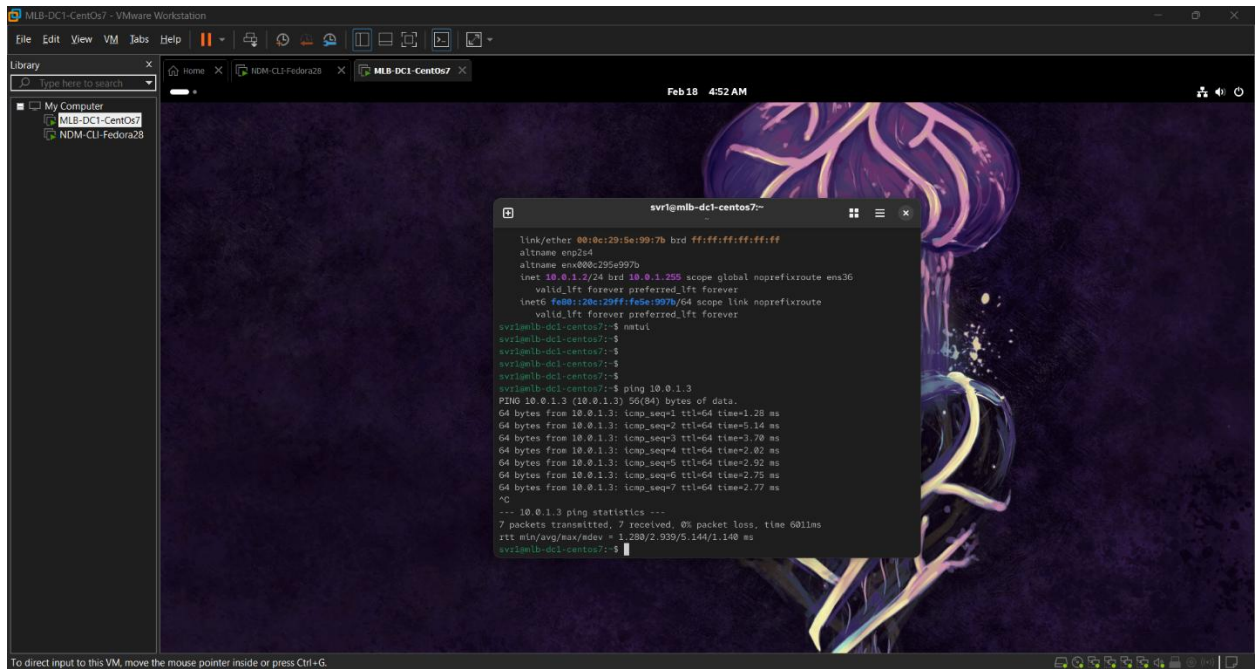
12. Deactivate the Ethernet connection, then click ‘OK’.



13. Click ‘Quit’.



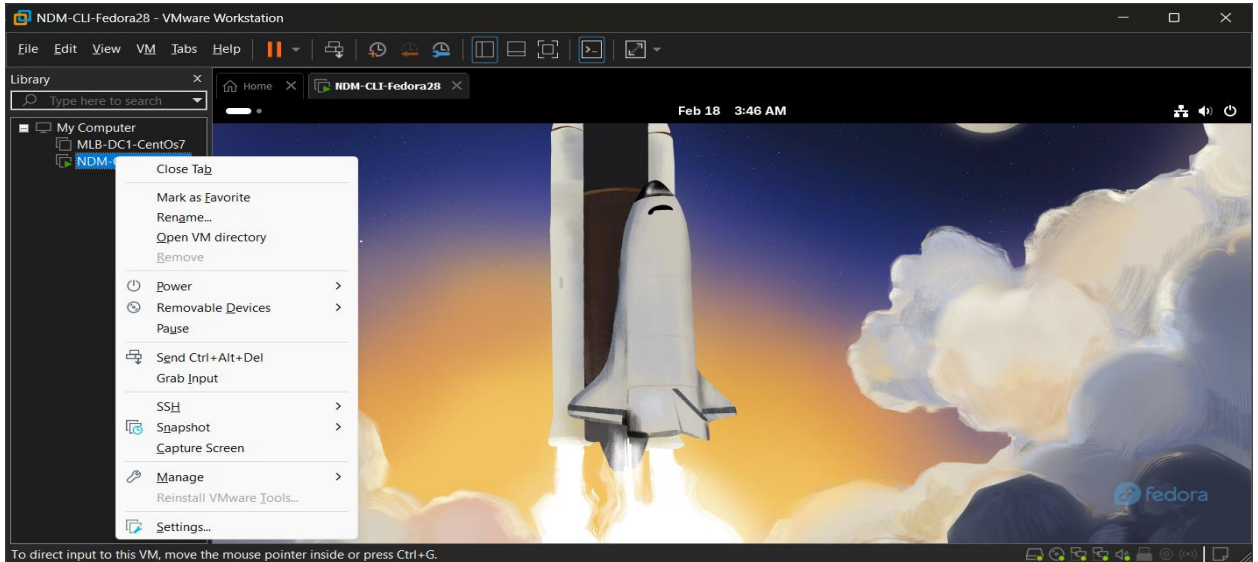
14. Open the terminal and type the ping command.(To stop the command in the terminal, press Ctrl+C)



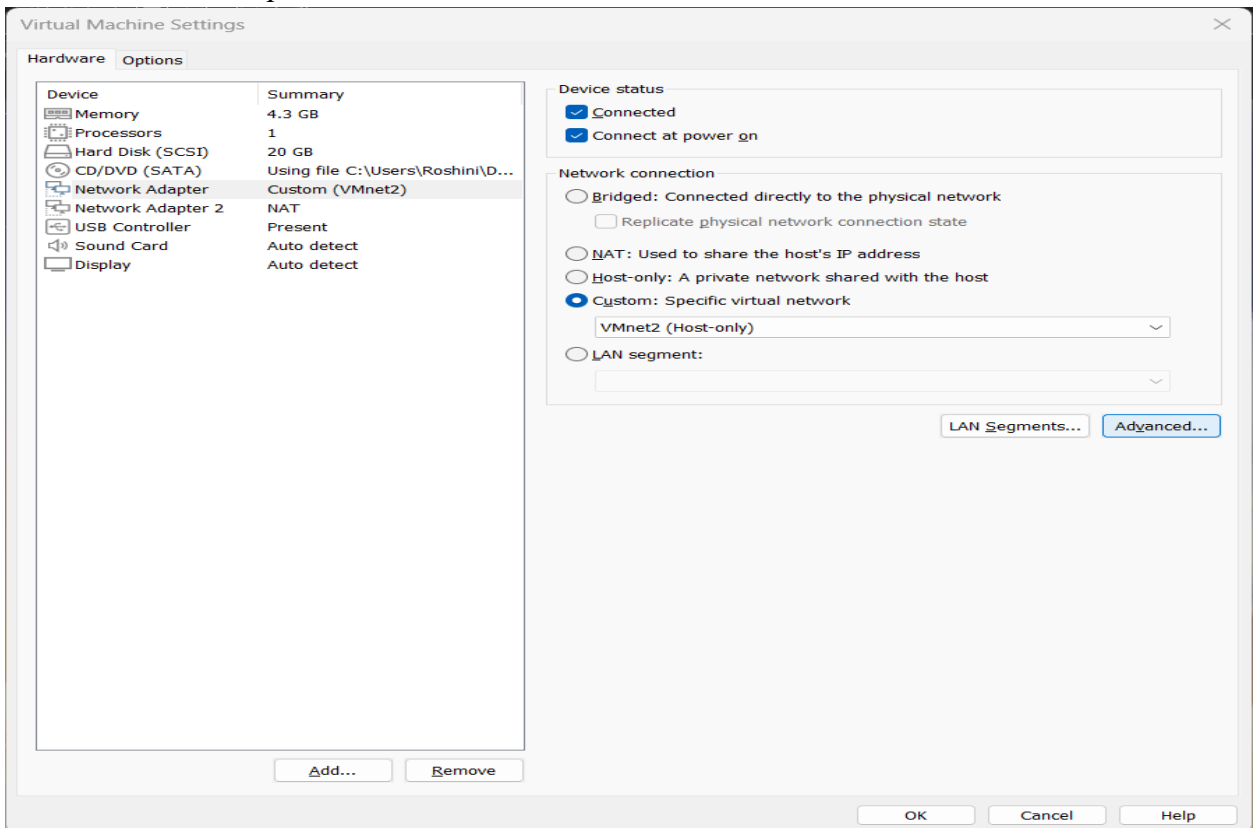
```
link/ether 00:0c:29:5e:99:7b brd ff:ff:ff:ff:ff:ff
aliasname ens254
aliasname ens800:295e997b
inet 10.0.1.2/24 brd 10.0.1.255 scope global noprefixroute ens36
    valid_lft forever preferred_lft forever
inet6 fe80::208:295f:fe99:7b/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
svr@mlb-dcl-centos7:~$ netui
svr@mlb-dcl-centos7:~$
svr@mlb-dcl-centos7:~$
svr@mlb-dcl-centos7:~$
svr@mlb-dcl-centos7:~$ ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data:
64 bytes from 10.0.1.3: icmp_seq=1 ttl=64 time=1.28 ms
64 bytes from 10.0.1.3: icmp_seq=2 ttl=64 time=5.14 ms
64 bytes from 10.0.1.3: icmp_seq=3 ttl=64 time=3.70 ms
64 bytes from 10.0.1.3: icmp_seq=4 ttl=64 time=2.02 ms
64 bytes from 10.0.1.3: icmp_seq=5 ttl=64 time=2.92 ms
64 bytes from 10.0.1.3: icmp_seq=6 ttl=64 time=2.75 ms
64 bytes from 10.0.1.3: icmp_seq=7 ttl=64 time=2.77 ms
^C
--- 10.0.1.3 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 1.280/2.930/5.144/1.140 ms
svr@mlb-dcl-centos7:~$
```

➤ Client

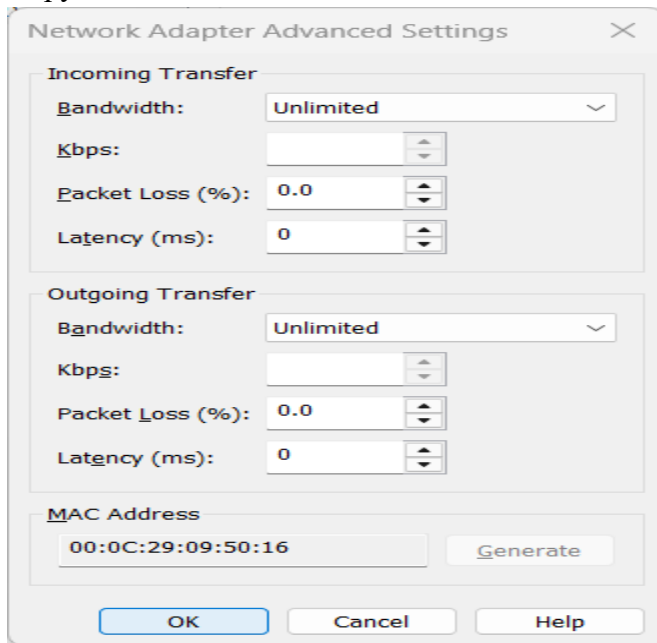
1. Power on the Fedora virtual machine.
2. Go to 'Settings'.



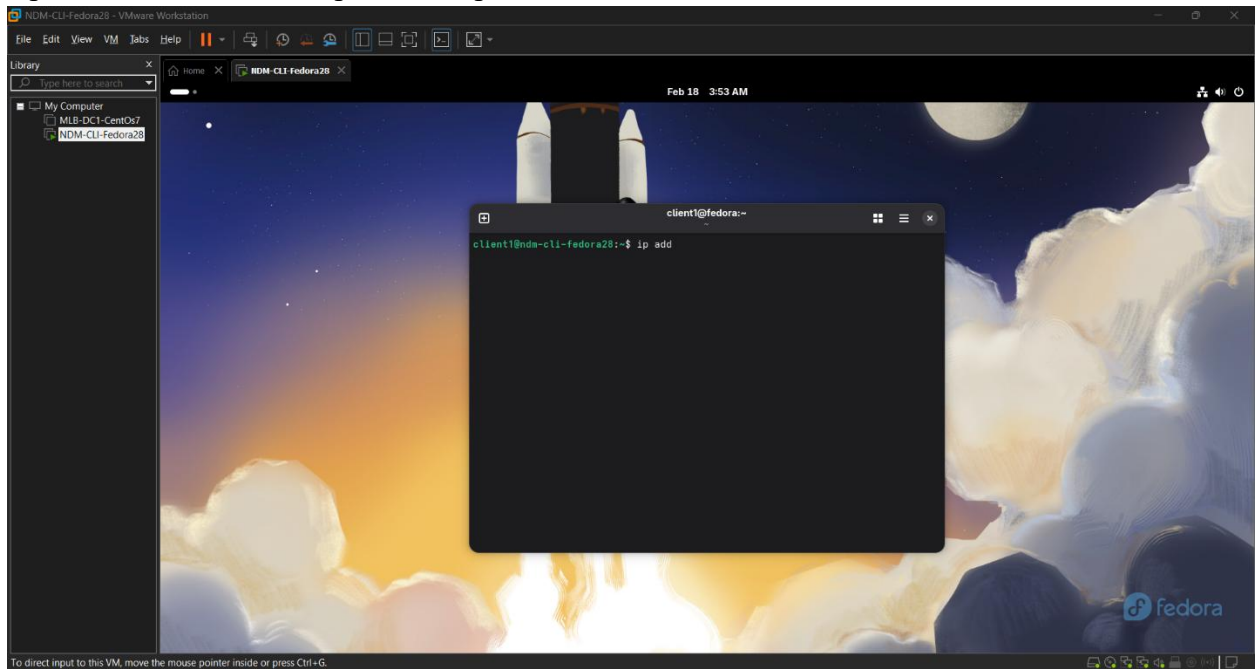
3. Go to 'Network Adapter', then click 'Advanced'.

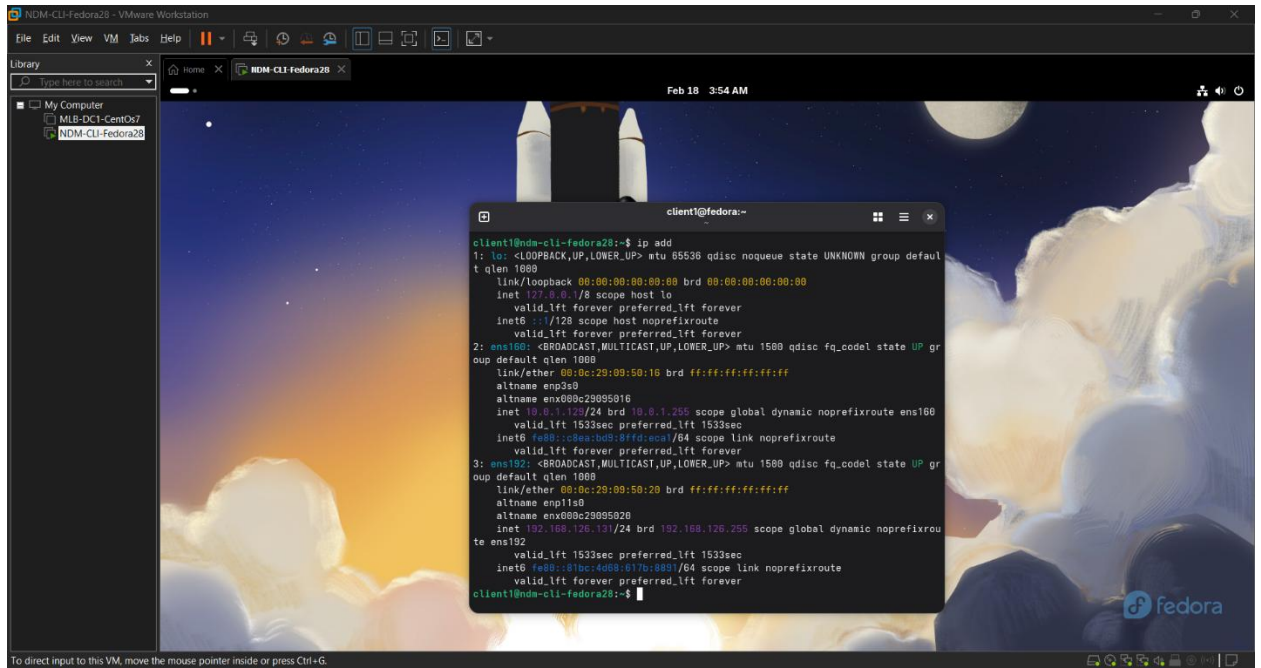


- Copy the MAC address, then click 'OK'.

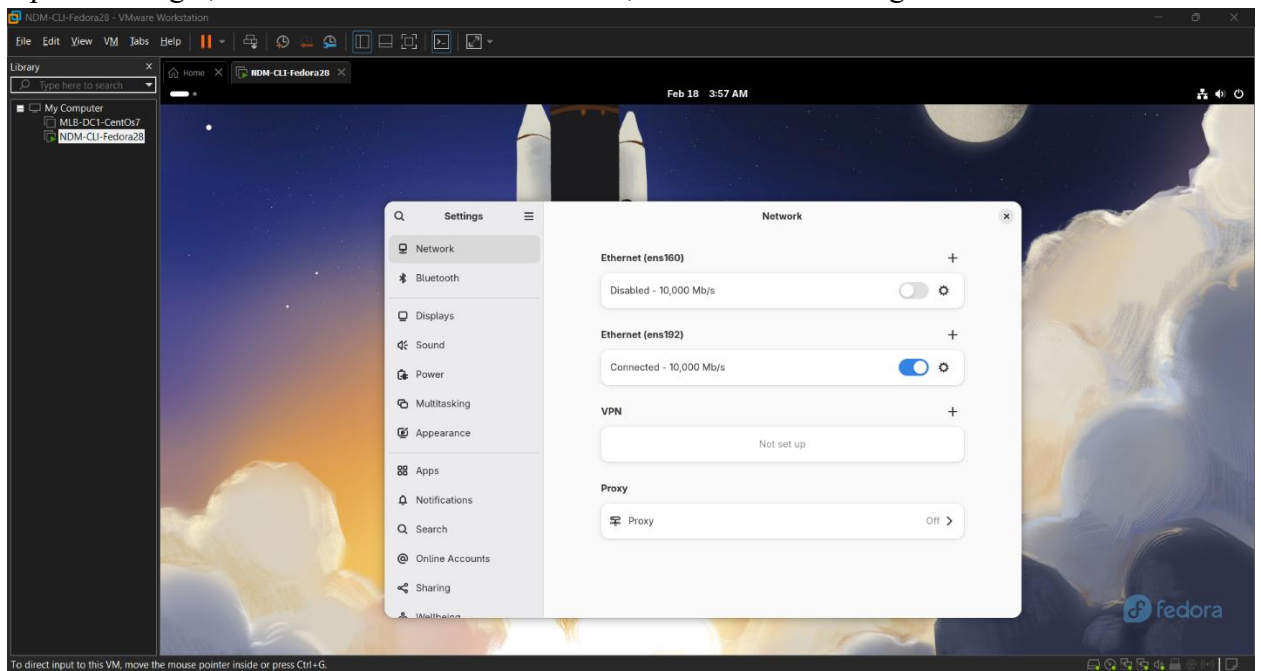


- Click 'OK' again.
- Open the terminal, enter ip add, and press Enter.

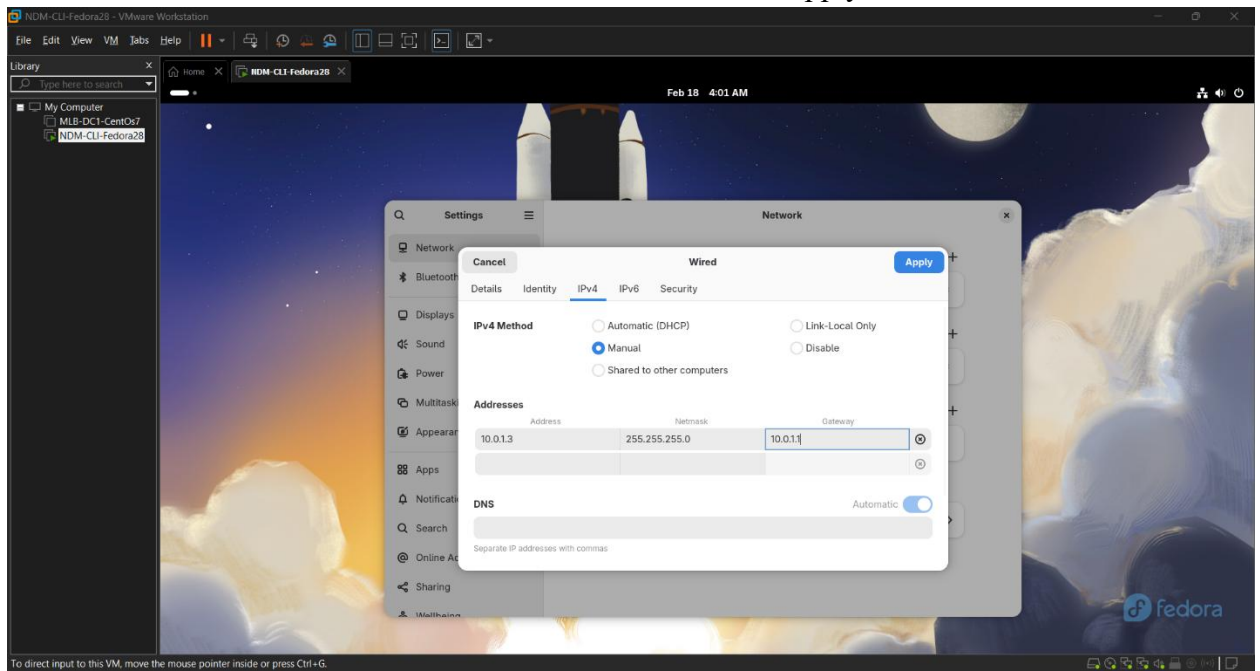




7. Open the 'Settings'.
8. Open 'Settings', disable the Ethernet connection, and click the Settings icon.



9. Go to 'IPv4', choose 'Manual', enter the address, and click 'Apply'.



10. Turn on Ethernet, open 'Settings', and check that the changes have been saved.

